



ALGEBRA

QUESTIONS

FROM 2010 TO 2020 MAINS

SSC CGL MAINS 2010

{01 Aug. 2010}

1. If $a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} = 4$, then the value of $a^2 + b^2$ will be-

यदि $a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} = 4$ हो, तो $a^2 + b^2$ का मान होगा?

- (a) 1 (b) $1\frac{1}{2}$
(c) 2 (d) $2\frac{1}{2}$

2. If $\left(x + \frac{1}{x}\right)^2 = 3$ then the value of

$\left(x^3 + \frac{1}{x^3}\right)$ is

यदि $\left(x + \frac{1}{x}\right)^2 = 3$ हो, तो $\left(x^3 + \frac{1}{x^3}\right)$ बराबर होगा?

- (a) 3 (b) 2
(c) 1 (d) 0

3. $\frac{0.1 \times 0.1 \times 0.1 + 0.02 \times 0.02 \times 0.02}{0.2 \times 0.2 \times 0.2 + 0.04 \times 0.04 \times 0.04} = ?$

- (a) 0.125 (b) 0.250
(c) 0.500 (d) 0.855

4. If $x + \frac{1}{x} = 2$, then the value of

$x^{100} + \frac{1}{x^{100}}$ is-

यदि $x + \frac{1}{x} = 2$ है, तो $x^{100} + \frac{1}{x^{100}}$ का मान होगा?

- (a) 2 (b) 0
(c) 1 (d) -2

5. If $x^3 + 3x^2 + 3x = 7$, then x is equal to

यदि $x^3 + 3x^2 + 3x = 7$ है, तो x बराबर होगा-

- (a) 2 (b) $\sqrt[3]{6}$
(c) 1 (d) -1

6. If $2x + \frac{2}{x} = 1$, then the value of

$x^3 + \frac{1}{x^3}$ is

यदि $2x + \frac{2}{x} = 1$ हो, तो $x^3 + \frac{1}{x^3}$ का मान होगा-

- (a) $\frac{13}{8}$ (b) $-\frac{11}{8}$
(c) $\frac{8}{11}$ (d) $-\frac{13}{8}$

7. If $2x + \frac{1}{3x} = 6$, then $3x + \frac{1}{2x}$ is equal to

यदि $2x + \frac{1}{3x} = 6$ हो, तो $3x + \frac{1}{2x}$ बराबर होगा-

- (a) 4 (b) 8
(c) 9 (d) 12

8. If $x = (\sqrt{2} - 1)^{-\frac{1}{2}}$ then the value of

$\left(x^2 - \frac{1}{x^2}\right)$ is

यदि $x = (\sqrt{2} - 1)^{-\frac{1}{2}}$ हो, तो $\left(x^2 - \frac{1}{x^2}\right)$ का मान होगा-

- (a) 2 (b) $-2\sqrt{2}$
(c) $2\sqrt{2}$ (d) $-\sqrt{2}$

9. $\frac{(0.87)^3 + (0.13)^3}{(0.87)^2 + (0.13)^2 - (0.87) \times (0.13)} = ?$

- (a) $\frac{1}{2}$ (b) 2
(c) 1 (d) $2\frac{1}{2}$

10. If $x^2 + y^2 - 2x + 6y + 10 = 0$ then the value of the $(x^2 + y^2)$ is-

यदि $x^2 + y^2 - 2x + 6y + 10 = 0$ हो, तो $(x^2 + y^2)$ का मान होगा-

- (a) 4 (b) 6
(c) 8 (d) 10

11. If $x^{1/3} + y^{1/3} = z^{1/3}$, then $(x + y - z)^3 + 27xyz$ is equal to

यदि $x^{1/3} + y^{1/3} = z^{1/3}$ हो, तो $(x + y - z)^3 + 27xyz$ बराबर होगा?

- (a) 0 (b) 1
(c) -1 (d) 27

12. If $a^2 = 2$, then $(a + 1)$ is equal to

यदि $a^2 = 2$ हो, तो $(a + 1)$ बराबर होगा-

- (a) $a - 1$ (b) $\frac{2}{a - 1}$
(c) $\frac{a + 1}{3 - 2a}$ (d) $\frac{a - 1}{3 - 2a}$

13. If $a + b + c = 1$ and $ab + bc + ca = 1/3$ then $a : b : c$ is

यदि $a + b + c = 1$ और $ab + bc + ca = 1/3$ हो, तो $a : b : c$ होगा-

- (a) 1 : 2 : 2 (b) 2 : 1 : 2
(c) 1 : 1 : 1 (d) 1 : 2 : 1

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{04 Sept. 2011}

14. If $a^2 + b^2 + c^2 = 2(a - b - c) - 3$, then the value of $(a - b + c)$ is

यदि $a^2 + b^2 + c^2 = 2(a - b - c) - 3$ तो $(a - b + c)$ का मान है-

- (a) -1 (b) 3
(c) 1 (d) -2

15. If $x^2 + 3x + 1 = 0$, then the value

of $x^3 + \frac{1}{x^3}$ is

यदि $x^2 + 3x + 1 = 0$ तो $x^3 + \frac{1}{x^3}$ का मान है-

- (a) -18 (b) 18
(c) 36 (d) -36

16. If $x^a \cdot x^b \cdot x^c = 1$ then the value of $a^3 + b^3 + c^3$ is

यदि $x^a \cdot x^b \cdot x^c = 1$ तो $a^3 + b^3 + c^3$ का मान है-

- (a) 9 (b) abc
(c) $a + b + c$ (d) $3abc$

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{16 Sept. 2012}

17. If $a + \frac{1}{a} + 2 = 0$, then the value

of $a^{37} - \frac{1}{a^{100}}$ is

यदि $a + \frac{1}{a} + 2 = 0$ तो $a^{37} - \frac{1}{a^{100}}$ का मान है-

- (a) 0 (b) -2
(c) 1 (d) 2

18. The value of k for which the graph of $(k-1)x + y - 2 = 0$ and $(2-k)x - 3y + 1 = 0$ are parallel is

k का मान जिसके लिए $(k-1)x + y - 2 = 0$ और $(2-k)x - 3y + 1 = 0$ के ग्राफ समानांतर हो, है-

- (a) $1/2$ (b) $-1/2$
(c) 2 (d) -2

19. If $x + \frac{1}{16x} = 1$, then the value of

$64x^3 + \frac{1}{64x^3}$ is

यदि $x + \frac{1}{16x} = 1$ तो $64x^3 + \frac{1}{64x^3}$ का मान है-

- (a) 4 (b) 52
(c) 64 (d) 76

20. If a, b, c are three non-zero real numbers such that $a + b + c = 0$, and $b^2 \neq ca$, then the value of $\frac{a^2 + b^2 + c^2}{b^2 - ca}$ is

यदि a, b और c तीन शून्येतर वास्तविक संख्याएँ इस प्रकार हैं कि $a + b + c = 0$

तथा $b^2 \neq ca$ तो $\frac{a^2 + b^2 + c^2}{b^2 - ca}$ का मान है-

- (a) 3 (b) 2
(c) 0 (d) 1

21. If $a^4 + a^2b^2 + b^4 = 8$ and $a^2 + ab + b^2 = 4$, then the value of ab is-

यदि $a^4 + a^2b^2 + b^4 = 8$ और $a^2 + ab + b^2 = 4$ है, तो ab का मान है-

- (a) -1 (b) 0
(c) 2 (d) 1

22. If $a = 25, b = 15, c = -10$, then the value of

$\frac{a^3 + b^3 + c^3 - 3abc}{(a-b)^2 + (b-c)^2 + (c-a)^2}$ is

यदि $a = 25, b = 15, c = -10$ तो

$\frac{a^3 + b^3 + c^3 - 3abc}{(a-b)^2 + (b-c)^2 + (c-a)^2}$ का मान है-

- (a) 30 (b) -15
(c) -30 (d) 15

23. If $x = \frac{\sqrt{3}}{2}$, then the value of

$\frac{1+x}{1+\sqrt{1+x}} + \frac{1-x}{1-\sqrt{1-x}}$ is equal to

यदि $x = \frac{\sqrt{3}}{2}$ है, तो $\frac{1+x}{1+\sqrt{1+x}} +$

$\frac{1-x}{1-\sqrt{1-x}}$ का मान बराबर है?

- (a) $\frac{\sqrt{3}}{2}$ (b) $\sqrt{3}$

- (c) 0 (d) 1

24. The real value of x , that satisfies the equation $\sqrt{4x-9} + \sqrt{4x+9} = 5 + \sqrt{7}$ is:-

x का वास्तविक मान, जो समीकरण $\sqrt{4x-9} + \sqrt{4x+9} = 5 + \sqrt{7}$ को संतुष्ट करता है।

- (a) $\frac{3}{\sqrt{7}}$ (b) 4

- (c) $\sqrt{5}$ (d) $2\sqrt{3}$

25. Find the value of $a^3 + b^3 + c^3 - 3abc$, If $a = 225, b = 226, c = 227$

जब $a = 225, b = 226, c = 227$ दिए हैं, तो $a^3 + b^3 + c^3 - 3abc$ का मान ज्ञात करें?

- (a) 2034 (b) 2340

- (c) 2304 (d) 2430

26. Number of solutions of the equation $\sqrt{x^2 - x + 1} + \frac{1}{\sqrt{x^2 + x + 1}} = 2 - x^2$ is-

$\sqrt{x^2 - x + 1} + \frac{1}{\sqrt{x^2 + x + 1}} = 2 - x^2$

समीकरण के हलों की संख्या है।

- (a) 2 (b) 4

- (c) 0 (d) 1

27. If $x + \frac{a}{x} = 1$, then the value of

$\frac{x^2 + x + a}{x^3 - x^2}$ is-

यदि $x + \frac{a}{x} = 1$ है, तो $\frac{x^2 + x + a}{x^3 - x^2}$ का मान है-

- (a) $\frac{2}{a}$ (b) $-\frac{2}{a}$

- (c) -2 (d) $-\frac{a}{2}$

28. If $\sqrt{28-6\sqrt{3}} = \sqrt{3}a + b$, (where a, b are rational), value of $(a+b)$ is-

यदि $\sqrt{28-6\sqrt{3}} = \sqrt{3}a + b$, (जहाँ a तथा b परिमेय हैं) $(a+b)$ का मान है:-

- (a) 1 (b) -1

- (c) -2 (d) 2

29. If $x = \sqrt{\frac{\sqrt{5}+1}{\sqrt{5}-1}}$, then $x^2 - x - 1$ is equal to

यदि $x = \sqrt{\frac{\sqrt{5}+1}{\sqrt{5}-1}}$, तब $x^2 - x - 1$ किसके बराबर होगा?

- (a) 2 (b) 5

- (c) 0 (d) 1

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{29 Sept. 2013}

30. If $x = y = 333$ and $z = 334$, then the values of $x^3 + y^3 + z^3 - 3xyz$ is-

यदि $x = y = 333$ और $z = 334$ हो, तो $x^3 + y^3 + z^3 - 3xyz$ का मान क्या होगा?

- (a) 0 (b) 667

- (c) 1000 (d) 2334

31. If $\frac{x-a^2}{b+c} + \frac{x-b^2}{c+a} + \frac{x-c^2}{a+b} = 4$ ($a+b+c$), then x is equal to-

$\frac{x-a^2}{b+c} + \frac{x-b^2}{c+a} + \frac{x-c^2}{a+b} = 4$ ($a+b+c$), हो, तो x किसके बराबर होगा?

- (a) $(a+b+c)^2$

- (b) $a^2 + b^2 + c^2$

- (c) $ab + bc + ca$

- (d) $a^2 + b^2 + c^2 - ab - bc - ca$

32. If $(x-a)(x-b) = 1$ and $a-b+5 = 0$, then the value of $(x-a)^3 -$

$\frac{1}{(x-a)^3}$ is-

यदि $(x-a)(x-b) = 1$ तथा $a-b+5 = 0$ हो, तो $(x-a)^3 -$

$\frac{1}{(x-a)^3}$ का मान क्या होगा?

- (a) -125 (b) 1

- (c) 125 (d) 140

33. If $\sqrt{x} = \sqrt{3} - \sqrt{5}$, then the value of $x^2 - 16x + 6$ is-

यदि $\sqrt{x} = \sqrt{3} - \sqrt{5}$, हो, तो $x^2 - 16x + 6$ का मान क्या होगा?

- (a) 0 (b) -2

- (c) 2 (d) 4

34. If $a^2 + b^2 + c^2 = 2(a - b - c) - 3$ then the value of $4a - 3b + 5c$ is—
यदि $a^2 + b^2 + c^2 = 2(a - b - c) - 3$ हो, तो $4a - 3b + 5c$ का मान क्या होगा?

(a) 2 (b) 3
(c) 5 (d) 6

35. If $2x + \frac{2}{x} = 3$, then the value of

$$x^3 + \frac{1}{x^3} + 2 \text{ is:}$$

यदि $2x + \frac{2}{x} = 3$ हो, तो $x^3 + \frac{1}{x^3} + 2$ का मान क्या होगा?

(a) $-\frac{9}{8}$ (b) $-\frac{25}{8}$
(c) $\frac{7}{8}$ (d) 11

36. Out of the given responses, one of the factors of $(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3$ is

नीचे दिए गए प्रत्युत्तरों में से $(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3$ का एक घटक कौन-सा होगा?

(a) $(a + b)(a - b)$
(b) $(a + b)(a + b)$
(c) $(a - b)(a - b)$
(d) $(b - c)(b - c)$

37. If $x = \sqrt[3]{5} + 2$ then, the value of $x^3 - 6x^2 + 21x - 13$ is—

यदि $x = \sqrt[3]{5} + 2$ हो, तो $x^3 - 6x^2 + 21x - 13$ का मान क्या होगा—

(a) -1 (b) 1
(c) 2 (d) 0

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38. If $a^2 + b^2 + c^2 = ab + bc + ac$

then the value of $\frac{a+c}{b}$ is -

यदि $a^2 + b^2 + c^2 = ab + bc + ac$ हो,

तो $\frac{a+c}{b}$ का मान क्या होगा ?

(a) 0 (b) 2
(c) 1 (d) -1

39. If $ab+bc+ca=0$, then the value of

$$\left(\frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab} \right) \text{ is -}$$

यदि $ab+bc+ca=0$ हो, तो

$$\left(\frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab} \right) \text{ का मान क्या होगा?}$$

(a) 0 (b) 1
(c) 3 (d) $a + b + c$

40. If $(2 + \sqrt{3})a = (2 - \sqrt{3})b = 1$, then

the value of $\frac{1}{a} + \frac{1}{b}$ is -

यदि $(2 + \sqrt{3})a = (2 - \sqrt{3})b = 1$, तो

$\frac{1}{a} + \frac{1}{b}$ का मान क्या होगा?

(a) 1 (b) 2
(c) $2\sqrt{3}$ (d) 4

41. If $3x + \frac{3}{x} = 1$, then $x^3 + \frac{1}{x^3} + 1$ is?

यदि $3x + \frac{3}{x} = 1$ तो $x^3 + \frac{1}{x^3} + 1$ का मान क्या होगा?

(a) $-1/29$ (b) $1/27$
(c) $-1/36$ (d) $1/24$

42. The factors of $(a^2 + 4b^2 + 4b - 4ab - 2a - 8)$ will be

$(a^2 + 4b^2 + 4b - 4ab - 2a - 8)$ के गुणखण्ड क्या होंगे।

(a) $(a - 2b - 4)(a - 2b + 2)$
(b) $(a - b + 2)(a - 4b - 4)$
(c) $(a + 2b - 4)(a + 2b + 2)$
(d) $(a + 2b - 1)(a - 2b + 1)$

43. The value of $\frac{1}{a^2 + ax + x^2} -$

$$\frac{1}{a^2 - ax + x^2} + \frac{2ax}{a^4 + a^2x^2 + x^4} \text{ is -}$$

$$\frac{1}{a^2 + ax + x^2} - \frac{1}{a^2 - ax + x^2} +$$

$$\frac{2ax}{a^4 + a^2x^2 + x^4} \text{ का मान क्या होगा?}$$

(a) 2 (b) 1
(c) -1 (d) 0

44. If $4x + 5y = 83$ and $3x : 2y = 21 : 22$, then $y - x$ equals -

यदि $4x + 5y = 83$ और $3x : 2y = 21 : 22$, तो $y - x$ किसके बराबर होगा?

(a) 3 (b) 4
(c) 7 (d) 11

45. If $x = 11$, then the value of

$$x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1 \text{ is -}$$

यदि $x = 11$ हो तो $x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1$ का मान क्या होगा?

(a) 5 (b) 10
(c) 15 (d) 20

46. If $p = 99$, then the value of $p(p^2 + 3p + 3)$ is—

यदि $p = 99$, तो $p(p^2 + 3p + 3)$ का मान क्या होगा?

(a) 10000000 (b) 9990000
(c) 9999999 (d) 9900000

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{12 Apr. 2015} PAPER - I

47. If $\frac{3-5x}{2x} + \frac{3-5y}{2y} + \frac{3-5z}{2z} = 0$,

then the value of $\frac{2}{x} + \frac{2}{y} + \frac{2}{z}$ is

$$\text{यदि } \frac{3-5x}{2x} + \frac{3-5y}{2y} + \frac{3-5z}{2z} = 0 \text{ है,}$$

तो $\frac{2}{x} + \frac{2}{y} + \frac{2}{z}$ का मान ज्ञात करें।

(a) 5 (b) 15
(c) 20 (d) 10

48. If $2s = a + b + c$, then the value of $s(s - c) + (s - a)(s - b)$ is

यदि $2s = a + b + c$ है, तो $s(s - c) + (s - a)(s - b)$ का मान ज्ञात करें।

(a) abc (b) $\frac{a+b+c}{2}$

(c) ab (d) 0

49. If $p + m = 6$ and $p^3 + m^3 = 72$, then the value of pm is

यदि $p + m = 6$ और $p^3 + m^3 = 72$ है, तो pm का मान ज्ञात करें।

(a) 12 (b) 8
(c) 6 (d) 9

50. When x^m is multiplied by x^n , product is 1. The relation between m and n is

जब x^m को x^n से गुणा किया जाता है, तो गुणनफल 1 प्राप्त होता है। m और n के बीच क्या संबंध है?

(a) $m = n$ (b) $m = -n$
(c) $mn = 1$ (d) $m + n = 1$

51. If $\frac{2P}{p^2 - 2p + 1} = \frac{1}{4}$ then the value of

$$\left(p + \frac{1}{p} \right) \text{ is}$$

यदि $\frac{2P}{p^2 - 2p + 1} = \frac{1}{4}$ है, तो $\left(p + \frac{1}{p} \right)$ का मान ज्ञात करें।

(a) $\frac{2}{5}$ (b) 10
(c) 7 (d) 1

52. $\frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \frac{1}{\sqrt{4}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{6}} + \frac{1}{\sqrt{6}+\sqrt{7}} + \frac{1}{\sqrt{7}+\sqrt{8}} + \frac{1}{\sqrt{8}+\sqrt{9}} = ?$
 (a) 0 (b) 1
 (c) 2 (d) 4

SSC CGL MAINS 2014

{12 Apr. 2015} PAPER -II

53. If $2x = \sqrt{a} + \frac{1}{\sqrt{a}}$, ($a > 0$), then the value of $\frac{\sqrt{x^2-1}}{x-\sqrt{x^2-1}}$ is:-
 यदि $2x = \sqrt{a} + \frac{1}{\sqrt{a}}$, ($a > 0$) तो $\frac{\sqrt{x^2-1}}{x-\sqrt{x^2-1}}$ का मान होगा?
 (a) $\frac{1}{2}(a-1)$ (b) $a-1$

(c) $a+1$ (d) $\frac{1}{2}(a+1)$

54. If a, b, c are real and $a^2+b^2+c^2=2$ ($a-b-c$)-3, then the value of $a+b+c$ is:-
 यदि a, b, c वास्तविक हैं और $a^2+b^2+c^2=2$ ($a-b-c$)-3 तो $a+b+c$ का मान क्या होगा?
 (a) 3 (b) 0
 (c) -1 (d) 1

55. If $\frac{a+b-c}{a+b} = \frac{b+c-a}{b+c} = \frac{c+a-b}{c+a}$ and $a+b+c \neq 0$, then
 यदि $\frac{a+b-c}{a+b} = \frac{b+c-a}{b+c} = \frac{c+a-b}{c+a}$ और $a+b+c \neq 0$ तो-
 (a) $a=b=c$ (b) $a \neq b=c$
 (c) $a \neq b \neq c$ (d) $a=b=c$

56. If $a = 2 + \sqrt{3}$, then the value of $\frac{a^6 + a^4 + a^2 + 1}{a^3}$ is-
 यदि $a = 2 + \sqrt{3}$ तो $\frac{a^6 + a^4 + a^2 + 1}{a^3}$ का मान क्या होगा?
 (a) 42 (b) 56
 (c) 45 (d) 65

57. If $x = \sqrt{5} + \sqrt{3}$ and $y = \sqrt{5} - \sqrt{3}$, then the value of $x^4 - y^4$ is-
 यदि $x = \sqrt{5} + \sqrt{3}$ और $y = \sqrt{5} - \sqrt{3}$ तो $x^4 - y^4$ का मान क्या होगा?

(a) 544 (b) $32\sqrt{15}$

(c) $64\sqrt{15}$ (d) 16

58. If $x+y+z=6$, then the value of $(x-1)^3 + (y-2)^3 + (z-3)^3$ is:

यदि $x+y+z=6$ तो $(x-1)^3 + (y-2)^3 + (z-3)^3$ का मान क्या होगा?

(a) $(x-1)(y-2)(z-3)$
 (b) $2(x-1)(y-2)(z-3)$
 (c) $3(x-1)(y-2)(z-3)$
 (d) $3xyz$

59. If $bc+ab+ca=abc$, then the value of $\frac{b+c}{bc(a-1)} + \frac{a+c}{ac(b-1)} + \frac{a+b}{ab(c-1)}$ is:

यदि $bc+ab+ca=abc$ तो

$\frac{b+c}{bc(a-1)} + \frac{a+c}{ac(b-1)} + \frac{a+b}{ab(c-1)}$ का मान क्या होगा?

(a) $-\frac{2}{3}$ (b) 1

(c) 0 (d) $-\frac{1}{2}$

60. If $\frac{a^2-bc}{a^2+bc} + \frac{b^2-ca}{b^2+ca} + \frac{c^2-ab}{c^2+ab} = 1$,

then the value of $\frac{a^2}{a^2+bc} + \frac{b^2}{b^2+ca} + \frac{c^2}{c^2+ab}$ is-

यदि $\frac{a^2-bc}{a^2+bc} + \frac{b^2-ca}{b^2+ca} + \frac{c^2-ab}{c^2+ab} = 1$ तो $\frac{a^2}{a^2+bc} + \frac{b^2}{b^2+ca} + \frac{c^2}{c^2+ab}$ का मान क्या होगा?

(a) -1 (b) 2
 (c) 0 (d) 1

61. If $p^4 = 119 - \frac{1}{p^4}$, then the value of $p^3 - \frac{1}{p^3}$ is-

यदि $p^4 = 119 - \frac{1}{p^4}$ तो $p^3 - \frac{1}{p^3}$ का मान क्या होगा?

(a) 36 (b) 18
 (c) 24 (d) 32

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62. If $a - \frac{1}{a-3} = 5$, then the value of $(a-3)^3 - \frac{1}{(a-3)^3}$ is-

(a) 544 (b) $32\sqrt{15}$
 (c) $64\sqrt{15}$ (d) 16

यदि $a - \frac{1}{a-3} = 5$, तो $(a-3)^3 - \frac{1}{(a-3)^3}$ का मान क्या होगा?

(a) 7 (b) 2
 (c) 5 (d) 14

63. If $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$, where $a \neq b \neq c \neq 0$, then the value of $a^2b^2c^2$ is

यदि $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$ है, जिसमें $a \neq b \neq c \neq 0$, तो $a^2b^2c^2$ का मान है

(a) 0 (b) 1
 (c) abc (d) -1

64. If $\left(\frac{p^{-1}q^2}{p^3q^{-2}}\right)^{\frac{1}{3}} \div \left(\frac{p^6q^{-3}}{p^{-2}q^3}\right)^{\frac{1}{3}} = p^a q^b$,

then the value of $a+b$ where p and q are different positive primes is-

यदि $\left(\frac{p^{-1}q^2}{p^3q^{-2}}\right)^{\frac{1}{3}} \div \left(\frac{p^6q^{-3}}{p^{-2}q^3}\right)^{\frac{1}{3}} = p^a q^b$ है,

तो $a+b$ का मान क्या है जिसमें p और q विभिन्न धनात्मक अभाज्य हैं?

(a) $2/3$ (b) $-2/3$
 (c) $3/4$ (d) 0

65. If $x^2 + y^2 + z^2 = xy + yz + zx$, then the value of

$\frac{3x^4 + 7y^4 + 5z^4}{5x^2y^2 + 7y^2z^2 + 3z^2x^2}$ is

यदि $x^2 + y^2 + z^2 = xy + yz + zx$, तो

$\frac{3x^4 + 7y^4 + 5z^4}{5x^2y^2 + 7y^2z^2 + 3z^2x^2}$ का मान क्या होगा?

(a) 0 (b) -1
 (c) 2 (d) 1

66. If $3(a^2+b^2+c^2) = (a+b+c)^2$, then the relation between a, b and c is -

यदि $3(a^2+b^2+c^2) = (a+b+c)^2$ है, तो a, b और c के बीच क्या संबंध है?

(a) $a \neq b \neq c$ (b) $a = b \neq c$
 (c) $a = b = c$ (d) $a \neq b = c$

67. If $(3x - 2y) : (2x + 3y) = 5 : 6$, then one of the value of

$$\left(\frac{\sqrt[3]{x} + \sqrt[3]{y}}{\sqrt[3]{x} - \sqrt[3]{y}} \right)^2 \text{ is:-}$$

यदि $(3x - 2y) : (2x + 3y) = 5 : 6$, तो

$$\left(\frac{\sqrt[3]{x} + \sqrt[3]{y}}{\sqrt[3]{x} - \sqrt[3]{y}} \right)^2 \text{ का मान क्या होगा?}$$

- (a) 5 (b) 25
(c) $\frac{1}{5}$ (d) $\frac{1}{25}$

68. If $x - \sqrt{3} - \sqrt{2} = 0$ and $y - \sqrt{3} + \sqrt{2} = 0$, then the value of $(x^3 - 20\sqrt{2}) - (y^3 - 2\sqrt{2})$

यदि $x - \sqrt{3} - \sqrt{2} = 0$ और $y - \sqrt{3} + \sqrt{2} = 0$, तो $(x^3 - 20\sqrt{2}) - (y^3 - 2\sqrt{2})$ का मान क्या होगा?

- (a) 1 (b) 0
(c) 2 (d) 3

69. Let, $x = \frac{\sqrt{13} + \sqrt{11}}{\sqrt{13} - \sqrt{11}}$ and $y = \frac{1}{x}$, then the value of $3x^2 - 5xy + 3y^2$ is-

मान लें कि $x = \frac{\sqrt{13} + \sqrt{11}}{\sqrt{13} - \sqrt{11}}$ और $y = \frac{1}{x}$, तो $3x^2 - 5xy + 3y^2$ का मान क्या होगा?

- (a) 1717 (b) 1171
(c) 1771 (d) 1177

70. Find the value of the following:

$$\frac{(0.67 \times 0.67 \times 0.67) - (0.33 \times 0.33 \times 0.33)}{(0.67 \times 0.67) + (0.67 \times 0.33) + (0.33 \times 0.33)}$$

निम्न का मान ज्ञात कीजिए:

- (a) 3.4 (b) 0.34
(c) 11 (d) 1.1

71. If $(x^3 - y^3) : (x^2 + xy + y^2) = 5 : 1$ and $(x^2 - y^2) : (x - y) = 7 : 1$, then the ratio $2x : 3y$ equal to:-

यदि $(x^3 - y^3) : (x^2 + xy + y^2) = 5 : 1$ और $(x^2 - y^2) : (x - y) = 7 : 1$, तो $2x : 3y$ किसके बराबर है?

- (a) 4 : 3 (b) 4 : 1
(c) 2 : 3 (d) 3 : 2

72. If $x = a^{\frac{1}{2}} + a^{-\frac{1}{2}}$, $y = a^{\frac{1}{2}} - a^{-\frac{1}{2}}$ then value of $(x^4 - x^2y^2 - 1) + (y^4 - x^2y^2 + 1)$ -

यदि $x = a^{\frac{1}{2}} + a^{-\frac{1}{2}}$, $y = a^{\frac{1}{2}} - a^{-\frac{1}{2}}$ है, तो $(x^4 - x^2y^2 - 1) + (y^4 - x^2y^2 + 1)$ का मान है

- (a) 12 (b) 14
(c) 16 (d) 13

73. If $a + b = 1$, find the value of $a^3 + b^3 - ab - (a^2 - b^2)^2$

यदि $a + b = 1$, तो $a^3 + b^3 - ab - (a^2 - b^2)^2$ का मान ज्ञात कीजिए।

- (a) 2 (b) 1
(c) 0 (d) -1

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74. If $P = 99$, then the value of $P(P^2 + 3P + 3)$ is

यदि $P = 99$ तो $P(P^2 + 3P + 3)$ का मान बताइए?

- (a) 9999 (b) 999999
(c) 99999 (d) 9999999

75. If $x + \frac{1}{x} = c + \frac{1}{c}$ then the value of x

यदि $x + \frac{1}{x} = c + \frac{1}{c}$ तो x का मान बताइए

- (a) $c, 1/c$ (b) c, c^2
(c) $c, 2c$ (d) $0, 1$

76. If the sum of squares of two real numbers is 41 and their sum is 9. Then the sum of cubes of these two numbers is

दो वास्तविक संख्याओं के वर्गों का जोड़ 41 है और उनका अपना जोड़ 9 है, तो इन दो संख्याओं के घनों का जोड़ क्या होगा?

- (a) 169 (b) 209
(c) 189 (d) 198

77. A complete factorisation of $x^4 + 64$ is

$x^4 + 64$ का सम्पूर्ण गुणनखण्डन करें:

- (a) $(x^2 + 8)^2$
(b) $(x^2 + 8)(x^2 - 8)$
(c) $(x^2 - 4x + 8)(x^2 - 4x - 8)$
(d) $(x^2 + 4x + 8)(x^2 - 4x + 8)$

78. If $a + b = 1$, then $a^4 + b^4 - a^3 - b^3 - 2a^2b^2 + ab$ is equal to

यदि $a + b = 1$ तो $a^4 + b^4 - a^3 - b^3 - 2a^2b^2 + ab$ किसके बराबर होगा?

- (a) 1 (b) 2
(c) 4 (d) 0

79. If $x^2 + y^2 + 6x + 5 = 4(x - y)$ then $x - y$ is

यदि $x^2 + y^2 + 6x + 5 = 4(x - y)$ हो, तो $x - y$ क्या होगा?

- (a) 1 (b) -1
(c) 0 (d) 4

80. If $a = 299$, $b = 298$, $c = 297$ then the value of $2a^3 + 2b^3 + 2c^3 - 6abc$ is

यदि $a = 299$, $b = 298$, $c = 297$ हो, तो $2a^3 + 2b^3 + 2c^3 - 6abc$ का मान बताएं?

- (a) 5154 (b) 5267
(c) 5364 (d) 5456

81. If $x + \frac{1}{x} = \sqrt{3}$ the value of $x^{18} + x^{12} + x^6 + 1$ is

यदि $x + \frac{1}{x} = \sqrt{3}$ हो, तो $x^{18} + x^{12} + x^6 + 1$ का मान बताएं?

- (a) 0 (b) 1
(c) 2 (d) 3

82. If $x = 1 + \sqrt{2} + \sqrt{3}$, then the value of $2x^4 - 8x^3 - 5x^2 + 26x - 28$ is

यदि $x = 1 + \sqrt{2} + \sqrt{3}$ हो, तो $2x^4 - 8x^3 - 5x^2 + 26x - 28$ का मान बताइए?

- (a) $2\sqrt{2}$ (b) $3\sqrt{3}$
(c) $5\sqrt{5}$ (d) $6\sqrt{6}$

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83. $\left(1 - \frac{2xy}{x^2 + y^2}\right) \div \left(\frac{x^3 - y^3}{x - y} - 3xy\right) = ?$

- (a) $\frac{1}{x^2 - y^2}$ (b) $\frac{1}{x^2 + y^2}$
(c) $\frac{1}{x - y}$ (d) $\frac{1}{x + y}$

84. If $a + b + c = 0$ then the value of

$$\frac{1}{(a+b)(b+c)} + \frac{1}{(b+c)(c+a)} + \frac{1}{(c+a)(a+b)} ?$$

यदि $a + b + c = 0$ तो

$$\frac{1}{(a+b)(b+c)} + \frac{1}{(b+c)(c+a)} + \frac{1}{(c+a)(a+b)} \text{ is}$$

का मान निकालें।

- (a) 0 (b) 1
(c) 3 (d) 2

85. If $x^2 + y^2 + 2x + 1 = 0$, then the value of $x^{31} + y^{35}$ is

यदि $x^2 + y^2 + 2x + 1 = 0$ हो, तो $x^{31} + y^{35}$ का मान निकालें।

- (a) -1 (b) 0
(c) 1 (d) 2

86. If $x = \frac{\sqrt{5}+1}{\sqrt{5}-1}$ and $y = \frac{\sqrt{5}-1}{\sqrt{5}+1}$ the

value of $\frac{x^2 + xy + y^2}{x^2 - xy + y^2}$ is

यदि $x = \frac{\sqrt{5}+1}{\sqrt{5}-1}$ और $y = \frac{\sqrt{5}-1}{\sqrt{5}+1}$ तो

$\frac{x^2 + xy + y^2}{x^2 - xy + y^2}$ का मान निकालें

- (a) $\frac{3}{4}$ (b) $\frac{4}{3}$
(c) $\frac{3}{5}$ (d) $\frac{5}{3}$

87. If $\left(x - \frac{1}{x}\right)^2 = 3$, then the value of $x^6 + \frac{1}{x^6}$ equals

यदि $\left(x - \frac{1}{x}\right)^2 = 3$ तो $x^6 + \frac{1}{x^6}$ का मान निकालें

- (a) 90 (b) 100
(c) 110 (d) 120

88. If $x^4 + 2x^3 + ax^2 + bx + 9$ is a perfect square, where a and b are positive real numbers, then the value of a and b are

यदि $x^4 + 2x^3 + ax^2 + bx + 9$ एक सम्पूर्ण वर्ग है जहाँ a तथा b धनात्मक वास्तविक संख्या है तो a और b का मान क्या होगा?

- (a) $a = 5, b = 6$
(b) $a = 6, b = 7$
(c) $a = 7, b = 6$
(d) $a = 7, b = 8$

89. If $a^2 + b^2 + c^2 = 16$, $x^2 + y^2 + z^2 = 25$ and $ax + by + cz = 20$, then

the value of $\frac{a+b+c}{x+y+z}$

यदि $a^2 + b^2 + c^2 = 16$, $x^2 + y^2 + z^2 = 25$ और $ax + by + cz = 20$ हो, तो

- $\frac{a+b+c}{x+y+z}$ का मान निकालें?
- (a) $3/5$ (b) $5/3$
(c) $4/5$ (d) $5/4$

90. The value of x which satisfies the equation

$$\frac{x+a^2+2c^2}{b+c} + \frac{x+b^2+2a^2}{c+a} + \frac{x+c+2b^2}{a+b} = 0$$

is x का मान बताइए जो भिन्न

$$\frac{x+a^2+2c^2}{b+c} + \frac{x+b^2+2a^2}{c+a} + \frac{x+c+2b^2}{a+b} = 0$$

को संतुष्ट करता है।

- (a) $(a^2 + b^2 + c^2)$
(b) $-(a^2 + b^2 + c^2)$
(c) $(a^2 + 2b^2 + c^2)$
(d) $-(a^2 + b^2 + 2c^2)$

91. If $a^3 = 117 + b^3$ and $a = 3 + b$, then the value of $a + b$ is:

यदि $a^3 = 117 + b^3$ और $a = 3 + b$ हो, तो $a + b$ का मान निकालें?

- (a) ± 7 (b) ± 49
(c) ± 13 (d) 0

92. If $a + \frac{1}{a} = -2$ then the value of $a^{1000} + a^{-1000}$ is

यदि $a + \frac{1}{a} = -2$ तो $a^{1000} + a^{-1000}$ का मान निकालें

- (a) 2 (b) 0
(c) 1 (d) $1/2$

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93. If $x^2 + \frac{1}{x^2} = 98$ ($x > 0$), then the value

of $x^3 + \frac{1}{x^3}$ is

यदि $x^2 + \frac{1}{x^2} = 98$ ($x > 0$) हो तो $x^3 + \frac{1}{x^3}$ का मान बताइए?

- (a) 970 (b) 1030
(c) -970 (d) -1030

94. If $a + \frac{1}{b} = 1$ and $b + \frac{1}{c} = 1$, then the

value of $c + \frac{1}{a}$ is

यदि $a + \frac{1}{b} = 1$ और $b + \frac{1}{c} = 1$ हो, तो $c +$

$\frac{1}{a}$ का मान क्या होगा?

- (a) 0 (b) 2
(c) 1 (d) 3

95. If $x = y + z$, then the value of $x^3 - y^3 - z^3$ is

यदि $x = y + z$ हो तो $x^3 - y^3 - z^3$ का मान बताइए?

- (a) 0 (b) $3xyz$
(c) $-3xyz$ (d) 1

96. If $a+b+c+d = 4$, then the value of

$$\frac{1}{(1-a)(1-b)(1-c)} + \frac{1}{(1-b)(1-c)(1-d)} +$$

$$\frac{1}{(1-c)(1-d)(1-a)} + \frac{1}{(1-d)(1-a)(1-b)}$$

is

यदि $a+b+c+d = 4$ हो तो

$$\frac{1}{(1-a)(1-b)(1-c)} + \frac{1}{(1-b)(1-c)(1-d)} +$$

$$\frac{1}{(1-c)(1-d)(1-a)} + \frac{1}{(1-d)(1-a)(1-b)}$$

का मान बताइए?

- (a) 0 (b) 1
(c) 4 (d) $1 + abcd$

97. If $x = 11$, the value of $x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1$ is

यदि $x = 11$ हो तो $x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1$ का मान बताइए?

- (a) 11 (b) 10
(c) 12 (d) -10

98. If $a = \frac{1}{a-5}$ ($a > 0$), then the value

of $a + \frac{1}{a}$ is

यदि $a = \frac{1}{a-5}$ ($a > 0$), हो तो $a + \frac{1}{a}$ का मान क्या होगा?

- (a) $2\sqrt{29}$ (b) $\sqrt{29}$
(c) $-\sqrt{29}$ (d) $\sqrt{27}$

99. If $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$ (where $a \neq b \neq c$), then abc is equal to

यदि $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$

(जहाँ $a \neq b \neq c$), तो abc किसके बराबर है?

- (a) +1
(b) -1
(c) +1 & -1
(d) None of these/ इनमे से कोई नहीं

100. If $ax+by=1$ and $bx+ay=\frac{2ab}{a^2+b^2}$

then $(x^2+y^2)(a^2+b^2)$ is equal to

यदि $ax+by=1$ और $bx+ay=\frac{2ab}{a^2+b^2}$ हो

तो $(x^2+y^2)(a^2+b^2)$ किसके बराबर है?

- (a) 1 (b) 2
(c) 0.5 (d) 0

101. If x, y, z are the three factors of $a^3 - 7a - 6$, then value of $x+y+z$ will be

यदि $a^3 - 7a - 6$ के तीन गुणखंड x, y, z हैं, तो $x+y+z$ का मान होगा?

- (a) 3a (b) 3
(c) 6 (d) a

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102. If $x + y + z = 0$, then what is the value of $(3y^2 + x^2 + z^2)/(2y^2 - xz)$?
 यदि $x + y + z = 0$, तो $(3y^2 + x^2 + z^2)/(2y^2 - xz)$ का मान क्या है ?
 (a) 2 (b) 1
 (c) 3/2 (d) 5/3
103. If $P = 7 + 4\sqrt{3}$ and $PQ = 1$, then what is the value of $1/P^2 + 1/Q^2$?
 यदि $P = 7 + 4\sqrt{3}$ तथा $PQ = 1$ है, तो $1/P^2 + 1/Q^2$ का मान क्या है ?
 (a) 196 (b) 194
 (c) 206 (d) 182
104. If $a^3 + 3a^2 + 9a = 1$, then what is the value of $a^3 + (3/a)$?
 यदि $a^3 + 3a^2 + 9a = 1$ हो तो $a^3 + (3/a)$ का मान क्या है ?
 (a) 31 (b) 26
 (c) 28 (d) 24
105. x, y and z are real numbers. If $x^3 + y^3 + z^3 = 13$, $x + y + z = 1$ and $xyz = 1$, then what is the value of $xy + yz + zx$?
 x, y तथा z वास्तविक संख्याएँ हैं। यदि $x^3 + y^3 + z^3 = 13$, $x + y + z = 1$ तथा $xyz = 1$ है, तो $xy + yz + zx$ का मान क्या है ?
 (a) -1 (b) 1
 (c) 3 (d) -3
106. If $(a + b)/c = 6/5$ and $(b + c)/a = 9/2$, then what is the value of $(a + c)/b$?
 यदि $(a + b)/c = 6/5$ तथा $(b + c)/a = 9/2$ है, तो $(a + c)/b$ का मान क्या है ?
 (a) 9/5 (b) 11/7
 (c) 7/11 (d) 7/4
107. If $x^3 + y^3 + z^3 = 3(1 + xyz)$, $P = y + z - x$, $Q = z + x - y$ and $R = x + y - z$, then what is the value of $P^3 + Q^3 + R^3 - 3PQR$?
 यदि $x^3 + y^3 + z^3 = 3(1 + xyz)$, $P = y + z - x$, $Q = z + x - y$ तथा $R = x + y - z$ है, तो $P^3 + Q^3 + R^3 - 3PQR$ का मान क्या है ?
 (a) 9 (b) 8
 (c) 12 (d) 6

108. If $x_1 x_2 x_3 = 4(4 + x_1 + x_2 + x_3)$, then what is the value of $[1/(2 + x_1)] + [1/(2 + x_2)] + [1/(2 + x_3)]$?
 यदि $x_1 x_2 x_3 = 4(4 + x_1 + x_2 + x_3)$ हो, तो $[1/(2 + x_1)] + [1/(2 + x_2)] + [1/(2 + x_3)]$ का मान क्या है ?
 (a) 1 (b) 1/2
 (c) 2 (d) 1/3
109. If α and β are the roots of equation $x^2 - x + 1 = 0$, then which equation will have roots α^3 and β^3 ?
 यदि α तथा β समीकरण $x^2 - x + 1 = 0$ के मूल हैं, तो किस समीकरण के मूल α^3 तथा β^3 होंगे ?
 (a) $x^2 + 2x + 1 = 0$
 (b) $x^2 - 2x - 1 = 0$
 (c) $x^2 + 3x - 1 = 0$
 (d) $x^2 - 3x + 1 = 0$
110. If $3x + 5y + 7z = 49$ and $9x + 8y + 21z = 126$, then what is the value of y ?
 यदि $3x + 5y + 7z = 49$ तथा $9x + 8y + 21z = 126$ है, तो y का मान क्या है ?
 (a) 4 (b) 2
 (c) 3 (d) 5
111. Cost of 4 pen, 6 note books and 9 files is ₹305. Cost of 3 pens, 4 notebooks and 2 files is ₹145. What is the cost (in ₹) of 5 pens, 8 notebooks and 16 files?
 4 कलम, 6 नोटबुक तथा 9 फाइल का मूल्य ₹305 है। 3 कलम, 4 नोटबुक तथा 2 फाइल का मूल्य ₹145 है। 5 कलम, 8 नोटबुक तथा 16 फाइल का मूल्य (₹ में) क्या है ?
 (a) 415 (b) 465
 (c) 440 (d) Cannot be determined

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112. If $P = 7 + 4\sqrt{3}$ and $PQ = 1$, then what is the value of $(1/P^2) + (1/Q^2)$?
 यदि $P = 7 + 4\sqrt{3}$ and $PQ = 1$ है तो, $(1/P^2) + (1/Q^2)$ का मान क्या है ?
 (a) 148 (b) 189
 (c) 194 (d) 204

113. If $x = (\sqrt{5}) + 1$ and $y = (\sqrt{5}) - 1$, then what is the value of $(x^2/y^2) + (y^2/x^2) + 4[(x/y) + (y/x)] + 6$?
 यदि $x = (\sqrt{5}) + 1$ और $y = (\sqrt{5}) - 1$ है, तो $(x^2/y^2) + (y^2/x^2) + 4[(x/y) + (y/x)] + 6$ का मान क्या है ?
 (a) 31 (b) $23\sqrt{5}$
 (c) $27\sqrt{5}$ (d) 25
114. If $x = 2 + \sqrt{3}$, $y = 2 - \sqrt{3}$ and $z = 1$, then what is the value of $(x/yz) + (y/xz) + (z/xy) + 2[(1/x) + (1/y) + (1/z)]$?
 यदि $x = 2 + \sqrt{3}$, $y = 2 - \sqrt{3}$ तथा $z = 1$ है, तो $(x/yz) + (y/xz) + (z/xy) + 2[(1/x) + (1/y) + (1/z)]$ का मान क्या है ?
 (a) 25 (b) 22
 (c) 17 (d) 43
115. A root of equation $ax^2 + bx + c = 0$ (where a, b and c are rational numbers) is $5 + 3\sqrt{3}$. What is the value of $(a^2 + b^2 + c^2)/(a + b + c)$?
 समीकरण $ax^2 + bx + c = 0$ (जहाँ a, b तथा c परिमेय संख्याएँ) का मूल $5 + 3\sqrt{3}$ $(a^2 + b^2 + c^2)/(a + b + c)$ का मान क्या है ?
 (a) 35/3 (b) 37/3
 (c) -105/11 (d) -105/13
116. If $x = (a/b) + (b/a)$, $y = (b/c) + (c/b)$ and $z = (c/a) + (a/c)$, then what is the value of $xyz - x^2 - y^2 - z^2$?
 यदि $x = (a/b) + (b/a)$, $y = (b/c) + (c/b)$ तथा $z = (c/a) + (a/c)$ है तो $xyz - x^2 - y^2 - z^2$ का मान क्या है ?
 (a) -4 (b) 2
 (c) -1 (d) -6
117. If $[a + (1/a)^2] - 2[a - (1/a)] = 12$, then which of the following is a value of 'a'?
 यदि $[a + (1/a)^2] - 2[a - (1/a)] = 12$, हो, तो निम्नलिखित में से कौन सा 'a' का एक मान है ?
 (a) $-8 + \sqrt{3}$
 (b) $-8 - \sqrt{3}$
 (c) $-8 + \sqrt{5}$
 (d) None of these/इनमें से कोई नहीं

118. If $x^2 - 4x + 1 = 0$, then what is the value of $x^9 + x^7 - 194x^5 - 194x^3$?
यदि $x^2 - 4x + 1 = 0$ हो, तो $x^9 + x^7 - 194x^5 - 194x^3$ का मान क्या है?

- (a) 4 (b) -4
(c) 1 (d) -1

119. If $x + y = 3$, then what is the value of $x^3 + y^3 + 9xy$?

यदि $x + y = 3$ हो, तो $x^3 + y^3 + 9xy$ का मान क्या है?

- (a) 15 (b) 81
(c) 27 (d) 9

120. $A = (x^8 - 1)/(x^4 + 1)$ and $B = (y^4 - 1)/(y^2 + 1)$. If $x = 2$ and $y = 9$, then what is the value of $A^2 + 2AB + AB^2$?

$A = (x^8 - 1)/(x^4 + 1)$ तथा $B = (y^4 - 1)/(y^2 + 1)$ है। यदि $x = 2$ तथा $y = 9$ है, तो $A^2 + 2AB + AB^2$ का मान क्या है?

- (a) 96475 (b) 98625
(c) 92425 (d) 89125

121. If $x - 4y = 0$ and $x + 2y = 24$, then what is the value of $(2x + 3y)/(2x - 3y)$?

यदि $x - 4y = 0$ तथा $x + 2y = 24$ है, तो $(2x + 3y)/(2x - 3y)$ का मान क्या है?

- (a) 9/5 (b) 11/5
(c) 13/7 (d) 9/7

122. If $(x/a) + (y/b) = 3$ and $(x/b) - (y/a) = 9$, then what is the value of x/y ?

यदि $(x/a) + (y/b) = 3$ तथा $(x/b) - (y/a) = 9$ है, तो x/y का मान क्या है?

- (a) $(b + 3a)/(a - 3b)$
(b) $(a + 3b)/(b - 3a)$
(c) $(1 + 3a)/(a + 3b)$
(d) $(a + 3b^2)/(b - 3a^2)$

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123. $p^3 + q^3 + r^3 - 3pqr = 4$. If $a = q + r$, $b = r + p$ and $c = p + q$, then what is the value of $a^3 + b^3 + c^3 - 3abc$?

- $p^3 + q^3 + r^3 - 3pqr = 4$ है। यदि $a = q + r$, $b = r + p$ तथा $c = p + q$ है, तो $a^3 + b^3 + c^3 - 3abc$ का मान क्या है?
- (a) 4 (b) 8
(c) 2 (d) 12

124. If α and β are the roots of the equation $x^2 + x - 1 = 0$, then what is the equation whose roots are α^5 and β^5 ?

यदि α तथा β समीकरण $x^2 + x - 1 = 0$ के मूल हैं, तो वह समीकरण क्या है जिसके मूल α^5 तथा β^5 हैं?

- (a) $x^2 + 7x - 1 = 0$
(b) $x^2 - 7x - 1 = 0$
(c) $x^2 - 11x - 1 = 0$
(d) $x^2 + 11x - 1 = 0$

125. If x and y are natural numbers such that $x + y = 2017$, then what is the value of $(-1)^x + (-1)^y$?

यदि x तथा y प्राकृतिक संख्याएँ इस प्रकार हैं कि $x + y = 2017$ है, तो $(-1)^x + (-1)^y$ का मान क्या है?

- (a) 2 (b) -2
(c) 0 (d) 1

126. If $x + (1/x) = (\sqrt{3} + 1)/2$, then what is the value of $x^4 + (1/x^4)$?

यदि $x + (1/x) = (\sqrt{3} + 1)/2$ है, तो $x^4 + (1/x^4)$ का मान क्या है?

- (a) $(4\sqrt{3} - 1)/4$
(b) $(4\sqrt{3} + 1)/2$
(c) $(-4\sqrt{3} - 1)/4$
(d) $(-4\sqrt{3} - 1)/2$

127. If $a + a^2 + a^3 - 1 = 0$, then what is the value of $a^3 + (1/a)$?

यदि $a + a^2 + a^3 - 1 = 0$ हो, तो $a^3 + (1/a)$ का मान क्या है?

- (a) 1 (b) 4
(c) 2 (d) 3

128. If $a - (1/a) = b$, $b - (1/b) = c$ and $c - (1/c) = a$, then what is the value of $(1/ab) + (1/bc) + (1/ca)$?

यदि $a - (1/a) = b$, $b - (1/b) = c$ तथा $c - (1/c) = a$ है, तो $(1/ab) + (1/bc) + (1/ca)$ का मान क्या है?

- (a) -3 (b) -6
(c) -1 (d) -9

129. If the roots of the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ are equal, then which of the following is true?

यदि समीकरण $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ के मूल बराबर हैं, तो निम्नलिखित में से कौन सा सही है?

- (a) $b = (a + c)/ac$
(b) $2/b = (1/a) + (1/c)$
(c) $2b = (1/a) + (1/c)$
(d) $abc = ab + bc + ca$

130. If $\left[\sqrt{a^2 + b^2 + ab}\right] + \left[\sqrt{a^2 + b^2 - ab}\right] = 1$, then what is the value of $(1 - a^2)(1 - b^2)$?

यदि $\left[\sqrt{a^2 + b^2 + ab}\right] + \left[\sqrt{a^2 + b^2 - ab}\right] = 1$ हो, तो $(1 - a^2)(1 - b^2)$ का मान क्या है?

- (a) 1/4 (b) 4/7
(c) 5/4 (d) 3/4

131. If $3x + 4y - 11 = 18$ and $8x - 6y + 12 = 6$, then what is the value of $5x - 3y - 9$?

यदि $3x + 4y - 11 = 18$ तथा $8x - 6y + 12 = 6$ है, तो $5x - 3y - 9$ का मान क्या है?

- (a) 18 (b) -9
(c) -27 (d) -18

132. If $a + b + c = 7/12$, $3a - 4b + 5c = 3/4$ and $7a - 11b - 13c = -7/12$, then what is the value of $a + c$?

यदि $a + b + c = 7/12$, $3a - 4b + 5c = 3/4$ तथा $7a - 11b - 13c = -7/12$ है, तो $a + c$ का मान क्या है?

- (a) 1/2 (b) 5/12
(c) 3/4 (d) 1/4

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133. If $x + y + z = 22$ and $xy + yz + zx = 35$, then what is the value of $(x - y)^2 + (y - z)^2 + (z - x)^2$?

यदि $x + y + z = 22$ तथा $xy + yz + zx = 35$ हैं, तो $(x - y)^2 + (y - z)^2 + (z - x)^2$ का मान क्या है?

- (a) 793 (b) 681
(c) 758 (d) 715

134. If $(x + y)/z = 2$, then what is the value of $[y/(y - z)] + [x/(x - z)]$?

यदि $(x + y)/z = 2$ है, तो $[y/(y - z)] + [x/(x - z)]$ का मान क्या है?

- (a) 0 (b) 1
(c) 2 (d) -1

135. If α and β are the roots of equation $x^2 - 2x + 4 = 0$, then what is the equation whose roots are α^3/β^2 and β^3/α^2 ?

यदि α तथा β समीकरण $x^2 - 2x + 4 = 0$ के मूल हैं, तो वह समीकरण क्या है जिसके मूल α^3/β^2 तथा β^3/α^2 हैं?

- (a) $x^2 - 4x + 8 = 0$
(b) $x^2 - 32x + 4 = 0$
(c) $x^2 - 2x + 4 = 0$
(d) $x^2 - 16x + 4 = 0$

136. If one root of the equation $Ax^2 + Bx + C = 0$ is two and a half times the others, then which of the following is TRUE?

यदि समीकरण $Ax^2 + Bx + C = 0$ का एक मूल दूसरे से ढाई गुणा है, तो निम्नलिखित में से कौन सा सत्य है?

- (a) $7B^2 = 3CA$
(b) $7B^2 = 4CA$
(c) $7B^2 = 36CA$
(d) $10B^2 = 49CA$

137. If $x^2 - 12x + 33 = 0$, then what is the value of $(x-4)^2 + [1/(x-4)^2]$?

यदि $x^2 - 12x + 33 = 0$ है, तो $(x-4)^2 + [1/(x-4)^2]$ का मान क्या है?

- (a) 16 (b) 14
(c) 18 (d) 20

138. If $a^4 + 1 = [a^2/b^2](4b^2 - b^4 - 1)$, then what is the value of $a^4 + b^4$?

यदि $a^4 + 1 = [a^2/b^2](4b^2 - b^4 - 1)$ हैं, तो $a^4 + b^4$ का मान क्या है?

- (a) 2 (b) 16
(c) 32 (d) 64

139. If $3\sqrt{\frac{1-a}{a}} + 9 = 19 - 3\sqrt{\frac{a}{1-a}}$, then what is the value of a ?

यदि $3\sqrt{\frac{1-a}{a}} + 9 = 19 - 3\sqrt{\frac{a}{1-a}}$ है,

तो a का मान क्या है?

- (a) $3/10, 7/10$
(b) $1/10, 9/10$
(c) $2/5, 3/5$
(d) $1/5, 4/5$

140. If $a + b = 10$ and $\sqrt{\frac{a}{b}} - 13 = -$

$\sqrt{\frac{b}{a}} - 11$, then what is the value of $3ab + 4a^2 + 5b^2$?

यदि $a + b = 10$ तथा $\sqrt{\frac{a}{b}} - 13 = -$

$\sqrt{\frac{b}{a}} - 11$ है, तो $3ab + 4a^2 + 5b^2$ का

मान क्या है?

- (a) 450 (b) 300
(c) 600 (d) 750

141. If $3x + 4y - 2z + 9 = 17$, $7x + 2y + 11z + 8 = 23$ and $5x + 9y + 6z - 4 = 18$, then what is the value of $x + y + z - 34$?

यदि $3x + 4y - 2z + 9 = 17$, $7x + 2y + 11z + 8 = 23$ तथा $5x + 9y + 6z - 4 = 18$ है, तो $x + y + z - 34$ का मान क्या है?

- (a) -28 (b) -14
(c) -31 (d) -45

142. If $x + 3y - \frac{2z}{4} = 6$, $x + \frac{2}{3}(2y + 3z)$

$= 33$ and $\frac{1}{7}(x + y + z) + 2z = 9$,

then what is the value of $46x + 131y$?

यदि $x + 3y - \frac{2z}{4} = 6$, $x + \frac{2}{3}(2y +$

$3z) = 33$ तथा $\frac{1}{7}(x + y + z) + 2z = 9$

हैं, तो $46x + 131y$ का मान क्या है?

- (a) 414 (b) 364
(c) 384 (d) 464

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143. If $a = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$ and $b = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$,

then what is the value of $a^2 + b^2 - ab$?

यदि $a = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$ तथा $b = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$ हैं, तो

$a^2 + b^2 - ab$ का मान क्या है?

- (a) 97 (b) $(2\sqrt{3}) + 2$
(c) $(4\sqrt{6}) + 1$ (d) 98

144. If the difference between the roots of the equation $Ax^2 - Bx + C = 0$ is 4, then which of the following is TRUE?

यदि समीकरण $Ax^2 - Bx + C = 0$ के मूलों का अंतर 4 है, तो निम्नलिखित में से कौन सा सत्य है?

- (a) $B^2 - 16A^2 = 4AC + 4B^2$
(b) $B^2 - 10A^2 = 4AC + 6A^2$
(c) $B^2 - 8A^2 = 4AC + 10A^2$
(d) $B^2 - 16A^2 = 4AC + 8B^2$

145. α and β are the roots of quadratic equation. If $\alpha + \beta = 8$ and $\alpha - \beta = 2\sqrt{5}$, then which of the following equation will have roots α^4 and β^4 ?

यदि α तथा β द्विघात समीकरण के मूल हैं। यदि $\alpha + \beta = 8$ तथा $\alpha - \beta = 2\sqrt{5}$ हैं, तो α^4 तथा β^4 निम्नलिखित में से किस समीकरण के मूल हैं?

- (a) $x^2 - 1522x + 14641 = 0$
(b) $x^2 + 1921x + 14641 = 0$
(c) $x^2 - 1764x + 14641 = 0$
(d) $x^2 + 2520x + 14641 = 0$

146. If a and b are the roots of the equation $Px^2 - Qx + R = 0$, then what is the value of $(1/a^2) + (1/b^2) + (a/b) + (b/a)$?

यदि a तथा b समीकरण $Px^2 - Qx + R = 0$ के मूल हैं, तो $(1/a^2) + (1/b^2) + (a/b) + (b/a)$ का मान क्या है?

- (a) $\frac{(Q^2 - 2P)(2R + P)}{PR^2}$
(b) $\frac{(Q^2 - 2PR)(R + P)}{PR^2}$
(c) $\frac{(Q^2 - 2R)(2P + R)}{P^2R^2}$
(d) $\frac{(Q^2 - 2PR)(2R + 2P)}{P^2R^2}$

147. If $x^2 - 16x + 59 = 0$, then what is the value of $(x-6)^2 + [1/(x-6)^2]$?

यदि $x^2 - 16x + 59 = 0$ है, तो $(x-6)^2 + [1/(x-6)^2]$ का मान क्या है?

- (a) 14 (b) 18
(c) 16 (d) 20

148. If A and B are the roots of the equation $Ax^2 - A^2x + AB = 0$, then what is the value of A and B respectively?

यदि A तथा B समीकरण $Ax^2 - A^2x + AB = 0$ के मूल हैं, क्रमशः A तथा B का मान क्या है?

- (a) 1, 0 (b) 1, 1
(c) 0, 2 (d) 0, 1

149. α and β are the roots of the quadratic equation $x^2 - x - 1 = 0$. What is the value of $\alpha^8 + \beta^8$?

यदि α तथा β द्विघात समीकरण $x^2 - x - 1 = 0$ के मूल हैं। $\alpha^8 + \beta^8$ का मान क्या है?

- (a) 47 (b) 54
(c) 59 (d) 68

150. If $a + b + c = 9$, $ab + bc + ca = 26$, $a^3 + b^3 + c^3 = 91$, $b^3 + c^3 = 72$ and $c^3 + a^3 = 35$, then what is the value of abc ?

यदि $a + b + c = 9$, $ab + bc + ca = 26$, $a^3 + b^3 + c^3 = 91$, $b^3 + c^3 = 72$ तथा $c^3 + a^3 = 35$ हैं, तो abc का मान क्या है?

- (a) 48 (b) 24
(c) 36 (d) 42

151. If $x^3 - 4x^2 + 19 = 6(x-1)$, then what is the value of $[x^2 + (1/x - 4)]$?

यदि $x^3 - 4x^2 + 19 = 6(x-1)$ हैं, तो $[x^2 + (1/x - 4)]$ का मान क्या है?

- (a) 3 (b) 5
(c) 6 (d) 8

152. Cost of 8 pencils, 5 pens and 3 erasers is ₹111. Cost of 9 pencils, 6 pens and 5 erasers is ₹130. Cost of 16 pencils, 11 pens and 3 erasers is ₹221. What is the cost (in ₹) of 39 pencils, 26 pens and 13 erasers?

8 पेंसिल, 5 कलम तथा 3 रबड़ का मूल्य ₹111 है। 9 पेंसिल, 6 कलम तथा 5 रबड़ का मूल्य ₹130 है। 16 पेंसिल, 11 कलम तथा 3 रबड़ का मूल्य ₹221 है। 39 पेंसिल, 26 कलम तथा 13 रबड़ का मूल्य (₹ में) क्या है?

- (a) 316 (b) 546
(c) 624 (d) 482

153. If $2x + 3y - 5z = 18$, $3x + 2y + z = 29$ and $x + y + 3z = 17$, then what is the value of $xy + yz + zx$?

यदि $2x + 3y - 5z = 18$, $3x + 2y + z = 29$ तथा $x + y + 3z = 17$ हैं, तो $xy + yz + zx$ का मान क्या है?

- (a) 32 (b) 52
(c) 64 (d) 46

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154. What is the value of

$$(2 + \sqrt{2}) + \left(\frac{1}{2 + \sqrt{2}} \right) + \frac{1}{2 - \sqrt{2}} + (2 - \sqrt{2})?$$

$$(2 + \sqrt{2}) + \left(\frac{1}{2 + \sqrt{2}} \right) + \frac{1}{2 - \sqrt{2}} + (2 - \sqrt{2})$$

का मान क्या है?

- (a) 2 (b) 4
(c) 8 (d) 6

155. If $a^2 + b^2 = 4b + 6a - 13$, then what is the value of $a + b$?

यदि $a^2 + b^2 = 4b + 6a - 13$ है, तो $a + b$ का मान क्या है?

- (a) 3 (b) 2
(c) 5 (d) 10

156. x and y are positive integers. If $x^4 + y^4 + x^2y^2 = 481$ and $xy = 12$, then what is the value of $x^2 - xy + y^2$?

x तथा y एक धनात्मक पूर्णांक है। यदि $x^4 + y^4 + x^2y^2 = 481$ तथा $xy = 12$ है तो $x^2 - xy + y^2$ का मान क्या है?

- (a) 16 (b) 13
(c) 11 (d) 15

157. If $A = 1 + 2^p$ and $B = 1 + 2^{-p}$, then what is the value of B ?

यदि $A = 1 + 2^p$ तथा $B = 1 + 2^{-p}$ है, तो B का मान क्या है?

- (a) $(A+1)/(A-1)$
(b) $(A+2)/(A+1)$
(c) $A/(A-1)$
(d) $(A-2)/(A+1)$

158. If a and b are roots of the equation $ax^2 + bx + c = 0$, then which equation will have roots $(ab+a+b)$ and $(ab-a-b)$?

यदि a तथा b समीकरण $ax^2 + bx + c = 0$ के मूल हैं, तो किस समीकरण के मूल $(ab+a+b)$ तथा $(ab-a-b)$ होंगे हैं?

- (a) $a^2x^2 + 2acx + c^2 + b^2 = 0$
(b) $a^2x^2 - 2acx + c^2 - b^2 = 0$
(c) $a^2x^2 - 2acx + c^2 + b^2 = 0$
(d) $a^2x^2 + 2acx + c^2 - b^2 = 0$

159. If $\sqrt{(1-p^2)(1-q^2)} = \frac{\sqrt{3}}{2}$, then what is the value of $\sqrt{2p^2 + 2q^2 + 2pq} + \sqrt{2p^2 + 2q^2 - 2pq}$?

यदि $\sqrt{(1-p^2)(1-q^2)} = \frac{\sqrt{3}}{2}$, है तो

$$\sqrt{2p^2 + 2q^2 + 2pq} + \sqrt{2p^2 + 2q^2 - 2pq}$$

का मान क्या है?

- (a) 2 (b) $\sqrt{2}$
(c) 1
(d) None of these/इनमें से कोई नहीं

160. If $(a+b)^2 - 2(a+b) = 80$ and $ab = 16$, then what is the value of $3a - 19b$?

यदि $(a+b)^2 - 2(a+b) = 80$ तथा $ab = 16$ हैं, तो $3a - 19b$ का मान क्या हो सकता है?

- (a) -16 (b) -14
(c) -18 (d) -20

161. If $x^{y+z} = 1$, $y^{x+z} = 1024$ and $z^{x+y} = 729$ (x, y and z are natural numbers), then what is the value of $(z+1)^{y+x+1}$?

यदि $x^{y+z} = 1$, $y^{x+z} = 1024$ तथा $z^{x+y} = 729$ (x, y तथा z प्राकृतिक संख्याएँ हैं), तो $(z+1)^{y+x+1}$ का मान क्या होगा है?

- (a) 6561 (b) 10000
(c) 4096 (d) 14641

162. If $x + y + z = 1$, $x^2 + y^2 + z^2 = 2$ and $x^3 + y^3 + z^3 = 3$, then what is value of xyz ?

यदि $x + y + z = 1$, $x^2 + y^2 + z^2 = 2$ तथा

$x^3 + y^3 + z^3 = 3$ हैं, तो xyz का मान क्या है?

- (a) $1/3$ (b) $1/6$
(c) $1/2$ (d) $1/4$

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{09 Mar 2018} Paper - II

163. If $f(x) = (x^2 + px + 4)$ and $(x-3)$ is a factor of $f(x)$ the what is the value of p ?

यदि $f(x) = (x^2 + px + 4)$ तथा $(x-3)$, $f(x)$ का गुणखण्ड है, तो p का मान क्या है?

- (a) 4 (b) -4
(c) $-\frac{13}{3}$ (d) $\frac{13}{3}$

164. If $y^2 = y + 7$, then what is the value of y^3 ?

यदि $y^2 = y + 7$ है, तो y^3 का मान क्या है?

- (a) $8y + 7$ (b) $y + 14$
(c) $y + 2$ (d) $4y + 7$

165. If $A = 125$ and $B = 8$, then what is the value of $(A+B)^3 - (A-B)^3 - 6B(A^2 - B^2)$?

यदि $A = 125$ तथा $B = 8$ है, तो $(A+B)^3 - (A-B)^3 - 6B(A^2 - B^2)$ का मान क्या है?

- (a) 4096 (b) 4608
(c) 4224 (d) 3456

166. If $x = \frac{1}{1 + \frac{1}{1+x}}$ and $y = \frac{1}{2 + \frac{1}{1+y}}$,

then which of the following can be the value of $x + y$?

यदि $x = \frac{1}{1 + \frac{1}{1+x}}$ तथा $y = \frac{1}{2 + \frac{1}{1+y}}$

है, तो निम्नलिखित में से कौन सा $x + y$ का मान हो सकता है?

- (a) $\frac{(-5 - \sqrt{17} + 3)}{4}$
(b) $\frac{(2\sqrt{5} + \sqrt{17} - 3)}{4}$
(c) $\frac{(-\sqrt{5} + \sqrt{17} + 1)}{4}$
(d) $\frac{(\sqrt{5} + \sqrt{17} - 1)}{4}$

167. If $[x - (1/x)] = 2$, then what is the value of $[x^6 - (1/x^6)]$?

यदि $[x - (1/x)] = 2$ है, तो $[x^6 - (1/x^6)]$ का मान क्या है?

- (a) $114\sqrt{3} + 1$ (b) $134\sqrt{2}$
(c) $142\sqrt{2} + 3$ (d) $140\sqrt{2}$

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168. If $3x+6y+9z=\frac{20}{3}$, $6x+9y+3z=$

$\frac{17}{3}$ and $18x+27y-z=\frac{113}{9}$, then what is the value of $75x+113y$?

यदि $3x+6y+9z=\frac{20}{3}$, $6x+9y+3z=$

$=\frac{17}{3}$ तथा $18x+27y-z=\frac{113}{9}$ है, तो $75x+113y$ का मान क्या है?

- (a) 163/3 (b) 143/6
(c) 218/9 (d) 311/3

169. If $x^{y^z}=1$, $y^{z^x}=125$ and $z^{y^x}=243$ (x, y and z are natural numbers), then what is the value of $9x+10y-18z$?

यदि $x^{y^z}=1$, $y^{z^x}=125$ तथा $z^{y^x}=243$ है (x, y तथा z प्राकृतिक संख्याएँ हैं) तो $9x+10y-18z$ का मान क्या है?

- (a) 18 (b) 15
(c) 12 (d) 5

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170. If $x = \sqrt{1+\frac{\sqrt{3}}{2}} - \sqrt{1-\frac{\sqrt{3}}{2}}$, then the

value of $\frac{\sqrt{2}-x}{\sqrt{2}+x}$ will be closest to:

यदि $x = \sqrt{1+\frac{\sqrt{3}}{2}} - \sqrt{1-\frac{\sqrt{3}}{2}}$ है, तो

$\frac{\sqrt{2}-x}{\sqrt{2}+x}$ का मान किसके निकटतम होगा?

- (a) 1.4 (b) 0.12
(c) 0.17 (d) 1.2

171. If $a^3+b^3=218$ and $a+b=2$, then the value of ab is?

यदि $a^3+b^3=218$ और $a+b=2$ तो ab का मान है

- (a) -35 (b) -31
(c) 34 (d) 32

172. The value of

$\frac{(253)^3+(247)^3}{25.3 \times 25.3 - 624.91 + 24.7 \times 24.7}$

is 50×10^k , where the value of k is:

$\frac{(253)^3+(247)^3}{25.3 \times 25.3 - 624.91 + 24.7 \times 24.7}$

का मान 50×10^k है, जहाँ का k मान है

- (a) 3 (b) -3
(c) 2 (d) 4

173. If $x+y+z=11$, $x^2+y^2+z^2=133$

and $x^3+y^3+z^3=881$, then the value of $\sqrt[3]{xyz}$ is:

यदि $x+y+z=11$, $x^2+y^2+z^2=133$ और $x^3+y^3+z^3=881$ तो

$\sqrt[3]{xyz}$ का मान है:

- (a) 6 (b) -8
(c) -6 (d) 8

174. If $\sqrt{86-60\sqrt{2}}=a-b\sqrt{2}$, then what will be the value of $\sqrt{a^2+b^2}$, correct to one decimal place?

यदि $\sqrt{86-60\sqrt{2}}=a-b\sqrt{2}$ है तो $\sqrt{a^2+b^2}$ का एक दशमलव स्थान सही मान क्या होगा?

- (a) 8.4 (b) 7.8
(c) 8.2 (d) 7.2

175. If $x^8-1442x^4+1=0$, then a possible value of $x-\frac{1}{x}$ is:

यदि $x^8-1442x^4+1=0$, तो $x-\frac{1}{x}$ का संभावित मान है:

- (a) 8 (b) 4
(c) 6 (d) 5

176. If $2\sqrt{2}x^3-3\sqrt{3}y^3=(\sqrt{2}x-\sqrt{3}y)(Ax^2+By^2+Cxy)$, then the value of $A^2+B^2-C^2$ is:

यदि $2\sqrt{2}x^3-3\sqrt{3}y^3=(\sqrt{2}x-\sqrt{3}y)(Ax^2+By^2+Cxy)$, है तो $A^2+B^2-C^2$ मान क्या है?

- (a) 7 (b) 10
(c) 11 (d) 19

177. If $a^2+b^2+c^2+96=8(a+b-2c)$, then $\sqrt{ab-bc+ca}$ is equal to:

यदि $a^2+b^2+c^2+96=8(a+b-2c)$ तो, $\sqrt{ab-bc+ca}$ निम्न में से किसके बराबर है?

- (a) 4 (b) $2\sqrt{3}$
(c) 6 (d) $2\sqrt{2}$

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{12 Sept. 2019}

178. If $x+\frac{1}{16x}=3$, then the value of

$16x^3+\frac{1}{256x^3}$ is:

यदि $x+\frac{1}{16x}=3$ है, तो $16x^3+$

$\frac{1}{256x^3}$ का मान होगा:

- (a) 423 (b) 441
(c) 432 (d) 414

179. If $x+y+z=2$, $xy+yz+zx=-11$, $xyz=-12$ then what is the value of $\sqrt{x^3+y^3+z^3-2}$

यदि $x+y+z=2$, $xy+yz+zx=-11$, और $xyz=-12$ है, तो

$\sqrt{x^3+y^3+z^3-2}$ का मान है:

- (a) 6 (b) 12
(c) 9 (d) 8

180. If $x^4-83x^2+1=0$, then a value of x^3-x^{-3}

यदि $x^4-83x^2+1=0$, तो x^3-x^{-3} का मान हो सकता है:

- (a) 758 (b) 756
(c) 739 (d) 737

181. If $(5x+1)^3+(x-3)^3+8(3x-4)^3$

$=6(5x+1)(x-3)(3x-4)$, then x is equal to:

यदि $(5x+1)^3+(x-3)^3+8(3x-4)^3=6(5x+1)(x-3)(3x-4)$ है, तो x का मान निम्नलिखित के बराबर होगा:

- (a) $\frac{5}{6}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

182. If $8x^3-27y^3=(Ax+By)(Cx^2-Dy^2+6xy)$, then $(A+B+C-D)$ is equal to:

यदि $8x^3-27y^3=(Ax+By)(Cx^2-Dy^2+6xy)$ तो, $(A+B+C-D)$ का मान ज्ञात करें।

- (a) -12 (b) 12
(c) 15 (d) 9

183. If $x = \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$ and y is the reciprocal of x , then what is the value of $(x^3 + y^3)$?

यदि $x = \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$ है और y , x का व्युत्क्रम

है, तो $(x^3 + y^3)$ का मान है:

- (a) 488 (b) 504
(c) 472 (d) 476

184. If $\sqrt{10-2\sqrt{21}} + \sqrt{8+2\sqrt{15}} =$

$\sqrt{a} + \sqrt{b}$ where a and b are positive integers, then the value of $\sqrt{ab}$ is closest to:

यदि $\sqrt{10-2\sqrt{21}} + \sqrt{8+2\sqrt{15}} =$

$\sqrt{a} + \sqrt{b}$, में a और b धनात्मक पूर्णांक हैं, तो $\sqrt{ab}$ का निकटतम मान है:

- (a) 4.6 (b) 5.9
(c) 6.8 (d) 7.2

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185. $ab(a-b) + bc(b-c) + ca(c-a)$ is equal to

$ab(a-b) + bc(b-c) + ca(c-a)$ निम्नलिखित में से किसके बराबर है?

- (a) $(a+b)(b-c)(c-a)$
(b) $(a-b)(b+c)(c-a)$
(c) $(a-b)(b-c)(c-a)$
(d) $(b-a)(b-c)(c-a)$

186. Given that $(5x-3)^3 + (2x+5)^3 + 27(4-3x)^3 = 9(3-5x)(2x+5)(3x-4)$, then the value of $(2x+1)$ is:

यदि $(5x-3)^3 + (2x+5)^3 + 27(4-3x)^3 = 9(3-5x)(2x+5)(3x-4)$ है, तो $(2x+1)$ का मान क्या होगा?

- (a) -13 (b) 15
(c) -15 (d) 13

187. If $5\sqrt{5}x^3 + 2\sqrt{2}y^2 = (Ax + \sqrt{2}y)(Bx^2 + 2y^2 + Cxy)$, then the value of $(A^2 + B^2 - C^2)$ is:

यदि $5\sqrt{5}x^3 + 2\sqrt{2}y^2 = (Ax + \sqrt{2}y)(Bx^2 + 2y^2 + Cxy)$

है, तो $A^2 + B^2 - C^2$ का मान क्या होगा?

- (a) 15 (b) 20
(c) 30 (d) 40

188. If $\frac{3(x^2+1)-7x}{3x} = 6, \neq 0$ then

the value of $\sqrt{x} + \frac{1}{\sqrt{x}}$ is:

यदि $\frac{3(x^2+1)-7x}{3x} = 6, \neq 0$ है, तो

$\sqrt{x} + \frac{1}{\sqrt{x}}$ का मान क्या होगा?

- (a) $\sqrt{\frac{25}{3}}$ (b) $\sqrt{\frac{11}{3}}$
(c) $\sqrt{\frac{35}{3}}$ (d) $\sqrt{\frac{31}{3}}$

189. Let, $x = \sqrt[3]{27} - \sqrt[3]{\frac{3}{4}}$ and $y = \frac{\sqrt{45} + \sqrt{605} + \sqrt{245}}{\sqrt{80} + \sqrt{125}}$.

then the value of $x^2 + y^2$ is:

यदि $x = \sqrt[3]{27} - \sqrt[3]{\frac{3}{4}}$ और $y =$

$\frac{\sqrt{45} + \sqrt{605} + \sqrt{245}}{\sqrt{80} + \sqrt{125}}$ है, तो $x^2 + y^2$ का मान क्या होगा?

- (a) $\frac{223}{36}$ (b) $\frac{221}{36}$
(c) $\frac{221}{9}$ (d) $\frac{227}{9}$

190. If $x + y + z = 6, xyz = -10$ and

$x^2 + y^2 + z^2 = 30$, then what is the value of $(x^3 + y^3 + z^3)$?

यदि $x + y + z = 6, xyz = -10$ तथा $x^2 + y^2 + z^2 = 30$ है, तो $(x^3 + y^3 + z^3)$ का मान क्या होगा?

- (a) 132 (b) 135
(c) 130 (d) 127

191. $\frac{(4.6)^4 + (5.4)^4 + (24.84)^2}{(4.6)^2 + (5.4)^2 + (24.84)} = ?$

- (a) 24.42 (b) 24.24
(c) 25.42 (d) 25.48

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{15 Nov. 2020}

192. If $27(x+y)^3 - 8(x-y)^3 = (x+5y)(Ax^2 + By^2 + Cxy)$, then what is the value of $(A+B-C)$?

यदि $27(x+y)^3 - 8(x-y)^3 = (x+5y)(Ax^2 + By^2 + Cxy)$ है, तो $(A+B-C)$ का मान ज्ञात कीजिए।

- (a) 13 (b) 16
(c) 18 (d) 11

193. Given that $x^8 - 34x^4 + 1 = 0, x >$

0. What is the value of $(x^3 + x^{-3})$?

दिया गया है $x^8 - 34x^4 + 1 = 0, x > 0$, तो $(x^3 + x^{-3})$ मान ज्ञात कीजिए।

(a) $5\sqrt{8}$ (b) $6\sqrt{6}$

(c) $5\sqrt{6}$ (d) $6\sqrt{8}$

194. If $3x^2 - 5x + 1 = 0$, then the value of

$(x^2 + \frac{1}{9x^2})$ is:

यदि $3x^2 - 5x + 1 = 0$ है, तो $(x^2 + \frac{1}{9x^2})$

का मान ज्ञात कीजिए।

(a) $1\frac{2}{3}$ (b) $1\frac{1}{3}$

(c) $2\frac{1}{3}$ (d) $2\frac{1}{9}$

195. If $\frac{1}{4-\sqrt{8}} + \frac{3+2\sqrt{2}}{3-2\sqrt{2}} - \frac{3-2\sqrt{2}}{3+2\sqrt{2}} =$

$a+b\sqrt{2}$, then what is the value of $(3a+4b)$?

यदि $\frac{1}{4-\sqrt{8}} + \frac{3+2\sqrt{2}}{3-2\sqrt{2}} - \frac{3-2\sqrt{2}}{3+2\sqrt{2}} =$

$a+b\sqrt{2}$ है, तो $(3a+4b)$ का मान ज्ञात कीजिए।

(a) $99\frac{1}{2}$ (b) 99

(c) $98\frac{1}{2}$ (d) 98

196. If $a+b+c=6, a^3+b^3+c^3-3abc=342$, then what is the value of $ab+bc+ca$?

यदि $a+b+c=6, a^3+b^3+c^3-3abc=342$ है, तो $ab+bc+ca$ का मान ज्ञात कीजिए।

(a) 8 (b) -7

(c) -5 (d) 5

197. If $9x^2 + y^2 = 37$ and $xy = 2, x, y > 0$, then the value of $(27x^3 + y^3)$ is:

यदि $9x^2 + y^2 = 37$ और $xy = 2, x, y > 0$ है, तो $(27x^3 + y^3)$ का मान ज्ञात कीजिए।

(a) 217 (b) 207

(c) 301 (d) 259

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{16 Nov. 2020}

198. If $a + b + c = 7$ and $a^3 + b^3 + c^3 - 3abc = 175$, then what is the value of $(ab + bc + ca)$?

यदि $a + b + c = 7$ और $a^3 + b^3 + c^3 - 3abc = 175$ है, तो $(ab + bc + ca)$ का मान ज्ञात कीजिए।

- (a) 7 (b) 8
(c) 6 (d) 9

199. If $x^2 + 4y^2 = 17$ and $xy = 2$, where $x > 0$, $y > 0$, then what is the value of $x^3 + 8y^3$?

यदि $x^2 + 4y^2 = 17$ और $xy = 2$ है, जहाँ $x > 0$, $y > 0$ है, तो $x^3 + 8y^3$ का मान ज्ञात कीजिए।

- (a) 85 (b) 76
(c) 65 (d) 95

200. If $(x + y)^3 + 8(x - y)^3 = (3x + Ay)(3x^2 + Bxy + Cy^2)$, then the value of $A + B + C$ is :

यदि $(x + y)^3 + 8(x - y)^3 = (3x + Ay)(3x^2 + Bxy + Cy^2)$ है, तो $A + B + C$ का मान ज्ञात कीजिए।

- (a) 0 (b) 4
(c) 2 (d) 3

201. If $2x^2 - 7x + 5 = 0$, then what is the value of $x^2 + \frac{25}{4x^2}$?

यदि $2x^2 - 7x + 5 = 0$ है, तो $x^2 + \frac{25}{4x^2}$ का मान ज्ञात कीजिए।

- (a) $9\frac{1}{2}$ (b) $7\frac{1}{4}$
(c) $9\frac{3}{4}$ (d) $5\frac{1}{2}$

202. If $x - \frac{1}{x} = 5$, $x \neq 0$, then what is

the value of $\frac{x^6 + 3x^3 - 1}{x^6 - 8x^3 - 1}$?

यदि $x - \frac{1}{x} = 5$, $x \neq 0$ है, तो

$\frac{x^6 + 3x^3 - 1}{x^6 - 8x^3 - 1}$ का मान ज्ञात कीजिए।

- (a) $\frac{13}{12}$ (b) $\frac{11}{13}$
(c) $\frac{3}{8}$ (d) $\frac{4}{9}$

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{18 Nov. 2020}

203. If $x^2 + \frac{1}{x^2} = 7$, then the value of

$x^3 + \frac{1}{x^3}$ where $x > 0$ is equal to:

यदि $x^2 + \frac{1}{x^2} = 7$ है, तो $x^3 + \frac{1}{x^3}$ का

मान ज्ञात करें, जहाँ $x > 0$ है।

- (a) 16 (b) 18
(c) 15 (d) 12

204. If $x\left(3 - \frac{2}{x}\right) = \frac{3}{x}$, then the value

of $x^3 - \frac{1}{x^3}$ is equal to:

यदि $x\left(3 - \frac{2}{x}\right) = \frac{3}{x}$ है, तो $x^3 - \frac{1}{x^3}$ का

मान ज्ञात कीजिए।

- (a) $\frac{8}{27}$ (b) $\frac{52}{27}$
(c) $\frac{62}{27}$ (d) $\frac{61}{27}$

205. If $x + \frac{16}{x} = 8$, then the value of

$x^2 + \frac{32}{x^2}$ is:

यदि $x + \frac{16}{x} = 8$ है, तो $x^2 + \frac{32}{x^2}$ का

मान ज्ञात कीजिए।

- (a) 24 (b) 16
(c) 20 (d) 18

206. If $x = \sqrt{-\sqrt{3} + \sqrt{3 + 8\sqrt{7 + 4\sqrt{3}}}}$

where $x > 0$, then the value of x is equal to:

यदि $x = \sqrt{-\sqrt{3} + \sqrt{3 + 8\sqrt{7 + 4\sqrt{3}}}}$

जहाँ $x > 0$ है, तो x का मान ज्ञात कीजिए।

- (a) 2 (b) 4
(c) 1 (d) 3

207. If $\sqrt{x} + \frac{1}{\sqrt{x}} = 3$, then the value of

$x^3 + \frac{1}{x^3}$ is:

यदि $\sqrt{x} + \frac{1}{\sqrt{x}} = 3$ है, तो $x^3 + \frac{1}{x^3}$ का मान ज्ञात कीजिए।

- (a) 324 (b) 326
(c) 322 (d) 422

208. If $x - \frac{3}{x} = 6$, $x \neq 0$, then the

value of $\frac{x^4 - 27}{x^2 - 3x - 3}$ is:

यदि $x - \frac{3}{x} = 6$, $x \neq 0$ है, तो

$\frac{x^4 - 27}{x^2 - 3x - 3}$ का मान ज्ञात कीजिए।

- (a) 90 (b) 80
(c) 270 (d) 54

209. If $\frac{8 + 2\sqrt{3}}{3\sqrt{3} + 5} = a\sqrt{3} - b$, then the value of $a + b$ is equal to:

यदि $\frac{8 + 2\sqrt{3}}{3\sqrt{3} + 5} = a\sqrt{3} - b$ है, तो $a + b$ का मान ज्ञात कीजिए।

- (a) 16 (b) 18
(c) 15 (d) 24

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{29 Jan. 2022}

210. If $2x - y = 2$ and $xy = \frac{3}{2}$, then

what is the value of $x^3 - \frac{y^3}{8}$

यदि $2x - y = 2$ और $xy = \frac{3}{2}$ तो $x^3 - \frac{y^3}{8}$ का मान ज्ञात करें?

- (a) $\frac{9}{2}$ (b) $-\frac{5}{4}$
(c) $\frac{5}{2}$ (d) $\frac{13}{4}$

211. If $2x^2 + 5x + 1 = 0$, then one of the values of $x - \frac{1}{2x}$?

यदि $2x^2 + 5x + 1 = 0$ है, तो $x - \frac{1}{2x}$ का मान क्या होगा?

- (a) $\frac{\sqrt{17}}{2}$ (b) $\frac{13}{2}$
(c) $\frac{5}{2}$ (d) $\frac{\sqrt{13}}{2}$

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{03 Feb. 2022}

212. If $a+b+c = 1$, $ab+bc+ca = -22$ and $abc = -40$, then what is the value of $a^3+b^3+c^3$?

यदि $a+b+c = 1$, $ab+bc+ca = -22$ और $abc = -40$ तो $a^3+b^3+c^3$ का क्या होगा?

- (a) 67 (b) -53
(c) -51 (d) 27

213. If $a+b = 8$, $ab = 10$ then the value of a^3+b^3 is:

यदि $a+b = 8$, $ab = 10$ तो a^3+b^3 का मान क्या होगा।

- (a) 312 (b) 215
(c) 272 (d) 111

214. If $x = \sqrt{1+\frac{\sqrt{3}}{2}} - \sqrt{1-\frac{\sqrt{3}}{2}}$ then the

value of $\frac{\sqrt{3}-x}{\sqrt{3}+x}$ (correct to one decimal place) is?

यदि $x = \sqrt{1+\frac{\sqrt{3}}{2}} - \sqrt{1-\frac{\sqrt{3}}{2}}$ तो

$\frac{\sqrt{3}-x}{\sqrt{3}+x}$ का मान क्या होगा (दशमलव के एक स्थान तक सही)?

- (a) 0.25 (b) 0.17
(c) 0.19 (d) 0.27

215. $\frac{1}{4-\sqrt{15}} - \frac{1}{\sqrt{15}-\sqrt{14}} + \frac{1}{\sqrt{14}-\sqrt{13}}$
 $-\frac{1}{\sqrt{13}-\sqrt{12}} + \frac{1}{\sqrt{12}-\sqrt{11}} - \frac{1}{\sqrt{11}-\sqrt{10}} + \frac{1}{\sqrt{10}-3} - \frac{1}{3-\sqrt{8}} = ?$

- (a) $2+2\sqrt{2}$ (b) $2-2\sqrt{2}$
(c) $4-2\sqrt{2}$ (d) $4+2\sqrt{2}$

216. If $x^2 - \sqrt{7}x + 1 = 0$, then what is

the value of $x^5 + \frac{1}{x^5}$?

यदि $x^2 - \sqrt{7}x + 1 = 0$ है, तो $x^5 + \frac{1}{x^5}$ का मान क्या होगा?

- (a) $19\sqrt{7}$ (b) $27\sqrt{7}$
(c) $21\sqrt{7}$ (d) $25\sqrt{7}$

217. If $\frac{22\sqrt{2}}{4\sqrt{2}-\sqrt{3}+\sqrt{5}} = a + \sqrt{5}b$, with $a, b > 0$, then what is the value of $(ab):(a+b)$?

यदि $\frac{22\sqrt{2}}{4\sqrt{2}-\sqrt{3}+\sqrt{5}} = a + \sqrt{5}b$ है, जहाँ $a, b > 0$ है, तो $(ab):(a+b)$ का मान क्या होगा?

- (a) 8 : 7 (b) 4 : 7
(c) 7 : 8 (d) 7 : 4

218. If $x^2-3x+1 = 0$, then the value of

$$\frac{x^4 + \frac{1}{x^2}}{x^2 + 5x + 1} \text{ is}$$

यदि $x^2-3x+1 = 0$ है तो $\frac{x^4 + \frac{1}{x^2}}{x^2 + 5x + 1}$

का मान क्या होगा?

- (a) $\frac{9}{4}$ (b) $\frac{27}{8}$
(c) $\frac{27}{8}$ (d) 2

219. If $x = 32.5$, $y = 34.6$ and $z = 30.9$, then the value of $x^3+y^3+z^3 - 3xyz$ is 0.98k, where k is equal to:

यदि $x = 32.5$, $y = 34.6$ और $z = 30.9$ है, तो $x^3+y^3+z^3 - 3xyz$ का मान 0.98k होता है, जहाँ k का मान है।

- (a) 1033 (b) 933
(c) 1026 (d) 921

220. If $27x^3 - 64y^3 = (Ax + By)(x^3 - Dy^2 + 12xy)$, then the value of $4A + B + 3C + 2D$ is:

यदि $27x^3 - 64y^3 = (Ax + By)(x^3 - Dy^2 + 12xy)$ तो $4A + B + 3C + 2D$ का मान क्या होगा?

- (a) 5 (b) 3
(c) -3 (d) -4

221. The value of $(0.3)^{\frac{((200-146)/(3 \times 3 \times 3)) - 3}{(0.3)^{\frac{((200-146)/(3 \times 3 \times 3)) - 3}}}}$

$(0.3)^{\frac{((200-146)/(3 \times 3 \times 3)) - 3}{(0.3)^{\frac{((200-146)/(3 \times 3 \times 3)) - 3}}}}$ का मान कितना होगा?

- (a) $\frac{10}{3}$ (b) $\frac{5}{3}$
(c) $\frac{7}{3}$ (d) $\frac{8}{3}$

222. If $\frac{\sqrt{26-7\sqrt{3}}}{\sqrt{14+5\sqrt{3}}} = \frac{b+a\sqrt{3}}{11}$ $b > 0$,

then what is the value of $\sqrt{(b-a)}$?

यदि $\frac{\sqrt{26-7\sqrt{3}}}{\sqrt{14+5\sqrt{3}}} = \frac{b+a\sqrt{3}}{11}$ $b > 0$ हो,

तो $\sqrt{(b-a)}$ का मान कितना होगा?

- (a) 5 (b) 25
(c) 12 (d) 9

223. If $\frac{\sqrt{38-5\sqrt{3}}}{\sqrt{26+7\sqrt{3}}} = \frac{a+b\sqrt{3}}{23}$, $b > 0$,

then the value of $(b-a)$, is:

यदि $\frac{\sqrt{38-5\sqrt{3}}}{\sqrt{26+7\sqrt{3}}} = \frac{a+b\sqrt{3}}{23}$, $b >$

0 हो तो $(b-a)$ का मान कितना होगा?

- (a) 7 (b) 18
(c) 29 (d) 11

224. If $x + \frac{1}{x} = 3$, $x \neq 0$, then the value

of $x^7 + \frac{1}{x^7}$ is:

यदि $x + \frac{1}{x} = 3$, $x \neq 0$ है तो $x^7 + \frac{1}{x^7}$ का

मान क्या होगा?

- (a) 749 (b) 843
(c) 746 (d) 849

225. If $x^4+y^4+x^2y^2 = 17 \frac{1}{16}$ and x^2-

$xy+y^2 = 5 \frac{1}{4}$, then one of the values of $(x-y)$ is:

यदि $x^4+y^4+x^2y^2 = 17 \frac{1}{16}$ और x^2-

$xy+y^2 = 5 \frac{1}{4}$ तो $(x-y)$ का मान है:

- (a) $\frac{5}{2}$ (b) $\frac{3}{4}$
(c) $\frac{5}{4}$ (d) $\frac{3}{2}$

ANSWER KEY - ALGEBRA

- | | | | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 1. (c) | 2. (d) | 3. (a) | 4. (a) | 5. (c) | 6. (b) | 7. (c) | 8. (a) | 9. (c) | 10. (d) |
| 11. (a) | 12. (d) | 13. (c) | 14. (c) | 15. (a) | 16. (d) | 17. (b) | 18. (a) | 19. (b) | 20. (b) |
| 21. (d) | 22. (d) | 23. (d) | 24. (b) | 25. (a) | 26. (d) | 27. (b) | 28. (d) | 29. (c) | 30. (c) |
| 31. (a) | 32. (d) | 33. (c) | 34. (a) | 35. (c) | 36. (a) | 37. (d) | 38. (b) | 39. (a) | 40. (d) |
| 41. (b) | 42. (a) | 43. (d) | 44. (b) | 45. (b) | 46. (c) | 47. (d) | 48. (c) | 49. (b) | 50. (b) |
| 51. (b) | 52. (c) | 53. (a) | 54. (c) | 55. (d) | 56. (b) | 57. (c) | 58. (c) | 59. (b) | 60. (b) |
| 61. (a) | 62. (d) | 63. (b) | 64. (b) | 65. (d) | 66. (c) | 67. (b) | 68. (b) | 69. (a) | 70. (b) |
| 71. (b) | 72. (c) | 73. (c) | 74. (b) | 75. (a) | 76. (c) | 77. (d) | 78. (d) | 79. (a) | 80. (c) |
| 81. (a) | 82. (d) | 83. (b) | 84. (a) | 85. (a) | 86. (b) | 87. (c) | 88. (c) | 89. (c) | 90. (b) |
| 91. (a) | 92. (a) | 93. (a) | 94. (c) | 95. (b) | 96. (a) | 97. (b) | 98. (b) | 99. (c) | 100. (a) |
| 101. (a) | 102. (a) | 103. (b) | 104. (c) | 105. (d) | 106. (d) | 107. (c) | 108. (b) | 109. (a) | 110. (c) |
| 111. (b) | 112. (c) | 113. (d) | 114. (a) | 115. (c) | 116. (a) | 117. (d) | 118. (b) | 119. (c) | 120. (b) |
| 121. (b) | 122. (a) | 123. (b) | 124. (d) | 125. (c) | 126. (c) | 127. (c) | 128. (a) | 129. (b) | 130. (d) |
| 131. (b) | 132. (b) | 133. (c) | 134. (c) | 135. (c) | 136. (d) | 137. (b) | 138. (a) | 139. (b) | 140. (b) |
| 141. (c) | 142. (a) | 143. (a) | 144. (b) | 145. (a) | 146. (b) | 147. (b) | 148. (a) | 149. (a) | 150. (b) |
| 151. (c) | 152. (b) | 153. (b) | 154. (d) | 155. (c) | 156. (b) | 157. (c) | 158. (b) | 159. (b) | 160. (b) |
| 161. (b) | 162. (b) | 163. (c) | 164. (a) | 165. (a) | 166. (b) | 167. (d) | 168. (a) | 169. (d) | 170. (c) |
| 171. (a) | 172. (a) | 173. (c) | 174. (b) | 175. (c) | 176. (a) | 177. (a) | 178. (a) | 179. (a) | 180. (b) |
| 181. (a) | 182. (b) | 183. (a) | 184. (b) | 185. (d) | 186. (b) | 187. (b) | 188. (d) | 189. (a) | 190. (a) |
| 191. (d) | 192. (b) | 193. (a) | 194. (d) | 195. (c) | 196. (b) | 197. (a) | 198. (b) | 199. (c) | 200. (a) |
| 201. (b) | 202. (a) | 203. (b) | 204. (c) | 205. (d) | 206. (a) | 207. (c) | 208. (a) | 209. (b) | 210. (d) |
| 211. (a) | 212. (b) | 213. (c) | 214. (d) | 215. (c) | 216. (a) | 217. (a) | 218. (a) | 219. (a) | 220. (b) |
| 221. (a) | 222. (a) | 223. (c) | 224. (b) | 225. (a) | | | | | |

Gagan

Hints & Solutions {ALGEBRA}

SSC CGL MAINS 2010

{01 Aug. 2010}

1. (c) $a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} = 4$

$$\left(a^2 + \frac{1}{a^2}\right) + \left(b^2 + \frac{1}{b^2}\right) = 4$$

$$\downarrow \quad \quad \downarrow \quad \quad = 4$$

$$a = 1, b = 1$$

$$a^2 + b^2 = 1 + 1 = 2$$

2. (d) $\left(x + \frac{1}{x}\right)^2 = 3 \Rightarrow x + \frac{1}{x} = \sqrt{3}$

Cube both side:-

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3x \times \frac{1}{x} \left(x + \frac{1}{x}\right)$$

$$x^3 + \frac{1}{x^3} = (\sqrt{3})^3 - 3\sqrt{3}$$

$$x^3 + \frac{1}{x^3}$$

$$= 3\sqrt{3} - 3\sqrt{3} = 0$$

3. (a) $\frac{0.1 \times 0.1 \times 0.1 + 0.02 \times 0.02 \times 0.02}{0.2 \times 0.2 \times 0.2 + 0.04 \times 0.04 \times 0.04}$

$$= \frac{[(0.1)^3 + (0.02)^3]}{8[(0.1)^3 + (0.02)^3]}$$

$$= \frac{1}{8} = 0.125$$

4. (a) $x + \frac{1}{x} = 2$

$$\text{Put, } x = 1$$

$$\Rightarrow 1 + \frac{1}{1} = 2$$

$$x^{100} + \frac{1}{x^{100}}$$

$$= (1)^{100} + \frac{1}{(1)^{100}} = 2$$

5. (c) $x^3 + 3x^2 + 3x = 7$

$$\text{add } (+1, -1)$$

$$x^3 + 3x^2 + 3x + 1 - 1 = 7$$

$$(x+1)^3 - 1 = 7$$

$$(x+1)^3 = 8$$

$$x = 1$$

6. (b) $2x + \frac{2}{x} = 1$

$$x + \frac{1}{x} = \frac{1}{2}$$

cube both side-

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$$

$$x^3 + \frac{1}{x^3} = \left(\frac{1}{2}\right)^3 - 3 \times \frac{1}{2}$$

$$x^3 + \frac{1}{x^3} = \frac{1}{8} - \frac{3}{2}$$

$$= \frac{1-12}{8} = \frac{-11}{8}$$

7. (c) $2x + \frac{1}{3x} = 6$

On multiplying both side by $3/2$

$$\frac{3}{2} \left(2x + \frac{1}{3x}\right) = (6) \times \frac{3}{2}$$

$$3x + \frac{1}{2x} = 9$$

8. (a) $x^2 = \frac{1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$

$$x^2 = \sqrt{2}+1$$

$$\left(\frac{1}{x}\right)^2 = \sqrt{2}-1$$

$$x^2 - \frac{1}{x^2} = (\sqrt{2}+1) - (\sqrt{2}-1) = 2$$

9. (c) $\frac{(0.87)^3 + (0.13)^3}{(0.87)^2 + (0.13)^2 - (0.87) \times (0.13)}$

multiply and divide by $(a+b)$

$$\frac{a^3 + b^3}{a^2 + b^2 - ab} = a+b$$

$$= 0.87 + 0.13 = 1$$

10. (d) $x^2 + y^2 - 2x + 6y + 10 = 0$

$$(x-1)^2 + (y+3)^2 = 0$$

$$x = 1, y = -3$$

$$x^2 + y^2 = 1^2 + (-3)^2 = 10$$

11. (a) $x^{1/3} + y^{1/3} = z^{1/3}$

Cubing both sides

$$x + y + 3x^{1/3}y^{1/3}(x^{1/3} + y^{1/3}) = z$$

$$(x + y - z) = -3x^{1/3}y^{1/3}z^{1/3}$$

Cubing both sides

$$(x + y - z)^3 = -27xyz$$

so,

$$(x + y - z)^3 + 27xyz = 0$$

12. (d) $a^2 = 2, a+1 = ?$

$$a = \sqrt{2}$$

$$a + 1 = \sqrt{2} + 1$$

From option (d)

$$\Rightarrow \frac{a-1}{3-2a} = \frac{\sqrt{2}-1}{3-2\sqrt{2}} = \frac{\sqrt{2}-1}{(\sqrt{2}-1)^2}$$

$$= \frac{1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$$

$$= \sqrt{2}+1 = a+1$$

13. (c) $a + b + c = 1$

$$ab + bc + ca = 1/3$$

Put, $a = b = c = 1/3$,

both equation are satisfied

$$a = b = c = 1/3$$

Hence

$$a : b : c = 1 : 1 : 1$$

SSC CGL MAINS 2011

{04 Sept. 2011}

14. (c) $a^2 + b^2 + c^2 = 2(a-b-c) - 3$

$$(a^2 - 2a + 1) + (b^2 + 2b + 1) + (c^2 + 2c + 1) = 0$$

$$(a-1)^2 + (b+1)^2 + (c+1)^2 = 0$$

$$a = 1, b = -1, c = -1$$

Then,

$$a - b + c = 1 - (-1) - 1 = 1$$

15. (a) $x^2 + 3x + 1 = 0$

$$\frac{x^2 + 3x + 1}{x} = 0$$

$$x + \frac{1}{x} + 3 = 0$$

$$x + \frac{1}{x} = -3$$

now cube both side

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3x \times \frac{1}{x} \left(x + \frac{1}{x}\right)$$

$$x^3 + \frac{1}{x^3} = (-3)^3 - 3 \times (-3)$$

$$x^3 + \frac{1}{x^3} = -27 + 9 = -18$$

16. (d) $x^a \cdot x^b \cdot x^c = 1$

$$x^{a+b+c} = x^0$$

$$\text{If } a+b+c = 0$$

Then,

$$a^3 + b^3 + c^3 = 3abc$$

17. (b) $a + \frac{1}{a} + 2 = 0$

$$a + \frac{1}{a} = -2$$

$$a^2 + 1 = -2a$$

$$a^2 + 1 + 2a = 0$$

$$(a+1)^2 = 0$$

$$a = -1$$

Then,

$$a^{37} - \frac{1}{a^{100}} = (-1)^{37} - \frac{1}{(1)^{100}}$$

$$= -1 - 1 = -2$$

18. (a) For the parallel lines

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\frac{k-1}{2-k} = \frac{1}{-3}$$

$$-3k+3 = 2-k$$

$$-2k = -1$$

$$k = 1/2$$

19. (b) $x + \frac{1}{16x} = 1$... (i)

multiply by 4

$$4x + \frac{1}{4x} = 4$$

Cubing both side-

$$\left(4x + \frac{1}{4x}\right)^3 = 4^3$$

$$64x^3 + \frac{1}{64x^3} + 3.4x \cdot \frac{1}{4x} \left(4x + \frac{1}{4x}\right) = 64$$

$$64x^3 + \frac{1}{64x^3} + 3 \times 4 = 64$$

$$\therefore 64x^3 + \frac{1}{64x^3} = 52$$

20. (b) $a+b+c = 0$

Let, $a=1, b=-2, c=1$

Given,

$$\frac{a^2 + b^2 + c^2}{b^2 - ca}$$

$$= \frac{1^2 + (-2)^2 + (1)^2}{(-2)^2 - (1)(1)}$$

$$= \frac{1+4+1}{4-1} = \frac{6}{3} = 2$$

21. (d) $(a^4+b^4+a^2b^2) = (a^2+ab+b^2) (a^2-ab+b^2)$

$$8 = 4 (a^2+b^2-ab)$$

$$a^2+ab+b^2 = 4$$

$$a^2-ab+b^2 = 2$$

$$2ab = 2$$

$$ab = 1$$

Alternatively:-

$$a^2+ab+b^2 = 4 \quad \dots (i)$$

$$a^4+b^4+a^2b^2 = 8$$

$$(a^2+b^2)^2 - a^2b^2 = 8$$

$$(a^2+b^2+ab)(a^2+b^2-ab) = 8$$

$$a^2-ab+b^2 = 2 \quad \dots (ii)$$

From eqn. (i) and (ii)

$$a^2+b^2 = 3 \quad \dots (iii)$$

$$a^2+b^2+ab = 4$$

$$3 + ab = 4$$

$$ab = 1$$

22. (d) Given

$$a = 25, b = 15, c = -10$$

Then,

$$\frac{a^3 + b^3 + c^3 - 3abc}{(a-b)^2 + (b-c)^2 + (c-a)^2}$$

$$= \frac{1}{2} \frac{(a+b+c)[(a-b)^2 + (b-c)^2 + (c-a)^2]}{(a-b)^2 + (b-c)^2 + (c-a)^2}$$

$$[\therefore a^3 + b^3 + c^3 - 3abc = \frac{1}{2} \{(a+b+c)(a-b)^2 + (b-c)^2 + (c-a)^2\}]$$

$$= \frac{1}{2} (a+b+c)$$

$$= \frac{1}{2} [25+15-10] = 15$$

SSC CGL MAINS 2012

{16 Sept. 2012}

23. (d) $\frac{1+x}{1+\sqrt{1+x}} + \frac{1-x}{1-\sqrt{1-x}}$

$$x+1 = \frac{\sqrt{3}}{2} + 1 = \frac{\sqrt{3}+2}{2}$$

$$x+1 = \frac{\sqrt{3}+2}{2} \times \frac{2}{2} = \frac{(\sqrt{3}+1)^2}{4}$$

$$\sqrt{1+x} = \frac{\sqrt{3}+1}{2}$$

Similarly,

$$\sqrt{1-x} = \frac{\sqrt{3}-1}{2}$$

$$\frac{\sqrt{3}+2}{2} + \frac{2-\sqrt{3}}{2} = \frac{2}{2} = 1$$

$$= \frac{\sqrt{3}+2}{3+\sqrt{3}} + \frac{2-\sqrt{3}}{3-\sqrt{3}} = \frac{3\sqrt{3}-3+6-2\sqrt{3}+2\sqrt{3}+6-3-3\sqrt{3}}{9-3}$$

$$= \frac{6}{6} = 1$$

24. (b) $\sqrt{4x-9} + \sqrt{4x+9} = 5 + \sqrt{7}$

Go through option, Put, $x = 4$

$$\sqrt{4(4)-9} + \sqrt{4(4)+9} = 5 + \sqrt{7}$$

$$\sqrt{7} + \sqrt{25} = 5 + \sqrt{7}$$

$$\sqrt{7} + 5 = 5 + \sqrt{7}$$

Hence proved

Required answered = 4

25. (a) $a^3 + b^3 + c^3 - 3abc$

$$= \frac{1}{2} (a+b+c) [(a-b)^2 + (b-c)^2 + (c-a)^2]$$

$$= \frac{1}{2} (225 + 226 + 227) [(1)^2 + (1)^2 + (-2)^2]$$

$$= \frac{1}{2} \times 678 \times 6$$

$$= 2034$$

26. (d) $\sqrt{x^2 - x + 1} + \frac{1}{\sqrt{x^2 + x + 1}} = 2 - x^2$

Go through option,

Put $x = 0$

$$\sqrt{0-0+1} + \frac{1}{\sqrt{0+0+1}} = 2 - (0)^2$$

$$2 = 2$$

Hence proved $x = 0$

$\therefore$ number of solution = 1

27. (b) $x + \frac{a}{x} = 1$

$$\Rightarrow x^2 + a = x, x^2 - x = -a$$

$$\therefore \frac{x^2 + x + a}{x^3 - x^2} = \frac{2x}{x[x^2 - x]}$$

$$= -\frac{2}{a}$$

28. (d) $\sqrt{28-6\sqrt{3}}$
 $\swarrow \searrow$
 $3\sqrt{3} \times 2 \times 1$

$$\sqrt{(3\sqrt{3}-1)^2} = \sqrt{3} a + b$$

On comparing-

$$a = 3, b = -1$$

$$a + b = 3 - 1 = 2$$

Alternatively:-

$$\sqrt{28-6\sqrt{3}} = \sqrt{3} a + b$$

Squaring both sides

$$28-6\sqrt{3} = (\sqrt{3}a+b)^2$$

$$28-6\sqrt{3} = (\sqrt{3}a)^2 + (b)^2 + 2\sqrt{3}ab$$

$$28-6\sqrt{3} = 3a^2 + b^2 + 2\sqrt{3}ab$$

Compare the coefficient

$$2\sqrt{3}ab = 6\sqrt{3}$$

$$2ab = -6$$

$$\therefore ab = -3,$$

$$3a^2 + b^2 = 28$$

$$a = 3, b = -1$$

Required answer

$$a + b = 3 - 1 = 2$$

29. (c) $x = \frac{\sqrt{5+1}}{\sqrt{5-1}}$

Rationalize it-

$$= \frac{\sqrt{5+1}}{\sqrt{5-1}} \times \frac{\sqrt{5+1}}{\sqrt{5+1}} = \frac{(\sqrt{5+1})^2}{(\sqrt{5})^2 - (1)^2}$$

$$= \frac{\sqrt{5+1}}{2}$$

$$\therefore x^2 - x - 1$$

$$= \left(\frac{\sqrt{5+1}}{2}\right)^2 - \left(\frac{\sqrt{5+1}}{2}\right) - 1$$

$$= \frac{6+2\sqrt{5}}{4} - \frac{(\sqrt{5+1})}{2} - 1$$

$$= \frac{6+2\sqrt{5}-2(\sqrt{5+1})-4}{4}$$

$$= \frac{6+2\sqrt{5}-2\sqrt{5}-2-4}{4} = 0$$

SSC CGL MAINS 2013

{29 Sept. 2013}

30. (c) $x^3+y^3+z^3-3xyz$

$$= \frac{1}{2} (x+y+z)[(x-y)^2+(y-z)^2+(z-x)^2]$$

$$= \frac{1}{2} (333+333+334)[(333-333)^2 + (333-334)^2 + (334-333)^2]$$

$$= \frac{1}{2} \times 1000 \times (0+1+1) = 1000$$

31. (a) Let, $a=1, b=0, c=1$

$$\frac{x-1}{1} + \frac{x-0}{2} + \frac{x-1}{1} = 4(1+0+1)$$

$$x-1 + \frac{x}{2} + x-1 = 4 \times 2$$

$$x + \frac{x}{2} + x = 4 \times 2 + 2$$

$$\frac{5x}{2} = 10 \Rightarrow x = 4$$

From option (i)

$$(1+0+1)^2 = (a+b+c)^2$$

32. (d) $(x-a)(x-b)=1$

$$(x-b) = \frac{1}{x-a}$$

$$b = x - \frac{1}{x-a}$$

$$a-b+5=0$$

$$a-x + \frac{1}{x-a} + 5 = 0$$

[From eq (i)]

$$(x-a) - \frac{1}{x-a} = 5$$

Cubing both sides

$$(x-a)^3 - \frac{1}{(x-a)^3} - 3 \times 5 = 125$$

$$\therefore (x-a)^3 - \frac{1}{(x-a)^3} = 140$$

33. (c) $\sqrt{x} = \sqrt{3} - \sqrt{5}$

square both side

$$x = 3+5-2\sqrt{15}$$

$$x = 8-2\sqrt{15}$$

Now,

$$x^2-16x+6 = (8-2\sqrt{15})^2 - 16(8-2\sqrt{15}) + 6$$

$$= 64 + 60 - 32\sqrt{15} - 128 + 32$$

$$\sqrt{15} + 6$$

$$= 130 - 128 = 2$$

34. (a) $a^2+b^2+c^2 = 2(a-b-c)+3 = 0$

$$a^2+b^2+c^2-2a+2b+2c+3 = 0$$

$$(a^2-2a+1)+(b^2+2b+1)+(c^2+2c+1) = 0$$

$$(a-1)^2+(b+1)^2+(c+1)^2 = 0$$

$$a = 1, b = -1, c = -1$$

$$\therefore 4a-3b+5c = 4+3-5 = 2$$

35. (c) $2x + \frac{2}{x} = 3$

$$x + \frac{1}{x} = \frac{3}{2}$$

Cubing both sides,

$$x^3 + \frac{1}{x^3} + 3 \times \frac{3}{2} = \frac{27}{8}$$

$$x^3 + \frac{1}{x^3} + \frac{9}{2} = \frac{27}{8}$$

$$x^3 + \frac{1}{x^3} + 2 = \frac{27}{8} - \frac{9}{2} + \frac{2}{1}$$

$$= \frac{27-36+16}{8} = \frac{7}{8}$$

36. (a) $(a^2-b^2)^3 + (b^2-c^2)^3 + (c^2-a^2)^3$

$$\therefore a^2-b^2 + b^2-c^2 + c^2-a^2 = 0$$

$$(a^2-b^2)^3 + (b^2-c^2)^3 + (c^2-a^2)^3$$

$$= 3(a^2-b^2)(b^2-c^2)(c^2-a^2)$$

$$\therefore \text{One factor} = (a+b)(a-b)$$

... (i) 37. (d) $x = \sqrt[3]{5} + 2$

$$x-2 = \sqrt[3]{5}$$

Cube both sides

$$x^3-8-3 \times x \times 2(x-2) = 5$$

$$x^3-8-6x^2+12x-5 = 0$$

$$x^3-6x^2+12x-13 = 0$$

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38. (b) Put, $a = b = c = 1$

$$\therefore \frac{a+c}{b} = \frac{1+1}{1} = 2$$

Alternatively:-

$$a^2 + b^2 + c^2 = ab + bc + ca$$

multiply by 2 both side

$$\Rightarrow 2(a^2 + b^2 + c^2) = 2(ab + bc + ca)$$

$$\Rightarrow (a^2 + b^2 - 2ab) + (b^2 + c^2 - 2bc) + (c^2 + a^2 - 2ca) = 0$$

$$\Rightarrow (a-b)^2 + (b-c)^2 + (c-a)^2 = 0$$

$$\Rightarrow a = b, b = c, c = a$$

$$\Rightarrow \therefore a = b = c$$

$$\therefore \frac{a+c}{b} = \frac{2b}{b} = 2$$

39. (a) $ab + bc + ca = 0$

$$-bc = ab + ac$$

$$-ca = ab + bc$$

$$-ab = bc + ca$$

$$\therefore \frac{1}{a^2 + ab + ac} + \frac{1}{b^2 + ab + bc}$$

$$+ \frac{1}{c^2 + bc + ca}$$

$$= \frac{1}{a(a+b+c)} + \frac{1}{b(a+b+c)}$$

$$+ \frac{1}{c(a+b+c)}$$

$$= \frac{bc + ac + ab}{abc(a+b+c)} = 0$$

Alternatively:-

Put, $a = -2, b = 1, c = -2$

40. (d) $(2 + \sqrt{3})a = (2 - \sqrt{3})b = 1$

$$(2 + \sqrt{3})a = 1, (2 - \sqrt{3})b = 1$$

$$\therefore \frac{1}{a} = 2 + \sqrt{3} \Rightarrow \frac{1}{b} = 2 - \sqrt{3}$$

$$\frac{1}{a} + \frac{1}{b} = 2 + \sqrt{3} + 2 - \sqrt{3} = 4$$

41. (b) $3x + \frac{3}{x} = 1$

$$x + \frac{1}{x} = \frac{1}{3}$$

Cubing both sides

$$x^3 + \frac{1}{x^3} + 3 \times \frac{1}{3} = \frac{1}{27}$$

$$x^3 + \frac{1}{x^3} + 1 = \frac{1}{27}$$

42. (a) Put, $a = b = 1$

$$1 + 4 + 4 - 4 - 2 - 8 = 9 - 14 = -5$$

only option (a) gives -5

Alternatively:-

$$a^2 + 4b^2 + 4b - 4ab - 2a - 8$$

$$= (a^2 + 4b^2 - 4ab) + (4b - 2a) - 8$$

$$\text{Add } (+2a - 2b + 4b - 4b)$$

$$= (a - 2b)^2 - 4(a - 2b) + 2(a - 2b) - 8$$

$$= (a - 2b) [(a - 2b - 4)] + 2[(a - 2b - 4)]$$

$$= (a - 2b - 4)(a - 2b + 2)$$

43. (d) Put, $a = 1, x = 1$

$$\therefore \frac{1}{1+1+1} - \frac{1}{1-1+1} + \frac{2}{1+1+1}$$

$$= \frac{1}{3} - 1 + \frac{2}{3} = 0$$

Alternatively:-

$$\frac{1}{a^2 + ab + x^2} - \frac{1}{a^2 - ax + x^2}$$

$$+ \frac{2ax}{a^4 + a^2x^2 + x^4}$$

$$= \frac{a^2 - ax + x^2 - a^2 - ax - x^2}{(a^2 + x^2)^2 - (ax)^2}$$

$$+ \frac{2ax}{a^4 + a^2x^2 + x^4}$$

$$= \frac{-2ax}{a^4 + x^4 + a^2x^2} + \frac{2ax}{a^4 + x^4 + a^2x^2} = 0$$

44. (b) $3x : 2y = 21 : 22$

$$\frac{3x}{2y} = \frac{21}{22} \Rightarrow \therefore \frac{x}{y} = \frac{7}{11}$$

$$4x + 5y = 83$$

When we put, $x = 7, y = 11$ it satisfies the equation

$$\therefore y - x = 11 - 7 = 4$$

45. (b) $x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1$

$$\Rightarrow x^5 - 11x^4 - x^4 + 11x^3 + x^3 - 11x^2 - x^2 + 11x + x - 1$$

$$\text{Put, } 11 = x$$

$$\Rightarrow x^5 - x \cdot x^4 + -x^4 + x^4 + x^3 - x^3 - x^2 + x^2 + x - 1$$

$$\Rightarrow x - 1 = 11 - 1 = 10$$

or, you can put, $x = 11$

$$= (11)^5 - 11 \cdot (11)^4 - (11)^4 + 11 \cdot (11)^3 + (11)^3 - 11 \cdot (11)^2 - (11)^2 + 11 \times 11 + 11 - 1$$

$$= (11)^5 - (11)^5 - (11)^4 + (11)^4 + (11)^3 - (11)^3 - (11)^2 + (11)^2 + 11 - 1$$

$$= 10$$

46. (c) $p(p^2 + 3p + 3)$

$$= p^3 + 3p^2 + 3p + 1 - 1 \text{ (add+1-1)}$$

$$= (p + 1)^3 - 1$$

$$= (99 + 1)^3 - 1$$

$$= (100)^3 - 1 = 1000000 - 1$$

$$= 999999$$

SSC CGL MAINS 2014

{12 Apr. 2015} Paper-I

47. (d) $\frac{3-5x}{2x} + \frac{3-5y}{2y} + \frac{3-5z}{2z} = 0$

equate all 3 parts separately with zero

$$\frac{3-5x}{2x} = 0$$

$$x = \frac{3}{5}$$

Similarly,

$$y = \frac{3}{5} \text{ and } z = \frac{3}{5}$$

$$= \frac{2}{x} + \frac{2}{y} + \frac{2}{z}$$

$$= \frac{2}{3} \times 5 + \frac{2}{3} \times 5 + \frac{2}{3} \times 5$$

$$= \frac{30}{3} = 10$$

Alternatively:-

$$\frac{yz(3-5x) + xz(3-5y) + xy(3-5z)}{2xyz}$$

$$\frac{3(xy + yz + zx) - 15xyz}{2xyz}$$

$$\frac{3 \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)}{2} = \frac{15}{2}$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{15}{2} \times \frac{2}{3} = 5$$

$$\therefore \frac{2}{x} + \frac{2}{y} + \frac{2}{z} = 5 \times 2 = 10$$

48. (c) Let, $a = 3, b = 4, c = 5$

$$2S = 3 + 4 + 5$$

$$S = 6$$

$$= s(s-c) + (s-a)(s-b)$$

$$= 6(6-5) + (6-3)(6-4)$$

$$= 6 + 3 \times 2 = 12$$

From option

$$ab = 3 \times 4 = 12$$

49. (b) $p^3 + m^3 = [p + m][p^2 + m^2 - pm]$
... (i)

$$P + m = 6$$

squaring both side

$$p^2 + m^2 + 2pm = 36$$

$$p^2 + m^2 = 36 - 2pm$$

From (i)

$$72 = 6[36 - 2pm - pm]$$

$$12 = 36 - 3pm$$

$$pm = 8$$

50. (b) $x^m \times x^n = 1$

this is possible only

when $m = -n$

$$x^{-n} \times x^n = 1$$

$$\frac{1}{x^n} \times x^n = 1$$

$$51. (b) \frac{p}{p^2 - 2p + 1} = \frac{1}{4}$$

$$8p = p^2 - 2p + 1$$

$$8 = p - 2 + \frac{1}{p}$$

$$p + \frac{1}{p} = 10$$

$$52. (c) \frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} +$$

$$\frac{1}{\sqrt{4}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{6}} + \frac{1}{\sqrt{6}+\sqrt{7}} +$$

$$\frac{1}{\sqrt{7}+\sqrt{8}} + \frac{1}{\sqrt{8}+\sqrt{9}}$$

Rationalize each part separately

$$= -1 + \sqrt{2} - \sqrt{2} + \sqrt{3} - \sqrt{3} + \sqrt{4} -$$

$$\sqrt{4} + \sqrt{5} - \sqrt{5} + \sqrt{6} - \sqrt{6} + \sqrt{7} -$$

$$\sqrt{7} + \sqrt{8} - \sqrt{8} + \sqrt{9}$$

$$= \sqrt{9} - 1$$

$$= 3 - 1 = 2$$

SSC CGL MAINS 2014**{12 Apr. 2015} Paper-II**

$$53. (a) \text{ Put, } a = 4$$

$$2x = 2 + \frac{1}{2} = \frac{5}{2}$$

$$x = \frac{5}{4}$$

$$\Rightarrow \frac{\sqrt{x^2-1}}{x-\sqrt{x^2-1}} = \frac{\sqrt{\frac{25}{16}-1}}{\frac{5}{4}-\sqrt{\frac{25}{16}-1}}$$

$$\frac{\frac{3}{4}}{\frac{5}{4}-\frac{3}{4}} = \frac{3}{2}$$

option (a) gives $\frac{3}{2}$ after putting $a = 4$

Alternatively:-

$$2x = \sqrt{a} + \frac{1}{\sqrt{a}} = \frac{a+1}{\sqrt{a}}$$

$$\Rightarrow x = \frac{a+1}{2\sqrt{a}}$$

square both side

$$x^2 = \frac{a^2+1+2a}{4a}$$

add (-1) both side

$$x^2 - 1 = \frac{a^2+1+2a}{4a} - 1$$

$$x^2 - 1 = \frac{a^2+1+2a}{4a}$$

$$x^2 - 1 = \frac{(a-1)^2}{4a}$$

make root both side

$$\sqrt{x^2-1} = \sqrt{\frac{(a-1)^2}{4a}} = \frac{a-1}{2\sqrt{a}}$$

$$\therefore \frac{\sqrt{x^2-1}}{x-\sqrt{x^2-1}} = \frac{\frac{a-1}{2\sqrt{a}}}{\frac{a+1}{2\sqrt{a}} - \frac{a-1}{2\sqrt{a}}}$$

$$= \frac{a-1}{2\sqrt{a}} \times \frac{2\sqrt{a}}{2} = \frac{a-1}{2}$$

$$54. (c) a^2 + b^2 + c^2 = 2(a-b-c) - 3$$

$$a^2 + b^2 + c^2 - 2a + 2b + 2c + 3 = 0$$

$$(a-1)^2 + (b+1)^2 + (c+1)^2 = 0$$

$$a = 1, b = -1, c = -1$$

$$\therefore a + b + c = 1 - 1 - 1 = -1$$

$$55. (d) \text{ Put, } a = b = c = 1$$

$$\frac{a+b-c}{a+b} = \frac{b+c-a}{b+c} = \frac{c+a-b}{c+a}$$

$$\frac{1}{2} = \frac{1}{2} = \frac{1}{2}$$

$$\text{So, } a = b = c$$

$$56. (b) a = 2 + \sqrt{3}$$

$$\frac{1}{a} = 2 - \sqrt{3}$$

$$a + \frac{1}{a} = 4$$

$$a^3 + \frac{1}{a^3} = (4)^3 - 3(4) = 52$$

$$\frac{a^6 + a^4 + a^2 + 1}{a^3} = a^3 + a + \frac{1}{a} + \frac{1}{a^3}$$

$$= 52 + 4 = 56$$

$$57. (c) x = \sqrt{5} + \sqrt{3}$$

square both side

$$x^2 = 8 + 2\sqrt{15}$$

again square both side

$$x^4 = 124 + 32\sqrt{15}$$

$$y = \sqrt{5} - \sqrt{3}$$

square both side

$$y^2 = 8 - 2\sqrt{15}$$

again square both side

$$y^4 = 124 - 32\sqrt{15}$$

$$\therefore x^4 - y^4 = 124 + 32\sqrt{15} - 124$$

$$+ 32\sqrt{15} = 64\sqrt{15}$$

$$58. (c) \text{ Put, } y \text{ \& } z = 0$$

$$\therefore x = 6$$

$$\therefore (6-1)^3 + (-2)^3 + (-3)^3 = 90$$

Put, $x = 6$, $y \text{ \& } z = 0$ in options
option (c) gives 90

Alternatively:-

$$x + y + z = 6$$

$$x = y = z = 2$$

$$(x-1)^3 + (y-2)^3 + (z-3)^3$$

↓

a

↓

b

↓

c = 0

We know that

$$a + b + c = 0$$

$$\begin{cases} a^3 + b^3 + c^3 - 3abc = 0 \\ a^3 + b^3 + c^3 = 3abc \end{cases}$$

$$\therefore (x-1)^3 + (y-2)^3 + (z-3)^3$$

$$= 3(x-1)(y-2)(z-3)$$

$$59. (b) \text{ Put, } a = b = c = 3$$

$$\therefore \frac{b+c}{bc(a-1)} + \frac{a+c}{ac(b-1)} + \frac{a+b}{ab(c-1)}$$

$$= \frac{6}{18} + \frac{6}{18} + \frac{6}{18} = 1$$

Alternatively:-

$$bc + ab + ca = abc$$

$$ab + ca + = abc - bc$$

$$a(b+c) = bc(a-1)$$

Similarly,

$$b(c+a) = ac(b-1)$$

$$c(a+b) = ab(c-1)$$

$$\therefore \frac{b+c}{bc(a-1)} + \frac{a+c}{ac(b-1)} + \frac{a+b}{ab(c-1)}$$

$$\frac{b+c}{a(b+c)} + \frac{a+c}{b(a-c)} + \frac{a+b}{c(a+b)}$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

$$= \frac{bc+ac+ab}{abc} = \frac{abc}{abc} = 1$$

$$60. (b) \frac{a^2-bc}{a^2+bc} + \frac{b^2-ca}{b^2+ca} + \frac{c^2-ab}{c^2+ab} = 1$$

Adding 3 both side

$$\frac{a^2-bc}{a^2+bc} + 1 + \frac{b^2-ca}{b^2+ca} + 1 + \frac{c^2-ab}{c^2+ab} = 4$$

$$\frac{2a^2}{a^2+bc} + \frac{2b^2}{b^2+ca} + \frac{2c^2}{c^2+ab} = 4$$

$$2 \left(\frac{a^2}{a^2+bc} + \frac{b^2}{b^2+ca} + \frac{c^2}{c^2+ab} \right) = 4$$

$$\frac{a^2}{a^2+bc} + \frac{b^2}{b^2+ca} + \frac{c^2}{c^2+ab} = 2$$

61. (a) $p^4 + \frac{1}{p^4} = 119$

make perfect square by adding 2

$$\left(p^2 + \frac{1}{p^2}\right)^2 = 119 + 2$$

$$p^2 + \frac{1}{p^2} = 11$$

again make perfect square by adding 2

$$\left(p - \frac{1}{p}\right)^2 = 11 - 2$$

$$p - \frac{1}{p} = 3$$

cube both side

$$p^3 - \frac{1}{p^3} = \left(p - \frac{1}{p}\right)^3 - 3 \times p \times \frac{1}{p} \left(p - \frac{1}{p}\right)$$

$$p^3 - \frac{1}{p^3} = 27 + 3(3) = 36$$

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{25 Aug. 2015}

62. (d) $a - \frac{1}{a-3} = 5$

add (-3) both side

$$(a-3) - \frac{1}{(a-3)} = 2$$

cube both side

$$\therefore (a-3)^3 - \frac{1}{(a-3)^3} = 2^3 + 3(2) = 14$$

63. (b) $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$

$$a \neq b \neq c \neq 0$$

$$\text{Put, } a = b = c = 1$$

$$1 + \frac{1}{1} = 1 + \frac{1}{1} = 1 + \frac{1}{1}$$

$$2 = 2 = 2$$

$$\text{So, } a^2 b^2 c^2 = 1 \times 1 \times 1 = 1$$

64. (b) $\left(\frac{q^4}{p^4}\right)^{1/3} + \left(\frac{p^8}{q^6}\right)^{1/3} = p^a q^b$

$$\left(\frac{q^4}{p^4} \times \frac{q^6}{p^8}\right)^{1/3} = p^a q^b$$

$$(p^{-12} q^{10})^{1/3} = p^a q^b$$

After comparing

$$a = -4, b = \frac{10}{3}$$

$$a + b = -4 + \frac{10}{3} = \frac{-2}{3}$$

65. (d) Degree of the equation is 4 in numerator and denominator

$$\therefore \frac{3+7+5}{5+7+3} = \frac{15}{15} = 1$$

Alternatively:-

$$x^2 + y^2 + z^2 = xy + yz + zx$$

$$\frac{3x^4 + 7y^4 + 5z^4}{5x^2y^2 + 7y^2z^2 + 3z^2x^2}$$

$$\text{Put, } x = y = z = 1$$

$$= \frac{3 \times 1 + 7 \times 1 + 5 \times 1}{5 \times 1 \times 1 + 7 \times 1 \times 1 + 3 \times 1 \times 1} = 1$$

66. (c) $3(a^2 + b^2 + c^2) = (a + b + c)^2$

$$3(a^2 + b^2 + c^2) = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$2(a^2 + b^2 + c^2) = 2(ab + bc + ca)$$

$$a^2 + b^2 + c^2 = ab + bc + ca$$

$$\therefore a = b = c$$

67. (b) $\frac{3x-2y}{2x+3y} = \frac{5}{6}$

$$18x - 12y = 10x + 15y$$

$$8x = 27y$$

$$\frac{x}{y} = \frac{27}{8}$$

$$\therefore \left(\frac{\sqrt[3]{x} + \sqrt[3]{y}}{\sqrt[3]{x} - \sqrt[3]{y}}\right)^2 = \left(\frac{\sqrt[3]{27} + \sqrt[3]{8}}{\sqrt[3]{27} - \sqrt[3]{8}}\right)^2$$

$$\left(\frac{3+2}{3-2}\right)^2 = 25$$

68. (b) $x = \sqrt{3} + \sqrt{2}$

$$y = \sqrt{3} - \sqrt{2}$$

$$x = \frac{1}{y}$$

$$\therefore (x^3 - 20\sqrt{2}) - (y^3 + 2\sqrt{2})$$

$$\Rightarrow \left(x^3 - 20\sqrt{2} - \frac{1}{x^3} - 2\sqrt{2}\right)$$

$$\Rightarrow \left(x^3 - \frac{1}{x^3} - 22\sqrt{2}\right)$$

$$x - \frac{1}{x} = \sqrt{3} + \sqrt{2} - \sqrt{3} + \sqrt{2} = 2\sqrt{2}$$

cube both side

$$x^3 - \frac{1}{x^3} = \left(x - \frac{1}{x}\right)^3 + 3\left(x - \frac{1}{x}\right)$$

$$(2\sqrt{2})^3 + 3 \times 2\sqrt{2}$$

$$16\sqrt{2} + 6\sqrt{2} = 22\sqrt{2}$$

$$x^3 - \frac{1}{x^3} - 22\sqrt{2} = 22\sqrt{2} - 22\sqrt{2} = 0$$

69. (a) $x = \frac{\sqrt{13} + \sqrt{11}}{\sqrt{13} - \sqrt{11}} \times \frac{\sqrt{13} + \sqrt{11}}{\sqrt{13} + \sqrt{11}}$

$$x = \frac{24 + 2\sqrt{143}}{2}$$

$$x = 12 + \sqrt{143}$$

$$\frac{1}{x} = 12 - \sqrt{143}$$

$$x + \frac{1}{x} = 24$$

$$x^2 + \frac{1}{x^2} = 576 - 2 = 574$$

$$\Rightarrow 3x^2 - 5xy + 3y^2$$

$$= 3x^2 - 5xy + \frac{3}{x^2}$$

$$= 3\left(x^2 + \frac{1}{x^2}\right) - 5$$

$$= 3(574) - 5 = 1717$$

Alternatively:-

$$\Rightarrow 3x^2 - 5xy + 3y^2$$

$$3(12 + \sqrt{143})^2 - 5x \times \frac{1}{x} + 3(12 - \sqrt{143})^2$$

$$= 3(144 + 143 + 24\sqrt{143}) - 5 + 3$$

$$(144 + 143 - 24\sqrt{143})$$

$$861 - 5 + 61 = 1717$$

70. (b)

$$\frac{(0.67 \times 0.67 \times 0.67) - (0.33 \times 0.33 \times 0.33)}{(0.67 \times 0.67) + (0.67 \times 0.33) + (0.33 \times 0.33)}$$

$$\therefore \frac{a^3 - b^3}{(a^2 + b^2 + ab)} = a - b$$

multiply and divide by (a-b)

$$= (0.67 - 0.33) = 0.34$$

71. (b) $\frac{x^3 - y^3}{x^2 + xy + y^2} = \frac{5}{1}$

multiply and divide by (x-y)

$$x - y = 5 \quad \dots(i)$$

$$\frac{(x^2 - y^2)}{x - y} = \frac{7}{1}$$

$$x + y = 7 \quad \dots(ii)$$

From eq. (i) and (ii)

$$x = 6, y = 1$$

$$\therefore 2x : 3y = (2 \times 6) : (3 \times 1) = 12 : 3 = 4 : 1$$

72. (c) $x = (a^{1/2} + a^{-1/2})$

$$y = (a^{1/2} - a^{-1/2})$$

$$x + y = a^{1/2} + a^{-1/2} + a^{1/2} - a^{-1/2} = 2a^{1/2}$$

$$x - y = a^{1/2} + a^{-1/2} - a^{1/2} + a^{-1/2} = 2a^{-1/2}$$

$$\begin{aligned} (x^4 - x^2y^2 - 1) + (y^4 - x^2y^2 + 1) \\ = x^4 - 2x^2y^2 + y^4 = (x^2 - y^2)^2 \\ = \{(x + y)(x - y)\}^2 \\ = \{2a^{1/2} \times 2a^{-1/2}\}^2 = (4)^2 = 16 \end{aligned}$$

73. (c) Put, $b = 0$

$$\therefore a = 1$$

$$\begin{aligned} a^3 + b^3 - ab - (a^2 - b^2)^2 \\ = 1 + 0 - 0 - 1 = 0 \end{aligned}$$

Alternatively:-

$$a + b = 1$$

$$a^3 + b^3 - ab - (a^2 - b^2)^2$$

$$= (a + b)^3 - 3ab(a + b) - ab$$

$$- \{(a + b)(a - b)\}^2$$

$$= (1)^3 - 3ab \times (1) - ab - \{1 \times (a - b)\}^2$$

$$= 1 - 4ab - (a^2 + b^2 - 2ab)$$

$$= 1 - 4ab - a^2 - b^2 + 2ab$$

$$= 1 - (a + b)^2 = 1 - 1 = 0$$

SSC CGL MAINS 2016

{30 Nov. 2016}

74. (b) $P(P^2 + 3P + 3)$

$$= P^3 + 3P^2 + 3P + 1 - 1(\text{add}+1-1)$$

$$= (P + 1)^3 - 1$$

$$= (99 + 1)^3 - 1$$

$$= 1000000 - 1 = 999999$$

75. (a) $x + \frac{1}{x} = c + \frac{1}{c}$

$$x - c = \frac{1}{c} - \frac{1}{x}$$

$$x - c = \frac{x - c}{xc}$$

$$(x - c) - \left(\frac{x - c}{xc}\right) = 0$$

take $(x - c)$ common

$$(x - c)\left(1 - \frac{1}{xc}\right) = 0$$

$$x - c = 0 \Rightarrow x = c$$

$$1 - \frac{1}{xc} = 0 \Rightarrow \frac{1}{xc} = 1$$

$$\Rightarrow xc = 1 \Rightarrow x = \frac{1}{c}$$

76. (c) Let, two real number = x and y
ATQ

$$x^2 + y^2 = 41 \text{ and } x + y = 9$$

$$\therefore (x + y)^2 = x^2 + y^2 + 2xy$$

$$\Rightarrow 81 = 41 + 2xy$$

$$\Rightarrow xy = 20$$

$$\therefore x^3 + y^3 = (x + y)^3 - 3xy(x + y)$$

$$= (9)^3 - 3 \times 20(9)$$

$$= 729 - 540 = 189$$

77. (d) $x^4 + 64$

$$\text{Put, } x = 1$$

$$1 + 64 = 65$$

Now put in option (d)

$$(x^4 + 4x + 8)(x^2 - 4x + 8)$$

$$13 \times 5 = 65$$

Alternatively:-

Solving from options

$$(x^4 + 4x + 8)(x^2 - 4x + 8)$$

$$= (x^4 - 4x^3 + 8x^2 + 4x^3 - 16x^2 + 32x + 8x^2 - 32x + 64)$$

$$= (x^4 + 64)$$

Required factors

$$= (x^2 + 4x + 8)(x^2 - 4x + 8)$$

78. (d) $a^4 + b^4 - a^3 - b^3 - 2a^2b^2 + ab$

$$= (a^2 - b^2)^2 - (a^3 + b^3 - ab)$$

use formula of $(a^3 + b^3)$

$$= \{(a + b)(a - b)\}^2 - \{(a + b)(a^2 + b^2 - ab) - ab\}$$

$$= (a - b)^2 - (a^2 + b^2 - 2ab)$$

$$[\because a + b = 1]$$

$$= (a - b)^2 - (a - b)^2$$

$$= 0$$

79. (a) $x^2 + y^2 + 6x + 5 = 4(x - y)$

$$x^2 + y^2 + 6x + 5 = 4x - 4y$$

$$x^2 + y^2 + 2x + 4y + 1 + 4 = 0$$

$$(x + 1)^2 + (y + 2)^2 = 0$$

$$x = -1$$

$$y = -2$$

$$\therefore x - y = -1 - (-2) = 1$$

80. (c) $2a^3 + 2b^3 + 2c^3 - 6abc$

$$= 2(a^3 + b^3 + c^3 - 3abc)$$

$$= 2\left(\frac{1}{2}(a+b+c)\{(a-b)^2 + (b-c)^2 + (c-a)^2\}\right)$$

$$= (299 + 298 + 297) [(1)^2 + (1)^2 + (-2)^2]$$

$$= 894 \times 6 = 5364$$

81. (a) $x + \frac{1}{x} = \sqrt{3}$

$$x^3 + \frac{1}{x^3} = 3\sqrt{3} - 3\sqrt{3}$$

$$x^6 + 1 = 0$$

$$\therefore x^{18} + x^{12} + x^6 + 1$$

$$= x^{12}(x^6 + 1) + x^6 + 1$$

$$= 0$$

82. (d) $x = 1 + \sqrt{2} + \sqrt{3}$

$$x - 1 = \sqrt{3} + \sqrt{2}$$

Squaring both sides

$$x^2 - 2x + 1 = 3 + 2 + 2\sqrt{6}$$

$$x^2 - 2x + 1 - 5 = 2\sqrt{6}$$

$$x^2 - 2x - 4 = 2\sqrt{6}$$

Again squaring both side

$$(x^2 - 2x - 4)^2 = (2\sqrt{6})^2$$

$$\Rightarrow x^4 + 4x^2 + 16 - 4x^3 + 16x - 8x^2 = 24$$

$$\Rightarrow x^4 - 4x^3 - 4x^2 + 16x - 8 = 0$$

$$\Rightarrow 2x^4 + 8x^3 - 8x^2 + 32x - 16 = 0$$

$$\therefore 2x^4 - 8x^3 - 5x^2 + 26x - 28$$

$$= 2x^4 - 8x^3 - 8x^2 + 32x - 116$$

$$+ 3x^2 - 6x - 12$$

$$= 0 + 3(x^2 - 2x - 4) = 3 \times 2\sqrt{6}$$

$$= 6\sqrt{6}$$

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83. (b) Put, $x = 2, y = 1$

$$\left(1 - \frac{4}{5}\right) \div (7 - 6) = \frac{1}{5}$$

only option (b) satisfies

Alternatively:-

$$\left(1 - \frac{2xy}{x^2 + y^2}\right) \div \left(\frac{x^3 - y^3}{x - y} - 3xy\right)$$

$$\frac{x^2 + y^2 - 2xy}{x^2 + y^2} \div \frac{x^3 - y^3 - 3xy(x - y)}{x - y}$$

$$= \frac{(x - y)^2}{x^2 + y^2} \div \frac{(x - y)^3}{(x - y)} = \frac{1}{x^2 + y^2}$$

84. (a) Let, $a = 1, b = 1, c = -2$

$$a + b + c = 0$$

$$= \frac{1}{(a + b)(b + c)} + \frac{1}{(b + c)(c + a)} +$$

$$\frac{1}{(c + a)(a + b)}$$

$$= \frac{1}{2 \times (-1)} + \frac{1}{(-1)(-1)} + \frac{1}{(-1)(2)}$$

$$= -\frac{1}{2} + 1 - \frac{1}{2} = 0$$

85. (a) $x^2 + y^2 + 2x + 1 = 0$

$$(x+1)^2 + y^2 = 0$$

$$x = -1, y = 0$$

$$\therefore x^{31} + y^{35} = (-1)^{31} + (0)^{35} = -1$$

86. (b) Make perfect square for easy calculation by adding $(+xy - xy)$

$$\frac{x^2 + xy + y^2}{x^2 - xy + y^2} = \frac{(x+y)^2 - xy}{(x-y)^2 + xy}$$

$$x + y = \frac{\sqrt{5} + 1}{\sqrt{5} - 1} + \frac{\sqrt{5} - 1}{\sqrt{5} + 1}$$

$$x + y = \frac{5 + 1 + 2\sqrt{5} + 5 + 1 - 2\sqrt{5}}{5 - 1} = 3$$

$$x - y = \frac{4\sqrt{5}}{4} = \sqrt{5}$$

$$xy = 1$$

$$\therefore \frac{(x+y)^2 - xy}{(x-y)^2 + xy} = \frac{9 - 1}{5 + 1} = \frac{8}{6} = \frac{4}{3}$$

87. (c) $\left(x - \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2$

$$x^2 + \frac{1}{x^2} - 2 = 3$$

$$x^2 + \frac{1}{x^2} = 3 + 2$$

$$x^2 + \frac{1}{x^2} = 5$$

Cubing both sides

$$x^6 + \frac{1}{x^6} + 3 \times 5 = 125$$

$$x^6 + \frac{1}{x^6} = 125 - 15$$

$$x^6 + \frac{1}{x^6} = 110$$

88. (c) Put, $x = -1$

$$1 - 2 + a - b + 9$$

$$(a - b) + 9$$

now, go through option (c)

$$(7 - 6) + 8 = 9$$

9 is a perfect square no.

Alternatively:-

$x^4 + 2x^3 + ax^2 + bx + 9$ is a perfect square.

Solving from option

Taking, $a = 7, b = 6$

$$x^4 + 2x^3 + 7x^2 + 6x + 9$$

$$= (x^2 + 2x + 3)^2$$

= A perfect square

89. (c) Let, $a = 4, b = 0, c = 0$ and

$$\text{Let, } x = 5, y = 0, z = 0$$

Now

$$\frac{a+b+c}{x+y+z} = \frac{4+0+0}{5+0+0} = \frac{4}{5}$$

90. (b) Put, $a = b = c = 1$

$$\frac{x+a^2+2c^2}{b+c} + \frac{x+b^2+2a^2}{c+a} + \frac{x+c^2+2b^2}{a+b} = 0$$

$$\frac{x+1+2}{2} + \frac{x+1+2}{2} + \frac{x+1+2}{2} = 0$$

$$3x + 9 = 0$$

$$3x = -9$$

$$x = -3$$

now, Put, $a = b = c = 1$ in option (b)

$$-(a^2 + b^2 + c^2) = -3$$

Alternatively:-

From option (B)

$$\Rightarrow x = -(a^2 + b^2 + c^2)$$

$$\frac{x+a^2+2c^2}{b+c} + \frac{x+b^2+2a^2}{c+a}$$

$$= \frac{-(a^2 + b^2 + c^2) + a^2 + 2c^2}{b+c}$$

$$= \frac{-a^2 - b^2 - c^2 + a^2 + 2c^2}{b+c}$$

$$= \frac{-b^2 + c^2}{b+c}$$

$$= \frac{c^2 - b^2}{b+c} = \frac{(c+b)(c-b)}{(c+b)} = (c-b)$$

...(i)

$$\text{Again, } \frac{x+b^2+2a^2}{c+a}$$

$$= \frac{-(a^2 + b^2 + c^2) + b^2 + 2a^2}{c+a}$$

$$= \frac{-a^2 - b^2 - c^2 + b^2 + 2a^2}{c+a}$$

$$= \frac{a^2 - c^2}{a+c} = \frac{(a+c)(a-c)}{(a+c)} = a-c$$

...(ii)

Again,

$$\frac{x+c^2+2b^2}{a+b}$$

$$= \frac{-(a^2 + b^2 + c^2) + c^2 + 2b^2}{a+b}$$

$$= \frac{-a^2 - b^2 - c^2 + c^2 + 2b^2}{a+b}$$

$$= \frac{(b+a)(b-a)}{(b+a)} = (b-a) \quad \text{...(iii)}$$

Adding equation (i), (ii) and (iii)

$$= c - b + a - c + b - a = 0$$

91. (a) $a^3 = 117 + b^3$

$$a^3 - b^3 = 117 \quad \text{...(i)}$$

$$a - b = 3 \quad \text{...(ii)}$$

From equation (i) & (ii)

$$(a-b)(a^2 + b^2 + ab) = 117$$

$$\text{Put, } a - b = 3$$

$$a^2 + b^2 + ab = 39$$

make perfect square of $(a-b)$ by adding $(+2ab-2ab)$

$$(a-b)^2 + 3ab = 39$$

$$ab = 10$$

$$(a+b)^2 = (a-b)^2 + 4ab$$

$$(a+b)^2 = 9 + 40 = 49$$

$$a + b = \pm 7$$

92. (a) Let, $a = -1$

$$\therefore a^{1000} + a^{-1000}$$

$$= a^{1000} + \frac{1}{a^{1000}} = 1 + 1 = 2$$

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{12 Jan. 2017}

93. (a) $x^2 + \frac{1}{x^2} = 98$

add $(2 \times x \times \frac{1}{x})$ both side for making perfect square

$$\left(x + \frac{1}{x}\right)^2 = 98 + 2$$

$$x + \frac{1}{x} = 10$$

cube both side

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$$

$$= 1000 - 3 \times 10 = 970$$

94. (c) $a + \frac{1}{b} = 1, b + \frac{1}{c} = 1$

$$\text{Put, } a = \frac{1}{2}, b = 2, c = -1$$

these values satisfy each equation

$$\therefore c + \frac{1}{a} = -1 + 2 = 1$$

95. (b) $x = y + z$

$$\text{Let, } y = z = 2$$

$$x = 4$$

$$x^3 - y^3 - z^3 = 4^3 - 2^3 - 2^3 = 48$$

Put the value of x, y, z in options

$$3xyz = 3 \times 4 \times 2 \times 2 = 48 \text{ satisfied.}$$

96. (a) $a + b + c + d = 4$

$$\begin{aligned} & \frac{1}{(1-a)(1-b)(1-c)} \\ & + \frac{1}{(1-b)(1-c)(1-d)} \\ & + \frac{1}{(1-c)(1-d)(1-a)} \\ & + \frac{1}{(1-d)(1-a)(1-b)} \\ & = \frac{1-d+1-a+1-b+1-c}{(1-a)(1-b)(1-c)(1-d)} \\ & = \frac{4-(a+b+c+d)}{(1-a)(1-b)(1-c)(1-d)} \\ & = \frac{4-4}{(1-a)(1-b)(1-c)(1-d)} = 0 \end{aligned}$$

97. (b) $x = 11$

$$\begin{aligned} & x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1 \\ & = x^5 - 11x^4 - x^4 + 11x^3 + x^3 - 11x^2 \\ & \quad - x^2 + 11x + x - 1 \\ & = x^5 - x^5 - x^4 + x^4 + x^3 - x^3 - x^2 \\ & \quad + x^2 + x - 1 \\ & = x - 1 = 11 - 1 = 10 \end{aligned}$$

98. (b) $a = \frac{1}{a-5}$

$$a^2 - 5a = 1$$

divide by a

$$a - 5 = \frac{1}{a} \Rightarrow a - \frac{1}{a} = 5$$

square both side

$$a^2 + \frac{1}{a^2} = 27$$

Add $\left(2 \times a \times \frac{1}{a}\right)$ both side and make perfect square

$$a + \frac{1}{a} = \sqrt{29}$$

99. (c) $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$

$$\Rightarrow a - b = \frac{1}{c} - \frac{1}{b} = \frac{b-c}{bc} \quad \dots(i)$$

$$\Rightarrow b - c = \frac{1}{a} - \frac{1}{c} = \frac{c-a}{ac} \quad \dots(ii)$$

$$\Rightarrow c - a = \frac{1}{b} - \frac{1}{a} = \frac{a-b}{ab} \quad \dots(iii)$$

Equation (i) $\times$ (ii) $\times$ (iii)

$$\begin{aligned} & (a-b)(b-c)(c-a) \\ & = \frac{(b-c)(c-a)(a-b)}{a^2b^2c^2} \end{aligned}$$

$$a^2b^2c^2 = 1$$

$$abc = \pm 1$$

100. (a) Let, $a = b = 2$

$$x = y = \frac{1}{4}$$

$$ax + by = 1$$

$$2 \times \frac{1}{4} + 2 \times \frac{1}{4} = 1 \text{ (satisfied)}$$

$$bx + ay = \frac{2ab}{a^2 + b^2}$$

$$2 \times \frac{1}{4} + 2 \times \frac{1}{4} = \frac{2 \times 2 \times 2}{4 + 4} \text{ (satisfied)}$$

$$(x^2 + y^2)(a^2 + b^2)$$

$$\left(\frac{1}{16} + \frac{1}{16}\right)(4+4)$$

$$\frac{1}{8} \times 8 = 1$$

you can't put $a=b=1$, $x = y = \frac{1}{2}$

101. (a) $a^3 - 7a - 6$

$$a^3 - a - 6a - 6$$

$$a(a^2 - 1) - 6(a + 1)$$

$$(a + 1)[a(a - 1) - 6]$$

$$(a + 1)[a^2 - a - 6]$$

$$(a + 1)[a^2 - 3a + 2a - 6]$$

$$(a + 1)[a(a - 3) + 2(a - 3)]$$

$$\Rightarrow x = a + 1$$

$$y = a + 2$$

$$z = a - 3$$

$$x + y + z$$

$$= a + 1 + a + 2 + a - 3$$

$$= 3a$$

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{17 Feb. 2018}

102. (a) $x + y + z = 0$,

$$\frac{(3y^2 + x^2 + z^2)}{2y^2 - zx} = ?$$

Put $x = 1, y = 1, z = -2$

$$\Rightarrow \frac{3(y)^2 + x^2 + z^2}{2y^2 - xz}$$

$$\Rightarrow \frac{3(1)^2 + (1)^2 + (-2)^2}{2(1)^2 - (1 \times (-2))}$$

$$\Rightarrow \frac{4+4+8}{2+2} = \frac{16}{4} = 4$$

You can also put,

$$z = 0, x = 1, y = -1$$

103. (b) $P = 7 + 4\sqrt{3}$

$$PQ = 1, Q = \frac{1}{P}$$

$$Q = \frac{1}{7 + 4\sqrt{3}} = 7 - 4\sqrt{3}$$

$$(\therefore (7)^2 - (4\sqrt{3})^2 = 1)$$

$$P + Q = 7 + 4\sqrt{3} + 7 - 4\sqrt{3} = 14$$

$$\Rightarrow \frac{1}{P^2} + \frac{1}{Q^2} = \frac{P^2 + Q^2}{P^2Q^2}$$

For easy calculation make perfect square of $(P+Q)$.

$$= \frac{(P+Q)^2 - 2PQ}{P^2Q^2}$$

$$= (14)^2 - 2$$

$$= 196 - 2 = 194$$

104. (c) $a^3 + 3a^2 + 9a = 1$

$$a(a^2 + 3a + 9) = 1$$

$$a^2 + 3a + 9 = \frac{1}{a}$$

$$(a-b)(a^2 + ab + b^2) = a^3 - b^3$$

On comparing:-

$$(b = 3)$$

$$(a-3)(a^2 + 3a + 9) = a^3 - 27$$

$$(a-3) \frac{1}{a} = a^3 - 27$$

$$a^3 + \frac{3}{a} = 27 + 1$$

$$a^3 + \frac{3}{a} = 28$$

Alternatively:-

$$a^3 + 3a^2 + 9a = 1$$

Multiply by 'a' & 3 in eq separately-

$$(a^3 + 3a^2 + 9a = 1) \times a$$

$$(a^3 + 3a^2 + 9a = 1) \times 3$$

$$a^4 + 3a^3 + 9a^2 = a$$

$$\frac{3a^3 + 9a^2 + 27a = 3}{a^4 - 27a - a - 3}$$

$$a^4 + 3 = 28a$$

$$a^4 + 3 = 28a$$

Divide by (a) -

$$a^3 + \frac{3}{a} = 28$$

105. (d) $x^3 + y^3 + z^3 = 13$

$$x + y + z = 1$$

$$xyz = 1$$

$$\Rightarrow x^3 + y^3 + z^3 - 3xyz = (x + y + z)$$

$$[(x + y + z)^3 - 3(xy + yz + zx)]$$

$$\Rightarrow 13 - 3(1) = [1 - 3(xy + yz + zx)]$$

$$\Rightarrow 10 - 1 = -3(xy + yz + zx)$$

$$\Rightarrow xy + yz + zx = \frac{9}{-3}$$

$$\Rightarrow xy + yz + zx = -3$$

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{18 Feb. 2018}

$$106.(d) \frac{(a+b)}{c} = \frac{6}{5}, \quad c = 5$$

$$\frac{(b+c)}{a} = \frac{9}{2}, \quad a = 2$$

$$\Rightarrow \frac{(b+5)}{2} = \frac{9}{2}, \quad b = 4$$

$$= \frac{(a+c)}{b} = \frac{2+5}{4} = \frac{7}{4}$$

Alternatively:-

$$\frac{(a+b)}{c} = \frac{6}{5} \Rightarrow a+b+c = 11 \text{ part}$$

$$\frac{(b+c)}{a} = \frac{9}{2} \Rightarrow a+b+c = 11 \text{ part}$$

By option-

$$a+b+c = 7+4 = 11 \text{ part ... (must)}$$

$$\therefore \frac{a+c}{b} = \frac{7}{4}$$

$$107.(c) x^3 + y^3 + z^3 = 3(1 + xyz)$$

$$P = y + z - x,$$

$$Q = z + x - y,$$

$$R = x + y - z$$

Put, $y = 0$ and $z = 0$

$$x^3 = 3, P = -x, Q = x, R = x$$

$$P^3 + Q^3 + R^3 - 3PQR$$

$$= (-x)^3 + (x)^3 + (x)^3 - 3(-x)(x)(x)$$

$$= x^3 + 3x^3 = 4x^3$$

$$= 4 \times 3 = 12$$

$$108.(b) x_1 x_2 x_3 = 4(4 + x_1 + x_2 + x_3)$$

$$\left[\frac{1}{(2+x_1)} + \frac{1}{(2+x_2)} + \frac{1}{(2+x_3)} \right] = ?$$

Assume value of x_1, x_2 & x_3

$$x_1 = 4$$

$$x_2 = 4$$

$$x_3 = 4$$

$$4 \times 4 \times 4 = 4(4 + 4 + 4 + 4)$$

$$64 = 64 \text{ ...value satisfied}$$

$$\therefore \left[\frac{1}{2+4} + \frac{1}{2+4} + \frac{1}{2+4} \right]$$

$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

$$109.(a) x^2 - x + 1 = 0$$

$$a = 1, b = -1, c = 1$$

$$\text{Sum of roots} = (\alpha + \beta) = \frac{-b}{a} = \frac{-(-1)}{1} = 1$$

$$\text{Product of roots} = \alpha\beta = \frac{c}{a} = \frac{1}{1} = 1$$

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

$$\alpha^3 + \beta^3 = (1)^3 - 3(1)(1)$$

$$\alpha^3 + \beta^3 = 1 - 3 = -2$$

$$\alpha^3 \beta^3 = 1^3 = 1$$

$$\therefore \text{eq. } x^2 - (-2)x + 1 = 0$$

$$x^2 + 2x + 1 = 0$$

$$110.(c) 3x + 5y + 7z = 49 \quad \times 3 \dots (i)$$

$$9x + 8y + 21z = 126 \quad \dots (ii)$$

Multiply by 3 in eq. (i)

$$9x + 15y + 21z = 147$$

$$9x + 8y + 21z = 126$$

$$7y = 21$$

$$y = 3$$

Alternatively:-

By eliminating variable 'z' as there are three unknown & only 2 equations.

By putting $z = 0$,

$$3x + 5y = 49 \quad \dots (i)$$

$$9x + 8y = 126 \quad \dots (ii)$$

Multiplying by 3 in eq. (i) and subtracting

$$9x + 15y = 147$$

$$9x + 8y = 126$$

$$7y = 21$$

$$y = 3$$

Alternatively:-

$$3x + 5y + 7z = 49$$

$$9x + 8y + 21z = 126$$

Assume value of x, y, z

$$x = 2$$

$$y = 3$$

$$z = 4$$

$$3(2) + 5(3) + 7(4) = 49$$

$$6 + 15 + 28 = 49$$

$$49 = 49 \quad \dots \text{satisfied}$$

$$9(2) + 8(3) + 21(4) = 126$$

$$18 + 24 + 84 = 126$$

$$126 = 126 \dots \text{value satisfied}$$

$$y = 3$$

$$111.(b) \text{ Let, pen} = x, \text{ Books} = y, \text{ files} = z$$

$$4x + 6y + 9z = 305 \quad \dots (i)$$

$$3x + 4y + 2z = 145 \quad \dots (ii)$$

Subtract (ii) From (i)

$$x + 2y + 7z = 160 \quad \dots (iii)$$

Adding eq. (i) and (iii)

$$5x + 8y + 16z = 465$$

$$112.(c) PQ = 1, P = \frac{1}{Q}$$

$$P = 7 + 4\sqrt{3} = \frac{1}{Q}$$

$$\frac{1}{P} = \frac{1}{7 + 4\sqrt{3}}$$

Rationalize it ($\because (7)^2 - (4\sqrt{3})^2 = 1$)

$$\frac{1}{P} = \frac{1}{7 - 4\sqrt{3}}$$

$$\frac{1}{P^2 + Q^2}$$

$$(7 - 4\sqrt{3})^2 + (7 + 4\sqrt{3})^2$$

$$49 + 48 - 56\sqrt{3} + 49 + 48 + 56\sqrt{3} = 194$$

$$113.(d) x = \sqrt{5} + 1$$

$$y = \sqrt{5} - 1$$

$$= \frac{x^2}{y^2} + \frac{y^2}{x^2} + 4 \left[\frac{x}{y} + \frac{y}{x} \right] + 6$$

add $(+2, -2)$

$$= \left(\frac{x}{y} + \frac{y}{x} \right)^2 - 2 + 4 \left[\frac{x}{y} + \frac{y}{x} \right] + 6$$

$$= \left(\frac{\sqrt{5}+1}{\sqrt{5}-1} + \frac{\sqrt{5}-1}{\sqrt{5}+1} \right)^2 + 4 \left[\frac{\sqrt{5}+1}{\sqrt{5}-1} + \frac{\sqrt{5}-1}{\sqrt{5}+1} \right]$$

$$+ \frac{\sqrt{5}-1}{\sqrt{5}+1} + 4$$

$$= \left[\frac{(\sqrt{5}+1)^2(\sqrt{5}-1)^2}{4} \right] + 4$$

$$\left[\frac{(\sqrt{5}+1)^2(\sqrt{5}-1)^2}{4} \right] + 4$$

$$\{(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)\}$$

$$= \left[\frac{2(5+1)}{4} \right] + 4 \left[\frac{2(5+1)}{4} \right] + 4$$

$$= (3)^2 + 12 + 4$$

$$= 25$$

$$114.(a) x = 2 + \sqrt{3}$$

$$y = 2 - \sqrt{3}$$

$$z = 1$$

$$\frac{x}{yz} + \frac{y}{xz} + \frac{z}{xy} + 2 \left[\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right]$$

$$= \frac{x^2 + y^2 + z^2 + 2xy + 2yz + 2zx}{xyz}$$

$$= \frac{(x+y+z)^2}{xyz}$$

$$= \frac{(2+\sqrt{3}+2-\sqrt{3}+1)^2}{(2+\sqrt{3})(2-\sqrt{3}) \cdot 1}$$

$$= \frac{25}{4-3} = 25$$

Alternatively:-

$$\frac{x^2+y^2+z^2}{xyz} + 2 \left[\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right]$$

$$= \frac{(2+\sqrt{3})^2 + (2-\sqrt{3})^2 + (1)^2}{(2+\sqrt{3})(2-\sqrt{3})(1)} + 2$$

$$\left[\frac{1}{2+\sqrt{3}} + \frac{1}{2-\sqrt{3}} + 1 \right]$$

$$= \frac{4+3+4\sqrt{3}+4+3-4\sqrt{3}+1}{1} + 2[2$$

$$- \sqrt{3} + 2 + \sqrt{3} + 1]$$

$$= 15+10$$

$$= 25$$

115.(c) If first root is $\alpha = 5+3\sqrt{3}$

then, second root will be

$$\beta = 5-3\sqrt{3}$$

Quadratic Equation = $x^2 - (\alpha + \beta)$

$x + \alpha \cdot \beta$

$$\Rightarrow x^2 - (5+3\sqrt{3} + 5-3\sqrt{3})x + (5+3$$

$$\sqrt{3})(5-3\sqrt{3})$$

$$\Rightarrow x^2 - 10x - 2 = 0$$

on comparing with $ax^2+bx+c=0$

$$a = 1$$

$$b = -10$$

$$c = -2$$

$$\therefore \frac{a^2+b^2+c^2}{a+b+c}$$

$$= \frac{(1)^2 + (-10)^2 + (-2)^2}{1-10-2}$$

$$= \frac{1+100+4}{-11}$$

$$= \frac{-105}{11}$$

116.(a) Let, $a = b = c = 1$

$$\therefore x = 2, y = 2, z = 2$$

$$xyz - x^2 - y^2 - z^2 = 2 \times 2 \times 2 - 4 - 4 - 4$$

$$= 8 - 12$$

$$= -4$$

$$\mathbf{117.(d)} \left(a + \frac{1}{a}\right)^2 - 2\left(a - \frac{1}{a}\right) = 12$$

make both term with (-) sign

$$\left(a - \frac{1}{a}\right)^2 + 4 - 2\left(a - \frac{1}{a}\right) = 12$$

$$\left(a - \frac{1}{a}\right)^2 - 2\left(a - \frac{1}{a}\right) - 8 = 0$$

$$\text{Let, } a - \frac{1}{a} = x$$

$$\therefore x^2 - 2x - 8 = 0$$

$$x^2 - 4x + 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$$x = 4, x = -2$$

$$a - \frac{1}{a} = -2, a - \frac{1}{a} = 4$$

$$\therefore a + \frac{1}{a} = \sqrt{4^2 + 4}$$

$$a + \frac{1}{a} = 2\sqrt{5}$$

...(i)

$$= a - \frac{1}{a} = 4$$

...(ii)

add both eq. (i) and (ii)

$$\therefore 2a = 2\sqrt{5} + 4$$

$$a = 2 + \sqrt{5}$$

118.(b) $x^2 - 4x + 1 = 0$

divide by x

$$x + \frac{1}{x} = 4$$

$$= x^5[x^4-194] + x^3[x^4-194]$$

$$= [x^5+x^3][x^4-194]$$

$$x + \frac{1}{x} = 4$$

square both side

$$x^2 + \frac{1}{x^2} = 14$$

again square both side

$$x^4 + \frac{1}{x^4} = 194$$

$$x^4 \cdot x^4 + 1 = 194 x^4$$

$$x^4(x^4-194) = -1$$

then

$$\Rightarrow [x^5+x^3][x^4-194]$$

$$\Rightarrow x^4 \left[x + \frac{1}{x} \right] [x^4-194]$$

$$\Rightarrow x^4(x^4-194)\left[x + \frac{1}{x}\right]$$

$$\Rightarrow (-1)(4) \Rightarrow -4$$

119.(c) $x + y = 3$

Put, $x = 1, y = 2$

$$x^3 + y^3 + 9xy$$

$$= (1)^3 + (2)^3 + 9 \times 1 \times 2$$

$$= 1 + 8 + 18$$

$$= 27$$

Alternatively:-

$$x + y = 3$$

...(i)

cube both sides

$$(x+y)^3 = (3)^3$$

$$x^3 + y^3 + 3xy(x+y) = 27$$

$$x^3 + y^3 + 3xy(3) = 27$$

$$x^3 + y^3 + 9xy = 27$$

$$\mathbf{120.(b)} A = \frac{(x^8-1)}{x^4+1}$$

$$= \frac{(x^4+1)(x^4-1)}{(x^4+1)} = x^4-1$$

$$B = \frac{y^4-1}{y^2+1}$$

$$= \frac{(y^2+1)(y^2-1)}{y^2+1} = y^2-1$$

$$x = 2, y = 9$$

$$A = x^4-1 = (2)^4-1 = 15$$

$$B = y^2-1 = (9)^2-1 = 80$$

$$A^2+2AB+AB^2$$

$$(15)^2 + 2 \times 15 \times 80 + 15 \times (80)^2$$

$$= 15(15+160+6400)$$

$$= 15 \times (6575)$$

$$= \mathbf{98625}$$

121.(b) $x - 4y = 0$

$$x = 4y \quad \frac{x}{y} = \frac{4}{1}$$

$$= \frac{2x+3y}{2x-3y} = \frac{2 \times 4 + 3 \times 1}{2 \times 4 - 3 \times 1} = \frac{11}{5}$$

Alternatively:-

$$x - 4y = 0$$

$$x = 4y$$

$$\text{and } x+2y = 24$$

put the value of x

$$4y + 2y = 24$$

$$6y = 24$$

$$y = 4, \quad x = 16$$

$$= \frac{2x+3y}{2x-3y}$$

$$= \frac{2 \times (16) + 3 \times (4)}{2 \times (16) - 3 \times (4)} = \frac{44}{20} = \frac{11}{5}$$

$$122. (a) \frac{x}{a} + \frac{y}{b} = 3 \quad \dots (i)$$

$$\frac{x}{b} - \frac{y}{a} = 9 \quad \dots (ii)$$

multiply by 3 in eq. (i)

$3\left(\frac{x}{a} + \frac{y}{b}\right) = 9$ and on comparing with eq. (ii), we get

$$3\left(\frac{x}{a} + \frac{y}{b}\right) = \frac{x}{b} - \frac{y}{a}$$

$$\frac{3x}{a} + \frac{3y}{b} = \frac{x}{b} - \frac{y}{a}$$

$$x\left(\frac{3b-a}{ab}\right) = y\left(\frac{-3a-b}{ab}\right)$$

$$\frac{x}{y} = \frac{b+3a}{a-3b}$$

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123. (b) Make any 2 variable zero

Let, $q = r = 0$

$$p^3 + q^3 + r^3 - 3pqr = 4$$

$$p^3 = 4$$

$$a = 0, b = p, c = p$$

$$a^3 + b^3 + c^3 - 3abc$$

$$= p^3 + p^3 + p^3 = 2 \times 4 = 8$$

Alternatively:-

$$p^3 + q^3 + r^3 - 3pqr = 4$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc$$

Put, $a = q + r$

$$b = r + p$$

$$c = p + q$$

$$\Rightarrow (q+r)^3 + (r+p)^3 + (p+q)^3 - 3(p+q)(q+r)(r+p)$$

$$\Rightarrow q^3 + r^3 + 3qr(q+r) + r^3 + p^3 + 3rp(r+p) + p^3 + r^3 + 3pq(p+q) - 3(p+q)(q+r)(r+p)$$

$$\Rightarrow 2p^3 + 2q^3 + 2r^3 + 3pq(p+q) + 3qr(q+r) + 3rp(r+p) - 3(p+q)(q+r)(r+p)$$

$$\Rightarrow 2(p^3+q^3+r^3) + 3p^2q+3pq^2+3p^2r + 3qr^2 + 3r^2p + 3rp^2 - 3[(p+q)(qr+qp+r^2+rp)]$$

Put, values of $(p^3+q^3+r^3)$

$$\Rightarrow 2(4+3pqr) + 3p^2q + 3pq^2 + 3q^2r + 3qr^2 + 3r^2p + 3rp^2 - 3pqr - 3p^2q - 3pr^2 - 3p^2r - 3q^2r - 3qr^2 - 3q^2p - 3pqr$$

$$\Rightarrow 8 + 6pqr - 6pqr = 8$$

$$124. (d) x^2 + x - 1 = 0$$

$$a = 1, b = 1, c = -1$$

$$\alpha + \beta = \frac{-b}{a} = \frac{-1}{1} \quad \dots (i)$$

$$\alpha\beta = \frac{c}{a} = -1$$

$$\beta = -\frac{1}{\alpha} \quad \text{Put, in eq (i)}$$

$$\alpha - \frac{1}{\alpha} = -1$$

$$\Rightarrow \alpha^5 + \beta^5 = \alpha^5 - \frac{1}{\alpha^5}$$

$$= \left(\alpha^2 + \frac{1}{\alpha^2}\right)\left(\alpha^3 - \frac{1}{\alpha^3}\right) - \left(\alpha - \frac{1}{\alpha}\right) \dots (ii)$$

$$\alpha^2 + \frac{1}{\alpha^2} = \left(\alpha - \frac{1}{\alpha}\right)^2 + 2 = (-1)^2 + 2 = 3$$

$$\alpha^3 - \frac{1}{\alpha^3} = \left(\alpha - \frac{1}{\alpha}\right)^3 + 3\left(\alpha - \frac{1}{\alpha}\right)$$

$$= (-1)^3 + 3(-1) = -4$$

put, the values in eq. (ii)

$$= 3 \times -4 = -12 + 1 = -11$$

$$\Rightarrow \alpha^5 + \beta^5 = -11 = \frac{-b}{a}$$

$$\alpha^5 \times \beta^5 = -1 = \frac{c}{a}$$

$$\Rightarrow b = -11, c = -1, a = 1$$

Required eqⁿ

$x^2 - (\text{sum of roots})x + \text{product of roots} = 0$

$$x^2 - (-11)x + (-1) = 0$$

$$x^2 + 11x - 1 = 0$$

125. (c) 2017 is an odd number which is made by adding 2 numbers means other number must be an even number.

$\therefore$ Even + odd = odd

$$x + y = 2017$$

$$(-1)^x + (-1)^y$$

$$\text{Put, } x = 2000 \text{ \& } y = 17$$

$$(-1)^{2000} + (-1)^{17}$$

$$= -1 + 1 = 0$$

$$126. (c) x + \frac{1}{x} = \frac{\sqrt{3}+1}{2}$$

square both side

$$x^2 + \frac{1}{x^2} = \left(\frac{\sqrt{3}+1}{2}\right)^2 - 2$$

$$= \frac{4+2\sqrt{3}}{4} - 2$$

$$= \frac{4+2\sqrt{3}-8}{4} = -\frac{4+2\sqrt{3}}{4}$$

$$x^2 + \frac{1}{x^2} = \frac{-2+\sqrt{3}}{4}$$

again square both side

$$x^4 + \frac{1}{x^4} = \left(\frac{-2+\sqrt{3}}{2}\right)^2 - 2$$

$$x^4 + \frac{1}{x^4} = \frac{4+3-4\sqrt{3}-8}{4}$$

$$x^4 + \frac{1}{x^4} = \frac{-1-4\sqrt{3}}{4}$$

$$127. (c) a + a^2 + a^3 - 1 = 0$$

$$a^3 = 1 - a^2 - a \quad \dots (i)$$

$$(a + a^2 + a^3 = 1) \times \frac{1}{a}$$

$$1 + a + a^2 = \frac{1}{a} \quad \dots (ii)$$

Add eq. (i) & (ii)

$$a^3 + \frac{1}{a} = 1 - a^2 - a + 1 + a + a^2 = 2$$

$$128. (a) a - \frac{1}{a} = b \quad \dots (i)$$

$$b - \frac{1}{b} = c \quad \dots (ii)$$

$$c - \frac{1}{c} = a \quad \dots (iii)$$

Add eq. (i) (ii) & (iii)

$$a + b + c - \left[\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right] = a + b + c$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0 \quad \dots (iv)$$

$$\Rightarrow a - \frac{1}{a} = b$$

Squaring both sides

$$a^2 + \frac{1}{a^2} - 2 = b^2 \quad \dots (v)$$

similarly,

$$b^2 + \frac{1}{b^2} - 2 = c^2 \quad \dots (vi)$$

$$\Rightarrow c^2 + \frac{1}{c^2} - 2 = a^2 \quad \dots (vii)$$

Add eq. (v), (vi) & (vii)

$$a^2 + b^2 + c^2 + \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} - 2 - 2 - 2$$

$$= a^2 + b^2 + c^2$$

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 6 \quad \dots (viii)$$

$$\Rightarrow \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)^2$$

$$= \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} + \frac{1}{2ab} + \frac{1}{2bc} + \frac{1}{2ca}$$

put, the values

$$0^2 = 6 + \frac{1}{2} \left[\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} \right]$$

$$\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} = \frac{-6}{2} = -3$$

129. (b) $a(b-c)x^2 + b(c-a)x + c(a-b)=0$

Roots of the equation are equal,

$$\therefore b^2 = 4ac$$

$$[b(c-a)]^2 = 4a(b-c) \times c(a-b)$$

$$b^2(c-a)^2 = 4ac(b-c)(a-b)$$

$$b^2(c^2 + a^2 - 2ac) = 4ac(ab - b^2 - ac + bc)$$

$$\Rightarrow b^2c^2 + b^2a^2 - 2ab^2c = 4a^2bc - 4ab^2c - 4a^2c^2 + 4abc^2$$

$$\Rightarrow b^2c^2 + b^2a^2 + 2ab^2c = 4a^2bc - 4a^2c^2 + 4abc^2$$

$$\Rightarrow b^2c^2 + b^2a^2 + 2ab^2c - 4a^2bc + 4a^2c^2 - 4abc^2 = 0$$

$$\Rightarrow a^2b^2 + b^2c^2 + 4a^2c^2 + 2ab^2c - 4abc^2$$

$$- 4a^2bc = 0$$

$$(ab+bc-2ac)^2 = 0$$

$$ab + bc = 2ac$$

Divide by abc

$$\frac{1}{a} + \frac{1}{c} = \frac{2}{b}$$

130. (d) Put, $b = 0$

$$a + a = 1$$

$$a = \frac{1}{2}$$

$$(1-a^2)(1-0) = 1 - a^2$$

$$= 1 - \frac{1}{4} = \frac{3}{4}$$

Alternatively:-

$$\sqrt{(a^2+b^2+ab)} + \sqrt{(a^2+b^2-ab)} = 1$$

Square both side-

$$a^2+b^2+ab+a^2+b^2-ab+$$

$$2\sqrt{(a^2+b^2)^2 - (ab)^2} = 1$$

$$2(a^2+b^2) + 2\sqrt{(a^2+b^2)^2 - (ab)^2} = 1$$

$$\sqrt{(a^2+b^2)^2 - (ab)^2} = \frac{1}{2} - (a^2+b^2)$$

again squaring both sides

$$a^4 + b^4 + 2a^2b^2 - a^2b^2 = \frac{1}{4}$$

$$+ (a^2 + b^2)^2 - (a^2 + b^2)$$

$$a^4 + b^4 + a^2b^2 = \frac{1}{4} + a^4 + b^4 + 2a^2b^2$$

$$b^2 - a^2 - b^2$$

$$a^2 + b^2 - a^2b^2 = \frac{1}{4}$$

$$\Rightarrow (1-a^2)(1-b^2)$$

$$= 1 - a^2 - b^2 + a^2b^2$$

$$= 1 - [a^2 + b^2 - a^2b^2]$$

$$= 1 - \frac{1}{4} = \frac{3}{4}$$

131. (b) $3x + 4y - 11 = 18$

$$8x - 6y + 12 = 6$$

$$3x + 4y = 29 \quad \times 3$$

$$8x - 6y = -6 \quad \times -2$$

$$9x + 12y = 87$$

$$-16x + 12y = 12$$

$$+ \quad - \quad -$$

$$25x = 75$$

$$x = 3$$

$$y = 5$$

$$5x - 3y - 9$$

$$\Rightarrow = 5 \times 3 - 3 \times 5 - 9$$

$$= 15 - 15 - 9 = -9$$

132. (b) $a + b + c = \frac{7}{12} \quad \dots(i)$

$3a - 4b + 5c = \frac{3}{4} \quad \dots(2)$

$7a - 11b - 13c = \frac{-7}{12} \quad \dots(3)$

Take eq. (i) & (ii) and solve

$$3a + 3b + 3c = \frac{7}{4}$$

$$3a - 4b + 5c = \frac{3}{4}$$

$$\begin{array}{r} - \quad + \quad - \quad - \\ 7b - 2c = 1 \end{array} \quad \dots(4)$$

now, take eq. (i) & (iii) and solve after multiply by 7 in eq. 1

$$7a + 7b + 7c = \frac{49}{12}$$

$$7a - 11b - 13c = \frac{-7}{12}$$

$$\begin{array}{r} - \quad + \quad + \quad + \\ 18b + 20c = \frac{14}{3} \end{array} \quad \dots(5)$$

$$18b + 20c = \frac{14}{3}$$

now, Solve eq. (4) & (5)

$$7b - 2c = 1 \times 10$$

$$18b + 20c = \frac{14}{3} \times 1$$

$$70b - 20c = 10$$

$$18b + 20c = \frac{14}{3}$$

$$88b = \frac{44}{3} \Rightarrow b = \frac{1}{6}$$

put value of b in eq. (i)

$$a + b + c = \frac{7}{12}$$

$$a + c = \frac{7}{12} - \frac{1}{6}$$

$$a + c = \frac{5}{12}$$

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133. (c) $(x-y)^2 + (y-z)^2 + (z-x)^2$

$$= 2 \times (x^2 + y^2 + z^2 - xy - yz - zx)$$

$$= 2 \times (\text{any integer})$$

$$= \text{any even number}$$

when we multiply by 2 in any number we get an even number.

$\therefore$ By option (C) $\Rightarrow$ **758**

Alternatively:-

$$xy + yz + zx = 35$$

$$x + y + z = 22$$

square both side

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$(22)^2 = x^2 + y^2 + z^2 + 2 \times 35$$

$$484 - 70 = x^2 + y^2 + z^2$$

$$x^2 + y^2 + z^2 = 414$$

$$(x-y)^2 + (y-z)^2 + (z-x)^2$$

$$= 2(x^2 + y^2 + z^2 - xy - yz - zx)$$

$$= 2(414 - 35) = 2 \times 379 = \mathbf{758}$$

134. (c) Take $x = 3, y = 1, z = 2,$

$$\frac{x+y}{2} = 2 \Rightarrow \frac{3+1}{2} = 2 \quad \dots \text{satisfied}$$

$$\therefore \frac{y}{y-z} + \frac{x}{x-z} \Rightarrow \frac{1}{-1} + \frac{3}{1} = 2$$

135. (c) $x^2 - 2x + 4 = 0$

$$\text{Sum of roots} = \frac{-b}{a} = \frac{-(-2)}{1} = 2$$

$$\text{Product of roots} = \frac{c}{a} = \frac{4}{1} = 4$$

$$\alpha + \beta = 2$$

$$\alpha\beta = 4$$

$$\text{Equation with roots } \frac{\alpha^3}{\beta^2} \text{ and } \frac{\beta^3}{\alpha^2},$$

$$\text{Product} = \frac{\alpha^3}{\beta^2} \times \frac{\beta^3}{\alpha^2} = \alpha\beta = 4 = \frac{c}{a}$$

$$\text{sum} = \frac{\alpha^3}{\beta^2} + \frac{\beta^3}{\alpha^2} = \frac{\alpha^5 + \beta^5}{\alpha^2 \beta^2}$$

$$= \frac{(\alpha^2 + \beta^2)(\alpha^3 + \beta^3) - \alpha^2 \beta^2 (\alpha + \beta)}{\alpha^2 \beta^2}$$

$$= \frac{(\alpha^2 + \beta^2)(\alpha^3 + \beta^3) - (\alpha + \beta)}{\alpha^2 \beta^2}$$

$$\alpha^2 + \beta^2 = 2^2 - 2 \times 4 = -4$$

$$\alpha^3 + \beta^3 = 2^3 - 3 \times 4 \times 2$$

$$= 8 - 24 = -16$$

$$\therefore \text{sum} = \frac{-4 \times -16}{4^2} - 2 = 2 = \frac{-b}{a}$$

$$\therefore a = 1, b = -2, c = 4$$

$$\therefore \text{equation} = ax^2 + bx + c = 0$$

$$x^2 - 2x + 4 = 0$$

$$136.(d) p = \frac{5}{2} q$$

$$\frac{p}{q} = \frac{5}{2}$$

$$p+q = 7k = -\frac{B}{A}$$

$$pq = 10k^2 = \frac{C}{A}$$

$$\frac{B^2}{A^2} = 49k^2 \quad \dots(i)$$

$$\frac{C}{A} = 10k^2 \quad \dots(ii)$$

Divide eqn (i) by (ii)

$$\frac{B^2}{CA} = \frac{49}{10}$$

$$10B^2 = 49CA$$

Alternatively:-

Let 2 roots = p and q

A.T.Q.,

$$p = \frac{5}{2} q \quad \dots(i)$$

$$2p = 5q$$

$$Ax^2 + Bx + C = 0$$

$$p+q = -\frac{B}{A}$$

$$pq = \frac{C}{A}$$

put, the value of p from eq. (i)

$$\frac{5}{2} q + q = -\frac{B}{A}$$

$$\frac{7q}{2} = -\frac{B}{A}$$

$$q = \frac{-2B}{7A}$$

$$\therefore p = \frac{5}{2} q = \frac{5}{2} \times \frac{-2B}{7A} = \frac{-5B}{7A}$$

$$pq = \frac{C}{A}$$

$$\frac{-5B}{7A} \times \frac{-2B}{7A} = \frac{C}{A}$$

$$\frac{10B^2}{49A^2} = \frac{C}{A}$$

$$10B^2 = 49AC$$

$$137.(b) x^2 - 12x + 33 = 0$$

add (+3, -3) for making perfect square

$$x^2 - 12x + 36 - 3 = 0$$

$$(x-6)^2 - 3 = 0$$

$$x = \sqrt{3} + 6$$

$$(x-4)^2 + \frac{1}{(x-4)^2}$$

put, the value of x

$$\therefore (\sqrt{3} + 2)^2 + \frac{1}{(\sqrt{3} + 2)^2}$$

$$= (\sqrt{3} + 2)^2 + (\sqrt{3} - 2)^2$$

$$= 2((\sqrt{3})^2 + (2)^2) = 2 \times 7 = 14$$

$$\therefore (a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$$

Alternatively:-

Let, $x - 4 = y$

$$x = y + 4$$

Put, in question-

$$(x-4)^2 + \frac{1}{(x-4)^2} = ?$$

$$y^2 + \frac{1}{y^2} = ?$$

$$x^2 - 12x + 33 = 0$$

$$(y+4)^2 - 12(y+4) + 33 = 0$$

$$y^2 + 16 + 8y - 12y - 48 + 33 = 0$$

$$y^2 - 4y + 1 = 0$$

$$y^2 + 1 = 4y$$

$$y + \frac{1}{y} = 4$$

$$y^2 + \frac{1}{y^4} = 16 - 2 = 14$$

Alternatively:-

$$x^2 - 12x + 33 = 0$$

Put, $(x-4)$ in place of x

$$x^2 - 4x - 8x + 32 + 1 = 0$$

$$x(x-4) - 8(x-4) + 1 = 0$$

$$(x-4)(x-8) = -1$$

$$(x-8) = -\frac{1}{x-4}$$

$$(x-4) - 4 = -\frac{1}{x-4}$$

$$(x-4) - \frac{1}{(x-4)} = 4$$

$$(x-4)^2 + \frac{1}{(x-4)^2} = 4^2 - 2 = 14$$

$$138.(a) a^4 + 1 = \left(\frac{a^2}{b^2}\right)(4b^2 - b^4 - 1)$$

take $a = b = 1$

$$1+1 = \frac{1}{1}(4-1-1) = 2 \quad \dots \text{satisfied}$$

$$\therefore a^4 + b^4 = 1 + 1 = 2$$

$$139.(b) 3\sqrt{\frac{1-a}{a}} + 9 = 19 - 3\sqrt{\frac{a}{1-a}}$$

$$3\sqrt{\frac{1-a}{a}} + 3\sqrt{\frac{a}{1-a}} = 10$$

$$\text{Let, } \sqrt{\frac{1-a}{a}} = 3, \text{ then } \sqrt{\frac{a}{1-a}} = \frac{1}{3}$$

$$3 \times 3 + 3 \times \frac{1}{3} = 10$$

$$10 = 10$$

$$\therefore \sqrt{\frac{1-a}{a}} = 3 \Rightarrow \frac{1-a}{a} = 9$$

$$1-a = 9a$$

$$a = \frac{1}{10}$$

By option (b)

$$a = \frac{1}{10}, \frac{9}{10}$$

Alternatively:-

$$3\sqrt{\frac{1-a}{a}} + 9 = 19 - 3\sqrt{\frac{a}{1-a}}$$

$$\text{Let, } \sqrt{\frac{1-a}{a}} = x$$

$$\therefore 3x + 9 = 19 - \frac{3}{x}$$

$$x + \frac{1}{x} = \frac{10}{3}$$

$$x - \frac{1}{x} = \sqrt{\left(\frac{10}{3}\right)^2 - 4}$$

$$\Rightarrow x - \frac{1}{x} = \pm \frac{8}{3}$$

First take positive value

$$x + \frac{1}{x} = \frac{10}{3}$$

$$x - \frac{1}{x} = +\frac{8}{3}$$

$$x = 3$$

$$\text{Now, } \sqrt{\frac{1-a}{a}} = 3$$

$$\frac{1-a}{a} = 9 \Rightarrow 10a = 1 \Rightarrow a = \frac{1}{10}$$

Now take negative value

$$x + \frac{1}{x} = \frac{10}{3}$$

$$x - \frac{1}{x} = -\frac{8}{3}$$

$$x = \frac{1}{3}$$

$$\text{Now, } \sqrt{\frac{1-a}{a}} = \frac{1}{3} \Rightarrow \frac{1-a}{a} = \frac{1}{9}$$

$$9 - 9a = a \Rightarrow 10a = 9$$

$$a = \frac{9}{10}$$

$$\therefore a = \frac{1}{10}, \frac{9}{10}$$

$$140.(b) \text{ Let, } \sqrt{\frac{a}{b}} = x$$

$$\therefore x - 13 = \frac{-1}{x} - 11$$

$$x + \frac{1}{x} = 2$$

$$\therefore x = 1$$

$$\sqrt{\frac{a}{b}} = 1$$

$$a+b = 10 \therefore a = b = 5$$

$$3ab + 4a^2 + 5b^2 = 3a^2 + 4a^2 + 5a^2 = 12a^2 = 12 \times 25 = 300$$

$$141.(c) 3x + 4y - 2z + 9 = 17 \quad \dots(i)$$

$$7x + 2y + 11z + 8 = 23 \quad \dots(ii)$$

$$5x + 9y + 6z - 4 = 18 \quad \dots(iii)$$

$$15x + 15y + 15z + 13 = 58$$

(By adding eq. (i), (ii) & (iii))

$$15(x + y + z) = 45$$

$$x + y + z = 3$$

$$x + y + z - 34 = 3 - 34 = -31$$

$$142.(a) x + 3y - \frac{2z}{4} = 6$$

$$2x + 6y - z = 12 \quad \dots(i)$$

$$x + \frac{2}{3}(2y + 3z) = 33$$

$$3x + 4y + 6z = 99 \quad \dots(ii)$$

$$\frac{1}{7}(x + y + z) + 2z = 9$$

$$x + y + z + 14z = 63$$

$$x + y + 15z = 63 \quad \dots(iii)$$

multiply by 21 in eq. (i) and

Add eq. (i)+(ii)+(iii) and remove (z)

$$21(2x + 6y - z) = 12 \times 21$$

$$42x + 126y - 21z = 252$$

$$3x + 4y + 6z = 99$$

$$x + y + 15z = 63$$

$$46x + 131y = 414$$

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{21 Feb. 2018}

$$143.(a) ab = 1 \quad a = \frac{1}{b}$$

$$\therefore a^2 + \frac{1}{a^2} - 1 = ?$$

$$a = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$$

$$\frac{1}{a} = \frac{2 \times 5}{1} = 10$$

$$a^2 + \frac{1}{a^2} = 10^2 - 2 = 98$$

$$\therefore a^2 + \frac{1}{a^2} - 1 = 98 - 1 = 97$$

Alternatively:-

$$a = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}, b = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$$

Rationalize both-

$$a = \frac{(\sqrt{3} + \sqrt{2}) \times (\sqrt{3} + \sqrt{2})}{(\sqrt{3} - \sqrt{2})(\sqrt{3} + \sqrt{2})}$$

$$a = \frac{(\sqrt{3} + \sqrt{2})^2}{3 - 2} = 5 + 2\sqrt{6}$$

$$b = \frac{(\sqrt{3} - \sqrt{2}) \times (\sqrt{3} - \sqrt{2})}{(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2})}$$

$$b = \frac{(\sqrt{3} - \sqrt{2})^2}{3 - 2} = 5 - 2\sqrt{6}$$

$$a + b = 5 + 2\sqrt{6} + 5 - 2\sqrt{6} = 10$$

$$a^2 + b^2 - ab$$

add (+2ab-2ab) for making perfect square for easy calculation

$$a^2 + b^2 - ab + 2ab - 2ab$$

$$(a + b)^2 - 3ab$$

$$(10)^2 - 3 \times 1 = 97 \quad (\because ab=1)$$

$$144.(b) \text{ Given eqn., } Ax^2 - Bx + C = 0$$

Let, α and β are the roots of this eqn.,

$$\therefore \alpha + \beta = \frac{-b}{a} = \frac{-(-B)}{A} = \frac{B}{A} \quad \dots(i)$$

$$\alpha \times \beta = \frac{c}{a} = \frac{C}{A}$$

$$\alpha - \beta = 4 \quad \dots(ii) \quad [\text{given}]$$

Using eqn. (1) & (2),

$$\alpha + \beta = \frac{B}{A}$$

$$\alpha - \beta = 4$$

$$2\alpha = \frac{B}{A} + 4, \Rightarrow \alpha = \frac{B}{2A} + 2$$

$$\text{Similarly, } \beta = \frac{B}{2A} - 2$$

$$\Rightarrow \left(\frac{B}{2A} + 2 \right) \left(\frac{B}{2A} - 2 \right) = \frac{C}{A}$$

$$\Rightarrow \frac{B^2}{4A^2} - 4 = \frac{C}{A}$$

$$\Rightarrow \frac{B^2 - 16A^2}{4A^2} = \frac{C}{A}$$

$$\Rightarrow B^2 - 16A^2 = 4AC$$

$$\Rightarrow B^2 - 10A^2 = 4AC + 6A^2$$

$$145.(a) \alpha + \beta = 8 \quad \dots(i)$$

$$\alpha - \beta = 2\sqrt{5} \quad \dots(ii)$$

From eqns. (i) & (ii)

$$\alpha = 4 + \sqrt{5}$$

$$\beta = 4 - \sqrt{5}$$

$$\alpha^4 + \beta^4 = (4 + \sqrt{5})^4 + (4 - \sqrt{5})^4$$

$$= \left[(4 + \sqrt{5})^2 \right]^2 + \left[(4 - \sqrt{5})^2 \right]^2$$

$$= (21 + 8\sqrt{5})^2 + (21 - 8\sqrt{5})^2$$

$$= 441 + 320 + 336\sqrt{5} + 441 +$$

$$320 - 336\sqrt{5}$$

$$= 1522$$

and if α^4 and β^4 are the roots of any quadratic eqn.

$$\alpha^4 + \beta^4 = \frac{-b}{a} = 1522$$

$$b = -1522$$

and $a = 1$ eqn. must be like-

$$x^2 - 1522x + c = 0$$

$\Rightarrow$ Option (i) is our answer.

To find c

$$\alpha \times \beta = (4 + \sqrt{5})(4 - \sqrt{5})$$

$$\alpha \times \beta = 16 - 5 = 11$$

$$\alpha^4 \beta^4 = (11)^4 = (11)^3 \times 11$$

$$\alpha^4 \beta^4 = 1331 \times 11 = 14641$$

$$\therefore x^2 - 1522x + 14641 = 0$$

146.(b) Let, roots $a = 1, b = 1$

now equation, $(x-1)^2 = 0$

$$x^2 - 2x + 1 = 0$$

compare with $Px^2 - Qx + R = 0$

$$P = 1, Q = 2, R = 1$$

now put, values in

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{a}{b} + \frac{b}{a} = 4 \quad (\because a=b=1)$$

by option (b)

$$\frac{(Q^2 - 2PR)(R + P)}{PR^2}$$

$$= \frac{(4 - 2 \times 1 \times 1)(1 + 1)}{1 \times 1} = 4 \text{ (satisfied)}$$

Alternatively:-

a and b are the root of the eqn.

$$Px^2 - Qx + R = 0$$

$$a + b = \frac{Q}{P}, \quad ab = \frac{R}{P}$$

$$(a + b)^2 = \frac{Q^2}{P^2}$$

$$a^2 + b^2 + 2ab = \frac{Q^2}{P^2}$$

$$(\text{put the value of } ab = \frac{R}{P})$$

$$a^2 + b^2 = \frac{Q^2}{P^2} - \frac{2R}{P} = \frac{Q^2 - 2PR}{P^2}$$

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{a}{b} + \frac{b}{a} = ?$$

$$\frac{a^2 + b^2}{a^2 b^2} + \frac{a^2 + b^2}{ab}$$

now put all values

$$\frac{Q^2 - 2PR}{P^2} + \frac{Q^2 - 2PR}{P^2} + \frac{R}{P^2} + \frac{R}{P^2}$$

$$= \frac{(Q^2 - 2PR)}{R^2} + \frac{Q^2 - 2PR}{RP}$$

$$= \frac{P(Q^2 - 2PR) + R(Q^2 - 2PR)}{R^2 P}$$

$$= \frac{(Q^2 - 2PR)(R + P)}{PR^2}$$

$$\mathbf{147.(b)} \quad (x-6)^2 + \frac{1}{(x-6)^2} = ?$$

Let, $x - 6 = t$

$$x = t + 6$$

$$t^2 + \frac{1}{t^2} = ?$$

$$x^2 - 16x + 59 = 0$$

$$\text{put, } x = t + 6$$

$$(t + 6)^2 - 16(t + 6) + 59 = 0$$

$$t^2 + 36 + 12t - 16t - 96 + 59 = 0$$

$$t^2 - 4t - 1 = 0$$

divide eq. by (t)

$$t - 4 - \frac{1}{t} = 0$$

$$t - \frac{1}{t} = 4$$

square both side

$$t^2 + \frac{1}{t^2} - 2 = 16$$

$$t^2 + \frac{1}{t^2} = \mathbf{18}$$

148.(a) A and B are the roots of $Ax^2 - A^2x + AB = 0$

$$A + B = \frac{-b}{a} = \frac{A^2}{A} = A$$

$$A + B = A \Rightarrow B = 0$$

$$A \times B = \frac{c}{a} = \frac{AB}{A} = B$$

$$A \times B = B \Rightarrow A = 1$$

$$\therefore A = \mathbf{1} \text{ and } B = \mathbf{0}$$

149.(a) $x^2 - x - 1 = 0$

$$a = 1, b = -1, c = -1$$

$$\alpha + \beta = \frac{-b}{a} = -1$$

$$\alpha \cdot \beta = -1 \Rightarrow \beta = \frac{-1}{\alpha}$$

$$\alpha - \frac{1}{\alpha} = 1$$

Square both sides

$$\alpha^2 + \frac{1}{\alpha^2} = (1)^2 + 2 = 3$$

again square both sides

$$\alpha^4 + \frac{1}{\alpha^4} = (3)^2 - 2 = 7$$

again square both sides

$$\alpha^8 + \frac{1}{\alpha^8} = (7)^2 - 2 = 49 - 2 = \mathbf{47}$$

150.(b) $a^3 + b^3 = 91 = 64 + 27 = (4)^3 + (3)^3$

$$b^3 + c^3 = 72 = 64 + 8 = (4)^3 + (2)^3$$

$$c^3 + a^3 = 35 = 27 + 8 = (3)^3 + (2)^3$$

on comparing-

$$a = 3, b = 4, c = 2$$

$$\therefore abc = 3 \times 4 \times 2 = \mathbf{24}$$

Alternatively:-

As we know

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \dots (i)$$

$$a^3 + b^3 = 91$$

$$b^2 + c^3 = 72$$

$$c^3 + a^3 = 35$$

after adding these eqⁿ, we get.

$$2(a^3 + b^3 + c^3) = 91 + 72 + 35$$

$$a^3 + b^3 + c^3 = \frac{198}{2} = 99$$

$$a + b + c = 9 \quad [\text{given}]$$

Squaring on both sides,

$$a^2 + b^2 + c^2 + 2[ab + bc + ca] = 81$$

$$a^2 + b^2 + c^2 = 81 - 2(ab + bc + ca)$$

$$a^2 + b^2 + c^2 = 81 - 26 \times 2$$

$$[ab + bc + ca = 26 \text{ (given)}]$$

$$a^2 + b^2 + c^2 = 29$$

put, all the values in eq. (i)

$$99 - 3abc = 9 \times [29 - 26]$$

$$3abc = 99 - 9 \times 3$$

$$3abc = 99 - 27 = 72$$

$$abc = \frac{72}{3} = \mathbf{24}$$

$$\mathbf{151.(c)} \quad \left(x^2 + \frac{1}{x-4}\right) \rightarrow \frac{x^2(x-4)+1}{x-4}$$

$$x^3 - 4x^2 + 19 = 6(x-1)$$

$$x^3 - 4x^2 + 19 = 6x - 6$$

$$x^3 - 4x^2 + 1 + 18 = 6x - 6$$

$$x^3 - 4x^2 + 1 = 6x - 6 - 18$$

$$x^3 - 4x^2 + 1 = 6x - 24$$

putting the value of $(x^3 - 4x^2 + 1)$ in que

$$= \frac{6x - 24}{x - 4} = \frac{6(x - 4)}{(x - 4)} = \mathbf{6}$$

152.(b) 8 Pencil + 5 Pen + 3 Era = 111

$$9 \text{ Pencil} + 6 \text{ Pen} + 5 \text{ Era} = 130$$

$$16 \text{ Pencil} + 11 \text{ Pen} + 3 \text{ Era} = 221$$

Adding all three eqns. we get

$$33 \text{ Pencil} + 22 \text{ Pen} + 11 \text{ Era} = 462$$

divide eq. by 11

$$3 \text{ Pencil} + 2 \text{ Pen} + 1 \text{ Era} = \frac{462}{11} = 42$$

multiply eq. by 13

$$13 \times [3 \text{ Pencil} + 2 \text{ Pen} + 1 \text{ Eraser}] = 13 \times 42$$

$$= 13 \times 42$$

$$39 \text{ Pencil} + 26 \text{ Pen} + 13 \text{ Eraser} =$$

$$\mathbf{₹546}$$

153.(b) $x + y + 3z = 17$

putting $x = 5, y = 6, z = 2$

$\therefore 3x + 2y + z = 29$

$15 + 12 + 2 = 29$

$29 = 29$

...satisfied

$\therefore xy + yz + zx = 30 + 12 + 10 = 52$

Alternatively:-

$2x + 3y - 5z = 18$... (i)

$2x + 2y - z = 29$... (ii)

$x + y + 3z = 17$... (iii)

Adding eqⁿ (i), (ii) & (iii)

$6x + 6y - z = 64$... (iv)

multiplying eqⁿ (3) by 6

$6x + 6y + 18z = 102$... (v)

eq. (iv) - (v)

$-19z = -38$

$z = 2$

from eqⁿ (iii) & (i)

$x + y + 6 = 17$

$x + y = 11$... (6)

$2x + 3y - 10 = 18$

$2x + 3y = 28$... (7)

multiplying eqⁿ (3) by 2 &

subtract by eqⁿ (7)

$2x + 2y = 22$

$2x + 3y = 28$

$y = 6$

eqⁿ (6), $x = 5$

$xy + yz + zx = 5 \times 6 + 6 \times 2 + 5 \times 2$

$= 30 + 12 + 10 = 52$

Alternatively:-

$2x + 3y - 5z = 18$... (i)

$3x + 2y + z = 29$... (ii)

$x + y + 3z = 17$... (iii)

Eq. (ii) - Eq. (i), we get

$3x + 2y + z = 29$

$2x + 3y - 5z = 18$

$$\begin{array}{r} - \quad - \quad + \quad - \\ x - y + 6z = 11 \end{array}$$
 ... (iv)

Adding (iv) & (iii),

$2x + 9z = 28$... (v)

and Eq. (ii) ... $2 \times$ Eq. (iii),

$3x + 2y + z = 29$

$2x + 2y - 6z = 34$

$$\begin{array}{r} - \quad - \quad - \quad - \\ x - 5z = -5 \end{array}$$
 ... (vi)

Eqn. (v) ... ii $\times$ Eqn. (vi)

$2x + 9z = 28$

$2x - 10z = -10$

$$\begin{array}{r} - \quad + \quad + \\ 19z = 38 \end{array}$$

$z = 2$

Now, from eqn. (6)

$x - 5 \times 2 = -5$

$x = 10 - 5 = 5$

and from eqn. (3),

$5 + y + 3 \times 2 = 17$

$y = 17 - 11 = 6$

$xy + yz + zx = 5 \times 6 + 6 \times 2 + 5 \times 2$

$= 30 + 12 + 10 = 52$

SSC CGL MAINS 2017

{09 Mar. 2018} Paper-I

154.(d) $(2 + \sqrt{2}) + \left(\frac{1}{2 + \sqrt{2}}\right) + \left(\frac{1}{2 - \sqrt{2}}\right)$

$+ (2 - \sqrt{2})$

rationalize fraction part

$= 2 + \sqrt{2} + \frac{2 - \sqrt{2}}{2} + \frac{2 + \sqrt{2}}{2} + 2 - \sqrt{2}$

$= \frac{4 + 2\sqrt{2} + 2 - \sqrt{2} + 2 + \sqrt{2} + 4 - 2\sqrt{2}}{2}$

$= \frac{12}{2} = 6$

155.(c) $a^2 + b^2 = 4b + 6a - 13$

$a^2 + 9 - 6a + b^2 + 4 - 4b = 0$

$(a - 3)^2 + (b - 2)^2 = 0$

$a - 3 = 0, b - 2 = 2$

$a = 3, b = 2$

$a + b = 5$

Alternatively:-

take 2 common

$a^2 + b^2 = 2(3a + 2b) - 13$

coefficient of $a = 3$

coefficient of $b = 2$

$a + b = 3 + 2 = 5$

156.(b) $x^4 + y^4 + x^2y^2 = 481, xy = 12$

add x^2y^2 both side

$x^4 + y^4 + 2x^2y^2 = 481 + x^2y^2$

$(x^2 + y^2)^2 = 481 + 144 = 625$

Do squareroot both side

$x^2 + y^2 = 25$

add $(-xy)$ both side

$25 - 12 = 13$

157.(c) $A = 1 + 2^P, B = 1 + 2^{-P}$

Put, $P = 1$

$A = 3, B = \frac{3}{2}$

by option (c)

$$\frac{A}{A-1} = \frac{3}{3-1} = \frac{3}{2}$$

$\therefore$ option (c) is correct.

Alternatively:-

$A = 1 + 2^P$

$B = 1 + 2^{-P} = 1 + \frac{1}{2^P}$

$= \frac{2^P + 1}{2^P} = \frac{A}{A-1} \quad (\because 1 + 2^P = 1)$

158.(b) Sum of roots $= a + b = \frac{-b}{a}$

Sum $= (ab + a + b + ab - a - b) = 2ab$

Sum $= \frac{2c}{a}$

Multiplication $= ab = \frac{c}{a}$

Product of roots $= (ab + a + b)(ab - a - b) = (ab)^2 - (a + b)^2$

$= \left(\frac{c}{a}\right)^2 - \left(\frac{-b}{a}\right)^2$

$= \frac{c^2 - b^2}{a^2}$

$x^2 - (\text{sum of roots})x +$

Multiplication of roots $= 0$

$x^2 - \frac{2c}{a}x + \frac{c^2 - b^2}{a^2} = 0$

$a^2x^2 - 2acx + c^2 - b^2 = 0$

159.(b) $\sqrt{(1 - p^2)(1 - q^2)} = \frac{\sqrt{3}}{2}$

put, $p = 0, q = \frac{1}{2}$

$\sqrt{(1 - 0^2)(1 - \frac{1}{4})} = \frac{\sqrt{3}}{2}$

Equation (i) is satisfied

$\sqrt{2p^2 + 2q^2 + 2pq} + \sqrt{2p^2 + 2q^2 - 2pq}$

$= \sqrt{0 + \frac{2}{4} + 0} + \sqrt{0 + \frac{2}{4} - 0}$

$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$

160.(b) $ab = 16$

Let, $a = 8, b = 2$

$(a + b)^2 - 2(a + B) = 80$

$(8 + 2)^2 - 2(8 + 2) = 80$

$8 \times 2 = 16$

...satisfied

So, $3a - 19b$

$3 \times 8 - 19 \times 2 = -14$

Alternatively:-

add 1 both sides

$(a + b)^2 - 2(a + b) + 1 = 80 + 1$

$[(a + b) - 1]^2 = 81$

$(a + b) - 1 = 9$

$a + b = 10$

$ab = 16$

Put, $a = 8, b = 2$

$3 \times 8 - 19 \times 2 = -14$

161.(b) Let, $x = 1, y = 2, z = 9$

$x^{y+z} = 1, y^{x+z} = 1024, z^{x+y} = 729$

Then,

$(z+1)^{y+x+1}$

$= (9+1)^4 = 10,000$

162.(b) $x^3 + y^3 + z^3 = 3, x^2 + y^2 + z^2 = 2,$

$x + y + z = 1$

square both side

$(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$

$\frac{-1}{2} = xy + yz + zx$

$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

$3 - 3xyz = (1) \left(2 + \frac{1}{2} \right)$

$-3xyz = \frac{5}{2} - 3$

$xyz = \frac{1}{6}$

SSC CGL MAINS 2017

{09 Mar. 2018} Paper-II

163.(c) If $(x-3)$ is a factor then for $x = 3$, $f(x)$ must be zero

put, $x = 3$ in eq.

$3^2 + 3p + 4$

$9 + 3p + 4 = 0$

$p = -\frac{13}{3}$

164.(a) $y^2 = y + 7$

multiply by y

$y^3 = y^2 + 7y$

putting, $y^2 = y + 7$

$y^3 = y + 7 + 7y = 8y + 7$

165.(a) $A = 125$ and $B = 8$

Let, $A + B = x$ $A - B = y$

$B = \frac{x - y}{2}$

$(A + B)^3 - (A - B)^3 - 6B(A^2 - B^2)$

$= x^3 - y^3 - 6 \frac{(x - y)}{2} \times xy$

$= x^3 - y^3 - 3xy(x - y)$

$= (x - y)^3$

$= (A + B - A + B)^3$

$= (2B)^3 = (2 \times 8)^3$

$= 16^3 = 4096$

166.(b) $x = \frac{1}{1 + \frac{1}{1 + x}}$

$x = \frac{x+1}{x+2}$

$x^2 + x - 1 = 0$

use formula $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \frac{-1 + \sqrt{5}}{2}$

$y = \frac{2}{2 + \frac{1}{1 + y}}$

$y = \frac{2y+2}{2y+2}$

$2y^2 + y - 2 = 0$

$y = \frac{-1 + \sqrt{17}}{4}$

$x + y = \frac{\sqrt{5} - 1}{2} + \frac{-1 + \sqrt{17}}{4}$

$= \frac{2\sqrt{5} + \sqrt{17} - 3}{4}$

167.(d) $x - \frac{1}{x} = 2$

$x^2 + \frac{1}{x^2} = 6$

$x + \frac{1}{x} = 2\sqrt{2}$

$x^3 + \frac{1}{x^3} = (2\sqrt{2})^3 - 3(2\sqrt{2})$

$x^3 + \frac{1}{x^3} = 16\sqrt{2} - 6\sqrt{2} = 10\sqrt{2}$

$x^3 - \frac{1}{x^3} = (2)^3 + 3(2) = 14$

$x^6 - \frac{1}{x^6} = (x^3 + \frac{1}{x^3})(x^3 - \frac{1}{x^3})$

$\Rightarrow x^6 - \frac{1}{x^6} = 14 \times 10\sqrt{2} = 140\sqrt{2}$

168.(a) $3x + 6y + 9z = \frac{20}{3}$... (i)

$6x + 9y + 3z = \frac{17}{3}$... (ii)

$18x + 27y - z = \frac{113}{9}$... (iii)

By $3 \times (ii) - (i)$

$15x + 21y = \frac{31}{3}$... (iv)

By $3 \times (iii) + (ii)$

$60x + 90y = \frac{130}{3}$

$6x + 9y = \frac{13}{3}$... (v)

solving (iv) and (v), we get

$x = \frac{2}{9}, y = \frac{1}{9}$

$75x + 113y = \frac{50}{3} + \frac{113}{3} = \frac{163}{3}$

169.(d) $x^{y^z} = 1, y^{z^x} = 125$ and z^{y^x}

$= 234$

$x = 1, y = 5, z = 3$ satisfies the condition

$9x + 10y - 18z$

$= 9 \times 1 + 10 \times 5 - 18 \times 3 = 5$

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{11 Sept. 2019}

170.(c) $x = \sqrt{1 + \frac{\sqrt{3}}{2}} - \sqrt{1 - \frac{\sqrt{3}}{2}}$

Multiply and divide by '4'

$x = \sqrt{\frac{4 + 2\sqrt{3}}{4}} - \sqrt{\frac{4 - 2\sqrt{3}}{4}}$

After making perfect square

$x = \frac{\sqrt{3} + 1}{2} - \frac{\sqrt{3} - 1}{2} \Rightarrow x = 1$

$\frac{\sqrt{2} - x}{\sqrt{2} + x} = \frac{\sqrt{2} - 1}{\sqrt{2} + 1}$

rationalize it

$= 3 - 2\sqrt{2} \quad (\because \sqrt{2} = 1.414)$

$= 0.17$

171.(a) $a^3 + b^3 = 218$

$a + b = 2$

$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$

$(b)^3 = 218 + 3ab(b)$

$8 = 218 + 6ab$

$ab = -35$

$$172.(a) \frac{(253)^3 + (247)^3}{25.3 \times 25.3 - 624.91 + 24.7 \times 24.7}$$

Multiply and divide by 100 and use

$$a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$$

$$= \frac{100(253+247)[(253)^2 + (247)^2 - 62491]}{(253)^2 - 62491 + (247)^2}$$

$$= 100 \times 500$$

$$= 50 \times 10^3 = 50 \times 10^K$$

After comparison $K = 3$

$$173.(c) x+y+z = 11$$

square both side

$$(x+y+z)^2 = x^2 + y^2 + z^2 + 2(xy+yz+zx)$$

$$121 = 133 + 2(xy+yz+zx)$$

$$xy+yz+zx = -6$$

$$x^3 + y^3 + z^3 - 3xyz = (x+y+z)$$

$$(x^2 + y^2 + z^2 - (xy+yz+zx))$$

$$881 - 3xyz = 11(133 - (-6))$$

$$881 - 3xyz = 11 \times 139$$

$$xyz = -216$$

$$\sqrt[3]{xyz} = -6$$

$$174.(b) \sqrt{86 - 60\sqrt{2}} = a - b\sqrt{2}$$

$$\begin{array}{cc} \sqrt{86 - 2\sqrt{1800}} & \\ \swarrow \quad \searrow & \swarrow \quad \searrow \\ 50 \times 36 & 50 \times 36 \end{array}$$

$$\sqrt{(\sqrt{50} - \sqrt{36})^2} = a - b\sqrt{2}$$

$$5\sqrt{2} - 6 = a - b\sqrt{2}$$

By comparing

$$a = -6, b = -5$$

$$\sqrt{a^2 + b^2} = \sqrt{36 + 25} = \sqrt{61} \approx 7.8$$

$$175.(c) x^3 - 1442x^4 + 1 = 0$$

divide by x^4

$$x^4 + \frac{1}{x^4} = 1442$$

$$x^2 + \frac{1}{x^2} = \sqrt{1442 + 2}$$

$$x^2 + \frac{1}{x^2} = 38$$

$$x - \frac{1}{x} = \sqrt{38 - 2} = 6$$

$$176.(a) 2\sqrt{2}x^3 - 3\sqrt{3}y^3 = (\sqrt{2}x - \sqrt{3}y)$$

$$(Ax^2 + By^2 + Cxy)$$

$$\Rightarrow (\sqrt{2}x)^3 - (\sqrt{3}y)^3 = (\sqrt{2}x - \sqrt{3}y)$$

$$(Ax^2 + By^2 + Cxy)$$

$$\Rightarrow (\sqrt{2}x - \sqrt{3}y)(2x^2 + 3y^2 + \sqrt{6}xy) =$$

$$(\sqrt{2}x - \sqrt{3}y)(Ax^2 + By^2 + Cxy)$$

After comparing

$$A = 2, B = 3, C = \sqrt{6}$$

$$A^2 + B^2 - C^2$$

$$(2)^2 + (3)^2 - (\sqrt{6})^2$$

$$= 13 - 6 = 7$$

$$177.(a) a^2 + b^2 + c^2 + 96 = 8(a+b+c)$$

$$a^2 + b^2 + c^2 - 8a - 8b + 16c + 96 = 0$$

make perfect square separately

$$(a-4)^2 + (b-4)^2 + (c+8)^2 = 0$$

$$a = 4, b = 4, c = -8$$

$$= \sqrt{ab - bc + ca}$$

$$= \sqrt{16 - (-32) + (-32)}$$

$$= \sqrt{16 + 32 - 32} = \sqrt{16} = 4$$

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{12 Sept. 2019}

$$178.(a) x + \frac{1}{16x} = 3$$

Cube both side-

$$\left(x + \frac{1}{16x}\right)^3 = (3)^3$$

$$x^3 + \frac{1}{(16x)^3} + 3 \times x \times \frac{1}{16x} \times (3) = 27$$

$$x^3 + \frac{1}{(16x)^3} = 27 - \frac{9}{16}$$

Multiply by 16 in the eq.

$$16x^3 + \frac{1}{(16x)^3} \times 16 = 27 \times 16 -$$

$$\frac{9}{16} \times 16$$

$$16x^3 + \frac{1}{256x^3} = 432 - 9 = 423$$

$$179.(a) x+y+z = 2$$

Square both side -

$$x^2 + y^2 + z^2 + 2(xy+yz+zx) = 4$$

$$x^2 + y^2 + z^2 + 2(-11) = 4$$

$$x^2 + y^2 + z^2 = 26$$

$$x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x^2 + y^2 + z^2)$$

$$-(xy+yz+zx)$$

$$x^3 + y^3 + z^3 - 3(-12) = 2(26 - (-11))$$

$$x^3 + y^3 + z^3 + 36 = 2(37)$$

$$x^3 + y^3 + z^3 = 38$$

$$\sqrt{x^3 + y^3 + z^3 - 2}$$

$$\sqrt{38 - 2} = 6$$

$$180.(b) x^4 - 83x^2 + 1 = 0$$

Divide eq by x^2

$$x^2 + \frac{1}{x^2} = 83$$

Subtract $\left(2 \times x \times \frac{1}{x}\right)$ both sides and make perfect square

$$x^2 + \frac{1}{x^2} - 2 \times x \times \frac{1}{x} = 83 - 2$$

$$x - \frac{1}{x} = 9$$

Cube both side

$$x^3 - \frac{1}{x^3} - 3 \times x \times \frac{1}{x} (9) = 729$$

$$x^3 - \frac{1}{x^3} = 729 + 27 = 756$$

$$181.(a) (5x+1)^3 + (x-3)^3 + 8(3x-4)^3 =$$

$$6(5x+1)(x-3)(3x-4)$$

$$(5x+1)^3 + (x-3)^3 + [2(3x-4)]^3 =$$

$$3(5x+1)(x-3)2(3x-4)$$

$$\text{If } a^3 + b^3 + c^3 = 3abc$$

$$a+b+c = 0$$

$$(5x+1) + (x-3) + 2(3x-4) = 0$$

$$12x = 10$$

$$x = \frac{5}{6}$$

$$182.(b) 8x^3 - 27y^3 = (Ax+By)(Cx^2 - Dy^2 + 6xy)$$

$$a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$$

$$(2x)^3 - (3y)^3 = (2x-3y)(4x^2 + 9y^2 + 6xy)$$

After comparing

$$A = 2, B = -3, C = 4, D = -9$$

$$A+B+C-D = 2-3+4-(-9) = 12$$

$$183.(a) x = \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}} \quad y = \frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} - \sqrt{3}}$$

$$x + y = \frac{2(5+3)}{5-3}$$

$$x + y = 8$$

$$xy = 1$$

$$x^3 + y^3 = (x+y)((x+y)^2 - 3xy)$$

$$= 8(64 - 3) = 8 \times 16 = 488$$

Alternatively:-

$$x = \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}}$$

Rationalization-

$$x = \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} - \sqrt{3}} \times \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} - \sqrt{3}}$$

$$x = \frac{5 + 3 - 2\sqrt{15}}{2}$$

$$x = 4 - \sqrt{15}$$

$$y = 4 + \sqrt{15} \text{ (y is reciprocal of x)}$$

$$x^3 + y^3 = (x+y)(x^2 + y^2 - xy)$$

$$\begin{aligned} &= (4 - \sqrt{15} + 4 + \sqrt{15}) [(4 - \sqrt{15})^2 + (4 + \sqrt{15})^2 + 45\sqrt{15}^2 - (4 - \sqrt{15})(4 + \sqrt{15})] \\ &= 8(16 + 15 - 8\sqrt{15} + 16 + 15 + 8\sqrt{15} - 1) \\ &= 8(62 - 1) = \mathbf{488} \end{aligned}$$

$$184.(b) \sqrt{10 - 2\sqrt{21}} + \sqrt{8 + 2\sqrt{15}} = \sqrt{a} + \sqrt{b}$$

Make perfect square

$$\sqrt{(\sqrt{7} - \sqrt{3})^2} + \sqrt{(\sqrt{5} + \sqrt{3})^2}$$

$$= \sqrt{a} + \sqrt{b}$$

$$\sqrt{7} - \sqrt{3} + \sqrt{5} + \sqrt{3} = \sqrt{a} + \sqrt{b}$$

$$\sqrt{7} + \sqrt{5} = \sqrt{a} + \sqrt{b}$$

Compare the equation.

$$a = 7 \quad b = 5$$

$$\sqrt{ab} = \sqrt{7 \times 5} = \sqrt{35} = \mathbf{5.9}$$

SSC CGL MAINS 2018**{13 Sept. 2019}**

$$185.(d) ab(a-b) + bc(b-c) + ca(c-a) = ?$$

Put,

$$c = 0, a = 2, b = 1$$

$$ab(a-b) + 0 + 0$$

$$2 \times 1(2-1) = 2$$

by option (d)

$$(b-a)(b-c)(c-a) = 2$$

$$186.(b) (5x-3)^3 + (2x+5)^3 + (3(4-3x))^3 =$$

$$9(3-5x)(2x+5)(3x-4)$$

$$(5x-3)^3 + (2x+5)^3 + (3(4-3x))^3 = 3(3-5x)(2x+5)(9x-12)$$

$$\text{If, } a^3 + b^3 + c^3 - 3abc = 0$$

$$\text{then, } a+b+c = 0$$

$$5x-3+2x+5+12-9x=0$$

$$2x=14 \Rightarrow x=7$$

$$(2x+1) = (2 \times 7 + 1) = \mathbf{15}$$

$$187.(b) 5\sqrt{3}x^3 + 2\sqrt{2}y^3 = (Ax + \sqrt{2}y)$$

$$(Bx^2 + 2y^2 + Cxy)$$

$$a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$$

$$(\sqrt{5}x)^3 + (\sqrt{2}y)^3$$

$$= (\sqrt{5}x + \sqrt{2}y)(5x^2 + 2y^2 - \sqrt{10}xy)$$

$$= (Ax + \sqrt{2}y)(Bx^2 + 2y^2 + Cxy)$$

After Comparing

$$A = \sqrt{5}, B = 5, C = -\sqrt{10}$$

$$\Rightarrow (A^2 + B^2 - C^2) = 5 + 25 - 10 = \mathbf{20}$$

$$188.(d) \frac{3x^2 + 3 - 7x}{3x} = 6$$

$$x + \frac{1}{x} - \frac{7}{3} = 6$$

$$x + \frac{1}{x} = 6 + \frac{7}{3}$$

$$x + \frac{1}{x} + 2 = 8 + \frac{7}{3}$$

$$\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = \frac{31}{3}$$

$$\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{\frac{31}{3}}$$

$$189.(a) x = \sqrt[5]{27} - \sqrt[6]{\frac{3}{4}}$$

$$y = \frac{\sqrt{45} + \sqrt{605} + \sqrt{245}}{\sqrt{80} + \sqrt{125}}$$

$$x = (3^3)^{\frac{1}{6}} - \sqrt[6]{\frac{27}{4}}$$

$$x = \sqrt{3} - \frac{3}{2}\sqrt{3} = \frac{-\sqrt{3}}{2}$$

$$y = \frac{\sqrt{5} \times \sqrt{9} + \sqrt{5} \times \sqrt{121} + \sqrt{5} \times \sqrt{49}}{\sqrt{5} \times \sqrt{16} + \sqrt{5} \times \sqrt{25}}$$

$$= \frac{3 + 11 + 7}{4 + 5} = \frac{21}{9} = \frac{7}{3}$$

$$x^2 + y^2 = \frac{3}{4} + \frac{49}{9} = \frac{27 + 196}{36} = \frac{\mathbf{223}}{\mathbf{36}}$$

$$190.(a) x + y + z = 6$$

square both side

$$x^2 + y^2 + z^2 + 2(xy + yz + zx) = 36$$

$$30 + 2(xy + yz + zx) = 36$$

$$xy + yz + zx = 3$$

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)[(x^2 + y^2 + z^2) - (xy + yz + zx)]$$

$$x^3 + y^3 + z^3 - 3(-10) = 6(30 - 3)$$

$$x^3 + y^3 + z^3 + 30 = 162$$

$$x^3 + y^3 + z^3 = \mathbf{132}$$

$$191.(d) \frac{(4.6)^4 + (5.4)^4 + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$$

$$4.6 = a, 5.4 = b, [a+b = 10]$$

$$ab = 24.84$$

$$\frac{(a^2)^2 + (b^2)^2 + 2(ab)^2 - (ab)^2}{a^2 + b^2 + ab}$$

$$\frac{(a^2 + b^2)^2 - (ab)^2}{a^2 + b^2 + ab} [a^2 - b^2 = (a+b)(a-b)]$$

$$\frac{(a^2 + b^2 + ab)(a^2 + b^2 - ab)}{(a^2 + b^2 + ab)}$$

$$= a^2 + b^2 - ab$$

Add and subtract (2ab)

$$= a^2 + b^2 + 2ab - 3ab$$

$$= (a+b)^2 - 3ab$$

$$= (10)^2 - 3 \times 24.84$$

$$= 100 - 74.52$$

$$= \mathbf{25.48}$$

SSC CGL MAINS 2019**{15 Nov. 2020}**

$$192.(b) 27(x+y)^3 - 8(x-y)^3$$

$$= (x+5y)(Ax^2 + By^2 + Cxy)$$

$$[A^3 - B^3 = (A-B)(A^2 + B^2 + AB)]$$

Take L.H.S.

$$(3x+3y)^3 - (2x-2y)^3$$

$$\Rightarrow (3x+3y-2x+2y)[(9x^2+9y^2+18xy) + (4x^2+4y^2-8xy) + (3x+3y)(2x-2y)]$$

$$\Rightarrow (x+5y)[9x^2+9y^2+18xy+4x^2+4y^2-8xy+6x^2-6xy+6xy-6y^2]$$

$$\Rightarrow (x+5y)(19x^2+7y^2+10xy)$$

$$\text{now compare with } (x+5y)(Ax^2 + By^2 + Cxy)$$

$$A = 19$$

$$B = 7$$

$$C = 10$$

Now the value of (A + B - C)

$$= 19 + 7 - 10 = \mathbf{16}$$

$$193.(a) x^8 - 34x^4 + 1 = 0$$

divide by x^4

$$x^4 + \frac{1}{x^4} = 34$$

Add 2 both side and make square root

$$x^2 + \frac{1}{x^2} = \sqrt{34+2}$$

$$x^2 + \frac{1}{x^2} = 6$$

Again add 2 both side and make square root

$$x + \frac{1}{x} = \sqrt{6+2} = 2\sqrt{2}$$

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$$

$$= (2\sqrt{2})^3 - 3(2\sqrt{2})$$

$$= 16\sqrt{2} - 6\sqrt{2} = 10\sqrt{2} = 5\sqrt{8}$$

194.(d) $3x^2 - 5x + 1 = 0$

divide by $3x$

$$x + \frac{1}{3x} = \frac{5}{3}$$

squaring both side

$$x^2 + \frac{1}{9x^2} + \frac{2}{3} = \frac{25}{9}$$

$$x^2 + \frac{1}{9x^2} = \frac{19}{9} = 2\frac{1}{9}$$

195.(c) $\frac{1}{4-\sqrt{8}} + \frac{3+2\sqrt{2}}{3-2\sqrt{2}} - \frac{3-2\sqrt{2}}{3+2\sqrt{2}}$

$$= a + b\sqrt{2}$$

$$\Rightarrow \frac{1}{4-\sqrt{8}} + \frac{(3+2\sqrt{2})^2 - (3-2\sqrt{2})^2}{(3-2\sqrt{2})(3+2\sqrt{2})}$$

$$= a + b\sqrt{2}$$

$$\Rightarrow \frac{1}{4-\sqrt{8}} + (9+8+12\sqrt{2}) - (9+8-12\sqrt{2}) = a + b\sqrt{2}$$

$$\Rightarrow \frac{4+\sqrt{8}}{8} + 24\sqrt{2} = a + b\sqrt{2}$$

$$\Rightarrow \frac{1}{2} + \frac{97\sqrt{2}}{4} = a + b\sqrt{2}$$

After comparing

$$a = \frac{1}{2}, b = \frac{97}{4}$$

$$\therefore 3a + 4b = 3 \times \frac{1}{2} + 4 \times \frac{97}{4}$$

$$= \frac{3}{2} + 97 = 98\frac{1}{2}$$

196.(b) $a^3 + b^3 + c^3 - 3abc$

$$= (a+b+c)[(a+b+c)^2 - 3(ab+bc+ca)]$$

$$342 = 6[36 - 3(ab+bc+ca)]$$

$$57 = [36 - 3(ab+bc+ca)]$$

$$ab+bc+ca = -7$$

197.(a) $(3x+y)^2 = 9x^2 + y^2 + 6xy$

$$(3x+y)^2 = 37+12$$

$$(3x+y)^2 = 49$$

$$3x+y = 7$$

$$27x^3 + y^3 = (3x+y)(9x^2 + y^2 - 3xy)$$

$$= 7[37 - 6] = 217$$

Alternatively:-

$$xy = 2 = 2 \times 1$$

$$x = 2, y = 1$$

Now verify the values in equation

$$9x^2 + y^2 = 37$$

$$9 \times (2)^2 + (1)^2 = 37$$

$$37 = 37$$

...satisfied

$$27x^3 + y^3 = 27 \times (2)^3 + (1)^3$$

$$= 216 + 1 = 217$$

SSC CGL MAINS 2019

{16 Nov. 2020}

198.(b) $a + b + c = 7$

$$a^3 + b^3 + c^3 - abc = 175$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = (a+b+c)[(a+b+c)^2 - 3(ab+bc+ca)]$$

$$\Rightarrow 175 = 7[(7)^2 - 3(ab+bc+ca)]$$

$$\Rightarrow 25 = 49 - 3(ab+bc+ca)$$

$$\Rightarrow -24 = -3(ab+bc+ca)$$

$$\Rightarrow ab + bc + ca = 8$$

199.(c) $x^2 + 4y^2 = 17$

$$xy = 2$$

$$\text{Put, } y = 2, x = 1$$

$$\therefore x^3 + 8y^3 = 1 + 64 = 65$$

Alternatively:-

$$(x+2y)^2 = x^2 + 4y^2 + 4xy$$

$$= 17 + 8 = 25$$

$$x + 2y = \sqrt{25} = 5$$

$$x^3 + 8y^3 = (x+2y)(x^2 + 4y^2 - 2xy)$$

$$= 5(17-4) = 5 \times 13 = 65$$

200.(a) $(x+y)^3 + 8(x-y)^3 = (3x+Ay)$

$$(3x^2 + Bxy + Cy^2)$$

$$[A^3 + B^3 = (A+B)(A^2 + B^2 - AB)]$$

$$\Rightarrow (x+y+2x-2y)[x^2 + y^2 + 2xy + 4x^2 + 4y^2 - 8xy - 2x^2 + 2y^2] = (3x+Ay)(3x^2 + Bxy + Cy^2)$$

$$\Rightarrow (3x-y)[3x^2 + 7y^2 - 6xy]$$

$$= (3x+Ay)(3x^2 + Bxy + Cy^2)$$

$$\Rightarrow (3x-y)[3x^2 + 7y^2 - 6xy]$$

$$= (3x+Ay)(3x^2 + Bxy + Cy^2)$$

After comparing-

$$A = -1, B = -6, C = 7$$

$$A + B + C = -1 - 6 + 7 = 0$$

201.(b) $2x^2 - 7x + 5 = 0$

Divide by $2x$

$$x + \frac{5}{2x} = \frac{7}{2}$$

Square both side

$$x^2 + \frac{25}{4x} + 5 = \frac{49}{4}$$

$$x^2 + \frac{25}{4x} = \frac{49}{4} - 5 = \frac{29}{4} = 7\frac{1}{4}$$

202.(a) $\frac{x^6 + 3x^3 - 1}{x^6 - 8x^3 - 1}$
Divide by x^3

$$\text{Que} \Rightarrow \frac{x^3 - \frac{1}{x^3} + 3}{x^3 - \frac{1}{x^3} - 8}$$

$$x - \frac{1}{x} = 5$$

Cube both side

$$x^3 - \frac{1}{x^3} = 125 + 3 \times 5 = 140$$

Put, the value in que.

$$\frac{140+3}{140-8} = \frac{143}{132} = \frac{13}{12}$$

SSC CGL MAINS 2019

{18 Nov. 2020}

203.(b) $x^2 + \frac{1}{x^2} = 7$

add 2 on both sides

$$x^2 + \frac{1}{x^2} + 2 = 7 + 2$$

Do square root-

$$x + \frac{1}{x} = 3$$

cube both side-

$$x^3 + \frac{1}{x^3} = 3^3 - 3 \times 3$$

$$= 27 - 9 = 18$$

204.(c) $x \left(3 - \frac{2}{x}\right) = \frac{3}{x} \Rightarrow 3x - 2 = \frac{3}{x}$

$$3 \left(x - \frac{1}{x}\right) = 2$$

$$x - \frac{1}{x} = \frac{2}{3}$$

cube both side

$$x^3 - \frac{1}{x^3} = \left(\frac{2}{3}\right)^3 + 3 \times \frac{2}{3}$$

$$= \frac{8}{27} + 2 = \frac{62}{27}$$

205.(d) $x + \frac{16}{x} = 8$

Put, $x = 4$

$$4 + \frac{16}{4} = 8$$

...satisfied

$$x^2 + \frac{32}{x^2} = 4^2 + \frac{32}{4^2}$$

$$= 16 + \frac{32}{16} = 16 + 2 = 18$$

$$\begin{aligned}
 206.(a) \quad x &= \sqrt{-\sqrt{3} + \sqrt{3+8\sqrt{7+4\sqrt{3}}}} \\
 &= \sqrt{-\sqrt{3} + \sqrt{3+8\sqrt{7+2 \times 2\sqrt{3}}}} \\
 &= \sqrt{-\sqrt{3} + \sqrt{3+8\sqrt{(2+\sqrt{3})^2}}} \\
 &= \sqrt{-\sqrt{3} + \sqrt{19+2 \times 4\sqrt{3}}} \\
 &= \sqrt{-\sqrt{3} + \sqrt{(4+\sqrt{3})^2}} \\
 &= \sqrt{4} = 2
 \end{aligned}$$

$$\begin{aligned}
 207.(c) \quad \sqrt{x} + \frac{1}{\sqrt{x}} &= 3 \\
 \text{squaring both sides, we get}
 \end{aligned}$$

$$x + \frac{1}{x} = 3^2 - 2 = 9 - 2$$

$$x + \frac{1}{x} = 7$$

cube both side

$$x^3 + \frac{1}{x^3} = 7^3 - 3 \times 7$$

$$x^3 + \frac{1}{x^3} = 343 - 21 = 322$$

$$[\text{If } x + \frac{1}{x} = P$$

$$\text{then, } x^3 + \frac{1}{x^3} = P^3 - 3P]$$

$$208.(a) \quad x - \frac{3}{x} = 6 \quad \dots(i)$$

$$\frac{x^4 - \frac{27}{x^2}}{x^2 - 3x - 3} = ?$$

[divide Numerator and Denominator by x]

$$\frac{x^3 - \frac{27}{x^3}}{x - \frac{3}{x} - 3} \Rightarrow \frac{x^3 - \left(\frac{3}{x}\right)^3}{\left(x - \frac{3}{x}\right) - 3}$$

make cube of eq. (i)

$$x^3 - \left(\frac{3}{x}\right)^3 = 6^3 + 3 \times 6 \times 3$$

$$= 216 + 54 = 270$$

Put the value in eq. (ii)

$$= \frac{270}{6-3} = \frac{270}{3} = 90$$

$$209.(b) \quad \text{LHS} = \frac{(8+2\sqrt{3})}{(3\sqrt{3}+5)}$$

Rationalize it-

$$= \frac{(8+2\sqrt{3})}{(3\sqrt{3}+5)} \times \frac{(3\sqrt{3}-5)}{(3\sqrt{3}-5)}$$

$$\begin{aligned}
 &= \frac{(8+2\sqrt{3}) \times (3\sqrt{3}-5)}{27-25} \\
 [\because (a+b)(a-b) &= a^2 - b^2] \\
 &= (4+\sqrt{3}) \times (3\sqrt{3}-5) \\
 &= 12\sqrt{3} - 20 + 9 - 5\sqrt{3} \\
 &= 7\sqrt{3} - 11 \\
 \Rightarrow 7\sqrt{3} - 11 &= a\sqrt{3} - b \\
 (\text{compare with R.H.S}) \\
 a &= 7, b = 11 \\
 \Rightarrow a + b &= 7 + 11 = 18
 \end{aligned}$$

SSC CGL MAINS 2020

{29 Jan. 2022}

$$210.(d) \quad 2x - y = 2 \quad xy = \frac{3}{2}$$

Divide by 2

$$\Rightarrow x - \frac{y}{2} = 1$$

Cube both sides

$$\left(x - \frac{y}{2}\right)^3 = 1^3$$

$$x^3 - \frac{y^3}{8} - 3 \times x \times \frac{y}{2} \left(x - \frac{y}{2}\right) = 1$$

$$x^3 - \frac{y^3}{8} - \frac{3}{2} \times \frac{3}{2} \times 1 = 1$$

$$x^3 - \frac{y^3}{8} = 1 + \frac{9}{4} = \frac{13}{4}$$

$$211.(a) \quad 2x^2 + 5x + 1 = 0$$

$$2x + \frac{1}{x} = -5$$

$$4x^2 + \frac{1}{x^2} = 25 - 2 \times 2 = 21$$

$$2x - \frac{1}{x} = \sqrt{21 - 2 \times 2} = \sqrt{17}$$

$$x - \frac{1}{2x} = \frac{\sqrt{17}}{2}$$

$$212.(b) \quad a+b+c = 1$$

$$ab + bc + ca = -22$$

$$a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca) \\ = 1 - 2(-22) = 45$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c) \\ (a^2 + b^2 + c^2 - ab - bc - ca)$$

$$a^3 + b^3 + c^3 - 3(-40) = (1)(45+22)$$

$$a^3 + b^3 + c^3 = -120 + 67 = -53$$

$$213.(c) \quad a+b = 8$$

$$ab = 10$$

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$64 = a^2 + b^2 + 20$$

$$a^2 + b^2 = 44$$

$$a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$$

$$= (8)(44-10)$$

$$= 34 \times 8 = 272$$

$$214.(d) \quad x = \sqrt{1 + \frac{\sqrt{3}}{2}} - \sqrt{1 - \frac{\sqrt{3}}{2}}$$

$$x = \sqrt{\frac{2+\sqrt{3}}{2}} - \sqrt{\frac{2-\sqrt{3}}{2}}$$

Multiply and divide by 2

$$x = \sqrt{\frac{4+2\sqrt{3}}{4}} - \sqrt{\frac{4-2\sqrt{3}}{4}}$$

$$x = \frac{\sqrt{3}+1}{2} - \left(\frac{\sqrt{3}-1}{2}\right)$$

$$x = \frac{1}{2} + \frac{1}{2} = 1$$

$$\Rightarrow \frac{\sqrt{3}-x}{\sqrt{3}+x} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)^2}{2}$$

$$= \frac{4-2\sqrt{3}}{2} = 2-\sqrt{3}$$

$$= 2-1.732$$

$$= 0.27$$

$$215.(c) \quad \frac{1}{4-\sqrt{15}} - \frac{1}{\sqrt{15}-\sqrt{14}} + \dots -$$

$$\frac{1}{3-\sqrt{8}}$$

Rationalize each part:-

$$= 4 + \sqrt{15} - \sqrt{15} + \sqrt{14} - \sqrt{14} + \sqrt{13}$$

$$\dots 9 - \sqrt{8}$$

$$= 4 - \sqrt{8}$$

$$= 4 - 2\sqrt{2}$$

$$216.(a) \quad x^2 - \sqrt{7}x + 1 = 0$$

$$x + \frac{1}{x} = \sqrt{7}$$

$$x^2 + \frac{1}{x^2} = 5$$

$$x^3 + \frac{1}{x^3} = 7\sqrt{7} - 3\sqrt{7} = 4\sqrt{7}$$

$$x^5 + \frac{1}{x^5} = \left(x^2 + \frac{1}{x^2}\right) \left(x^3 + \frac{1}{x^3}\right) - \left(x + \frac{1}{x}\right)$$

$$= (5)(4\sqrt{7}) - \sqrt{7}$$

$$= 19\sqrt{7}$$

$$217.(a) \frac{22\sqrt{2}}{4\sqrt{2}-\sqrt{3}+\sqrt{5}} = a + \sqrt{5}b$$

Multiply by $\sqrt{2}$ in numerator and denominator

$$\frac{44}{8-\sqrt{6}+2\sqrt{5}} = a + \sqrt{5}b$$

$$\frac{44}{8-(\sqrt{5}+1)} = a + \sqrt{5}b$$

$$\frac{44}{(7-\sqrt{5})} \times \frac{(7+\sqrt{5})}{(7+\sqrt{5})} = a + \sqrt{5}b$$

$$\frac{44 \times (7+\sqrt{5})}{44} = a + \sqrt{5}b$$

$$7 + \sqrt{5} = a + \sqrt{5}b$$

After comparing:-

$$a = 7, \quad b = 1$$

$$ab : a+b = 7 : 8$$

SSC CGL MAINS 2020

{03 Feb. 2022}

$$218.(a) x^2 - 3x + 1 = 0$$

divide by x

$$x + \frac{1}{x} = 3$$

$$\Rightarrow \frac{x^4 + \frac{1}{x^2}}{x^2 + 5x + 1}$$

Divide num. and den. by x

$$\frac{x^3 + \frac{1}{x^3}}{x + \frac{1}{x} + 5}$$

$$\Rightarrow \frac{x^3 + \frac{1}{x^3}}{x + \frac{1}{x} + 5}$$

$$\therefore \frac{(3)^3 - 3(3)}{3+5} = \frac{18}{8} = \frac{9}{4}$$

$$219.(a) x^3 + y^3 + z^3 - 3xyz = \frac{1}{2}(x+y+z)[x^2 - y^2 + (y-z)^2 + (z-x)^2]$$

$$= \frac{1}{2}(98)[(2.1)^2 + (3.7)^2 + (1.6)^2]$$

$$\Rightarrow \frac{98}{2} \times 20.66 = \frac{98K}{100}$$

$$\Rightarrow 10.33 \times 100 = K$$

$$K = 1033$$

$$220.(b) (3x)^3 - (4y)^3, \quad [a^3 - b^3 = (a-b)(a^2 + b^2 + ab)]$$

$$(3x-4y)(9x^2 + 16y^2 + 12xy)$$

$$= (Ax+By)(Cx^2-Dy^2 + 12xy)$$

After comparing:-

$$A = 3$$

$$B = -4$$

$$C = 9$$

$$D = -16$$

$$\Rightarrow 4A + B + 3C + 2D$$

$$= 12 - 4 + 27 - 32 = 3$$

$$221.(a) \text{ First solve the power}$$

$$\{[(200-146)/(3 \times 3 \times 3)] - 3\}$$

$$= \left(\frac{200-146}{27} \right) - 3$$

$$= \left(\frac{54}{27} \right) - 3$$

$$= 2 - 3 = -1$$

$$\therefore (0.3)^{-1}$$

$$= \frac{1}{(0.3)^1} = \frac{1}{0.3} = \frac{10}{3}$$

$$222.(a) \frac{\sqrt{26-7\sqrt{3}}}{\sqrt{14+5\sqrt{3}}} = \frac{b+a\sqrt{3}}{11}$$

Multiply and divide by 2

$$\frac{\sqrt{52-2.7\sqrt{3}}}{\sqrt{28+2.5\sqrt{3}}}$$

$$= \frac{\sqrt{(7-\sqrt{3})^2}}{\sqrt{(5+\sqrt{3})^2}}$$

$$= \frac{7-\sqrt{3}}{5+\sqrt{3}} \times \frac{5-\sqrt{3}}{5-\sqrt{3}}$$

$$= \frac{35-7\sqrt{3}-5\sqrt{3}+3}{25-3}$$

$$\Rightarrow \frac{38-12\sqrt{3}}{22} = \frac{b+a\sqrt{3}}{11}$$

$$\Rightarrow \frac{19-6\sqrt{3}}{11} = \frac{b-a\sqrt{3}}{11}$$

After comparing:-

$$b = 19 \text{ and } a = -6$$

$$\Rightarrow \sqrt{(b-a)} = \sqrt{19+6} = 5$$

$$223.(c) \frac{\sqrt{38-5\sqrt{3}}}{\sqrt{26+7\sqrt{3}}} = \frac{a+b\sqrt{3}}{23}$$

Multiply and divide by 2

$$\frac{\sqrt{76-2 \times 5\sqrt{3}}}{\sqrt{52+2 \times 7\sqrt{3}}} = \frac{\sqrt{(5\sqrt{3}-1)^2}}{\sqrt{(7+\sqrt{3})^2}}$$

$$= \frac{5\sqrt{3}-1}{7+\sqrt{3}} \times \frac{7-\sqrt{3}}{7-\sqrt{3}}$$

$$= \frac{35\sqrt{3}-15-7+\sqrt{3}}{46}$$

$$= \frac{36\sqrt{3}-22}{46}$$

$$\Rightarrow \frac{18\sqrt{3}-11}{23} = \frac{b\sqrt{3}+a}{23}$$

After comparing:-

$$a = -11, \quad b = 18$$

$$\Rightarrow b-a = 18-(-11) = 29$$

$$224.(b) x + \frac{1}{x} = 3$$

$$x^2 + \frac{1}{x^2} = 3^2 - 2 = 7$$

$$x^4 + \frac{1}{x^4} = 7^2 - 2 = 47$$

$$x^3 + \frac{1}{x^3} = (3)^3 - 3(3) = 18$$

$$\Rightarrow x^7 + \frac{1}{x^7} = \left(x^4 + \frac{1}{x^4}\right)\left(x^3 + \frac{1}{x^3}\right) - \left(x + \frac{1}{x}\right)$$

$$= (47 \times 18) - 3$$

$$= 846 - 3 = 843$$

$$225.(a) x^2 + y^2 + xy = \frac{17 \frac{1}{16}}{5 \frac{1}{4}} = \frac{13}{4}$$

$$(x^2 + y^2 + xy)(x^2 + y^2 - xy) = x^4 + y^4 + x^2y^2$$

$$x^2 + y^2 + xy = \frac{13}{4}$$

$$x^2 + y^2 - xy = \frac{21}{4} \quad (\text{Add})$$

$$2(x^2 + y^2) = \frac{34}{4}$$

$$x^2 + y^2 = \frac{17}{4}$$

$$\therefore xy = \frac{13}{4} - \frac{17}{4} = -1$$

$$\therefore (x-y)^2 = x^2 + y^2 - 2xy$$

$$= \frac{17}{4} - 2(-1) = \frac{25}{4}$$

$$\Rightarrow x-y = \sqrt{\frac{25}{4}} = \frac{5}{2}$$